```r
library(tidyverse)
```

```
## Warning: package 'dplyr' was built under R version 4.4.2
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```r
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.2
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```r
data <- read.csv("D:/First Semester/Project Learning from Data/final data.txt", sep=";")

head(data)
```

```
##       Region Crop Lutum Sand Silt   pH    C Cstock Years
## 1 Leemstreek   PG  0.23 0.25 0.52 6.12 2.43 118.38    15
## 2     Kempen   SM  0.26 0.54 0.20 4.95 2.15  84.99    16
## 3 Leemstreek   PG  0.22 0.23 0.55 6.17 2.41 118.28    14
## 4     Kempen SMGR  0.15 0.60 0.25 4.95 2.15  94.72    12
## 5     Kempen   GM  0.14 0.63 0.23 5.23 1.91  82.81    14
## 6 Leemstreek   GM  0.29 0.20 0.51 6.22 1.73  94.01    14
```

```r
print(data)
```

```
##         Region Crop Lutum Sand Silt   pH      C    Cstock Years
## 1   Leemstreek   PG  0.23 0.25 0.52 6.12   2.43    118.38    15
## 2       Kempen   SM  0.26 0.54 0.20 4.95   2.15     84.99    16
## 3   Leemstreek   PG  0.22 0.23 0.55 6.17   2.41    118.28    14
## 4       Kempen SMGR  0.15 0.60 0.25 4.95   2.15     94.72    12
## 5       Kempen   GM  0.14 0.63 0.23 5.23   1.91     82.81    14
## 6   Leemstreek   GM  0.29 0.20 0.51 6.22   1.73     94.01    14
## 7   Leemstreek SMGR  0.27 0.24 0.50 5.80   2.10     97.40    15
## 8       Kempen   SM  0.08 0.64 0.28 5.32   2.18     81.72    10
## 9   Leemstreek   GM  0.25 0.23 0.52 5.90   1.88    104.06    21
## 10  Leemstreek SMGR  0.25 0.26 0.49 6.06   1.97    102.18    17
## 11      Kempen   GM  0.23 0.62 0.14 5.14   2.14     78.95    19
## 12  Leemstreek   PG  0.34 0.26 0.41 5.91 252.00    122.77    12
## 13      Kempen   SM  0.15 0.63 0.22 5.15   2.13     85.81    10
## 14  Leemstreek   GM  0.24 0.25 0.51 5.75   1.70     99.11    13
## 15  Leemstreek   SM  0.25 0.24 0.52 5.97   1.76     97.24    10
## 16      Kempen   SM  0.29 0.47 0.24 5.27   1.97     88.42    24
## 17  Leemstreek   PG  0.26 0.25 0.49 5.57   2.45    115.76    17
## 18  Leemstreek   GM  0.16 0.26 0.57 5.70   1.82     96.06    22
## 19  Leemstreek   GM  0.29 0.27 0.44 5.77   1.83     93.58    18
## 20      Kempen   SM  0.24 0.57 0.18 4.78   2.10     86.64    12
## 21      Kempen   GM  0.13 0.69 0.18 5.18   2.00     82.63    10
## 22      Kempen SMGR  0.14 0.61 0.24 4.90   2.46     94.30    16
## 23      Kempen SMGR  0.18 0.61 0.22 4.67   2.25     97.85    13
## 24      Kempen   GM  0.09 0.66 0.24 4.97   2.16     82.63    22
## 25      Kempen   PG  0.22 0.56 0.21 5.02   2.72    104.07    18
## 26     Keempen SMGR  0.22 0.58 0.20 5.27   2.19     87.03    18
## 27  Leemstreek   GM  0.20 0.32 0.48 5.78   1.78     95.26    24
## 28  Leemstreek   GM  0.36 0.25 0.39 6.23   1.88     96.08    11
## 29  Leemstreek   SM  0.27 0.30 0.43 6.08   1.77    105.89    10
## 30  Leemstreek   PG  0.38 0.21 0.41 6.07   2.52    116.03    18
## 31      Kempen   PG  0.16 0.59 0.25 4.93   2.58    101.93    12
## 32      Kempen SMGR  0.18 0.62 0.20 5.49   2.16     88.65    14
## 33      Kempen   GM  0.14 0.62 0.24 5.07   1.94     84.35    12
## 34      Kempen   PG  0.18 0.55 0.27 4.74   2.83    109.17    12
## 35  Leemstreek   SM  0.26 0.25 0.49 5.96   1.79     94.84    19
## 36  Leemstreek SMGR  0.27 0.22 0.52 5.61   1.87    104.71    11
## 37      Kempen   SM  0.20 0.64 0.17 5.04   2.20     83.95    19
## 38  Leemstreek   GM  0.27 0.28 0.45 5.92   1.84    100.83    21
## 39  Leemstreek   SM  0.31 0.18 0.50 6.14   1.73    102.47    16
## 40  Leemstreek   SM  0.19 0.29 0.52 6.20   1.83     97.15    21
## 41  Leemstreek   SM  0.33 0.21 0.45 5.85   1.70     98.45    11
## 42  Leemstreek SMGR  0.08 0.26 0.66 6.16   1.86    106.81    19
## 43  Leemstreek SMGR  0.26 0.27 0.47 6.28   2.03     98.07    14
## 44  Leemstreek   SM  0.25 0.24 0.50 5.84   1.91     97.09    13
## 45  Leemstreek   PG  0.23 0.23 0.54 5.87   2.54    122.46    14
## 46  Leemstreek SMGR  0.25 0.25 0.50 6.13   1.85    102.34    19
## 47  Leemstreek SMGR  0.28 0.26 0.46 6.04   1.75    104.08    16
## 48  Leemstreek   SM  0.16 0.28 0.56 6.11   1.71     98.24    20
## 49  Leemstreek   GM  0.21 0.19 0.60 6.28   1.83     93.61    24
## 50      Kempen SMGR  0.26 0.52 0.22 5.18   2.17     86.36    18
```

```
## 51  Leemstreek   GM  0.17 0.29 0.53 6.21   1.58     95.58    14
## 52  Leemstreek   GM  0.22 0.28 0.49 6.12   1.74     95.01    21
## 53      Kempen   SM  0.13 0.62 0.25 5.19   2.21     87.31    18
## 54  Leemstreek   SM  0.30 0.27 0.43 6.08   1.85    104.13    12
## 55  Leemstreek   SM  0.08 0.28 0.64 6.26   1.73     98.31    14
## 56  Leemstreek   PG  0.37 0.17 0.46 5.80   2.32    121.78    12
## 57  Leemstreek   PG  0.30 0.28 0.41 5.78   2.38    119.99    12
## 58      Kempen SMGR  0.21 0.58 0.21 5.22   2.20     92.43    20
## 59      Kempen   SM  0.16 0.61 0.22 5.08   2.17     84.95    21
## 60  Leemstreek   SM  0.16 0.24 0.60 6.01   1.64     99.20    17
## 61      Kempen   PG  0.11 0.64 0.25 5.25   2.83    110.38    12
## 62  Leemstreek   GM  0.25 0.24 0.51 6.15   1.75     92.58    23
## 63  Leemstreek   GM  0.32 0.27 0.42 5.95   1.79    101.79    25
## 64  Leemstreek   SM  0.31 0.24 0.45 5.94   1.67     99.62    10
## 65      Kempen SMGR  0.24 0.55 0.22 5.11   2.14     90.81    18
## 66  Leemstreek   SM  0.06 0.29 0.65 6.21   1.76     96.21    22
## 67      Kempen   SM  0.20 0.64 0.16 5.19   2.18     83.52    18
## 68      Kempen SMGR  0.06 0.69 0.25 5.39   2.28     91.60    14
## 69      Kempen   SM  0.19 0.60 0.20 5.10   2.13     76.28    17
## 70  Leemstreek   PG  0.12 0.28 0.60 6.23   2.58    118.06    14
## 71      Kempen   SM  0.12 0.70 0.19 5.28   2.05     86.39    25
## 72      Kempen   PG  0.19 0.59 0.22 5.22   2.83    105.03    20
## 73      Kempen   GM  0.28 0.53 0.19 5.18   2.04     79.10    21
## 74  Leemstreek SMGR  0.20 0.24 0.55 5.94   1.85    101.88    10
## 75  Leemstreek SMGR  0.17 0.24 0.59 5.98   1.97    105.19    16
## 76  Leemstreek   GM  0.13 0.25 0.62 6.05   1.86     90.10    17
## 77  Leemstreek SMGR  0.15 0.30 0.55 6.01   1.97    107.62    12
## 78      Kempen   SM  0.14 0.58 0.28 5.29   2.06     75.92    14
## 79  Leemstreek   SM  0.16 0.26 0.57 5.98   1.76     96.39    12
## 80  Leemstreek SMGR  0.26 0.24 0.49 5.99   2.01    101.31    16
## 81  Leemstreek   SM  0.22 0.25 0.53 5.90   1.83     93.54    13
## 82      Kempen   PG  0.19 0.62 0.19 5.07   2.81    102.02    18
## 83  Leemstreek   GM  0.22 0.29 0.49 5.99   1.72     97.25    23
## 84      Kempen SMGR  0.14 0.61 0.26 4.64   2.29     89.57    18
## 85   Lemstreek   PG  0.20 0.27 0.53 5.73   2.38    119.11    16
## 86  Leemstreek   SM  0.20 0.27 0.53 5.99   1.85     89.48    14
## 87      Kempen   SM  0.17 0.65 0.19 5.05   2.03     79.08    13
## 88      Kempen SMGR  0.23 0.57 0.20 4.86   2.00     89.57    11
## 89      Kempen   GM  0.12 0.68 0.20 4.70   1.94     78.70    22
## 90  Leemstreek   PG  0.24 0.24 0.52 5.93   2.46    117.26    24
## 91  Leemstreek   PG  0.31 0.25 0.44 6.13   2.52    124.32    24
## 92      Kempen   SM  0.09 0.67 0.24 5.45   1.96     84.86    14
## 93        <NA>   GM  0.28 0.29 0.43 5.99   1.65     92.33    14
## 94      Kempen SMGR  0.18 0.55 0.27 5.40   2.34     88.40    15
## 95  Leemstreek   SM  0.25 0.22 0.53 6.02   1.76     97.22    15
## 96  Leemstreek   PG  0.33 0.21 0.46   NA   2.49    117.72    19
## 97      Kempen   SM  0.16 0.61 0.23 5.16   2.05     80.47    20
## 98  Leemstreek   SM  0.27 0.25 0.48 5.83   1.77     98.07    12
## 99  Leemstreek SMGR  0.22 0.26 0.52 5.99   1.91    107.12    17
## 100 Leemstreek SMGR  0.22 0.27 0.51 6.18   1.90    105.20    11
## 101 Leemstreek   GM  0.33 0.23 0.44 5.85   1.73     95.51    20
## 102 Leemstreek   GM  0.33 0.22 0.45 5.94   1.71     97.76    24
## 103 Leemstreek   SM  0.20 0.28 0.52 5.77   1.55    101.27    24
## 104     Kempen   SM  0.09 0.64 0.27 5.17   2.24     83.60    24
```

```
## 105 Leemstreek   GM  0.23 0.20 0.57 6.08   1.89     98.05    17
## 106     Kempen   SM  0.20 0.60 0.20 5.23   2.01     79.80    16
## 107     Kempen   GM  0.20 0.56 0.24 4.86   2.03     81.12    16
## 108     Kempen   GM  0.28 0.50 0.22 5.12   2.16     80.53    24
## 109 Leemstreek   PG  0.23 0.25 0.52 5.81   2.53    121.01    20
## 110 Leemstreek   GM  0.24 0.25 0.52 6.00   1.85     97.66    16
## 111     Kempen   GM  0.11 0.60 0.30 5.37   1.96     81.71    17
## 112 Leemstreek   SM  0.31 0.26 0.43 5.90   2.00     95.37    21
## 113 Leemstreek   PG  0.22 0.28 0.51 6.14   2.36    116.80    21
## 114     Kempen   GM  0.22 0.61 0.17 4.95   2.10     83.15    17
## 115     Kempen SMGR  0.19 0.57 0.24 5.00   2.21     95.26    16
## 116 Leemstreek   SM  0.21 0.23 0.57 6.09   1.77     95.33    16
## 117 Leemstreek   GM  0.17 0.25 0.59 5.86   1.72     94.81    20
## 118     Kempen SMGR  0.22 0.62 0.16 4.76   2.11     91.31    20
## 119 Leemstreek   SM  0.26 0.22 0.52 6.25   1.93     96.24    11
## 120 Leemstreek   PG  0.16 0.24 0.60 6.07   2.46    120.78    17
## 121     Kempen   GM  0.14 0.65 0.21 4.98   2.07     81.92    18
## 122 Leemstreek   GM  0.27 0.25 0.49 5.60   1.76     93.27    12
## 123 Leemstreek   GM  0.27 0.23 0.50 6.38   1.72     99.53    15
## 124 Leemstreek   SM  0.24 0.24 0.52 6.68   1.71    101.65    13
## 125 Leemstreek   GM  0.18 0.28 0.54 6.04   1.77    100.20    12
## 126     Kempen SMGR  0.11 0.63 0.26 5.27   2.03     94.98    23
## 127 Leemstreek   PG  0.22 0.29 0.49 6.24   2.34    118.42    15
## 128 Leemstreek   SM  0.23 0.25 0.51 5.86   1.71     96.51    11
## 129 Leemstreek   PG  0.09 0.26 0.65 5.93   2.36    121.28    20
## 130     Kempen   GM  0.19 0.57 0.23 4.75   2.05     84.44    21
## 131     Kempen   PG  0.14 0.63 0.23 4.95   2.87    104.80    18
## 132     Kempen SMGR  0.25 0.57 0.18 5.22   2.07     88.54    18
## 133     Kempen   GM  0.25 0.53 0.22 5.04 208.00     79.76    21
## 134     Kempen   PG  0.18 0.61 0.21 5.06   2.82    111.75    23
## 135 Leemstreek   SM  0.19 0.31 0.50 5.98   1.72     88.77    22
## 136     Kempen SMGR  0.16 0.64 0.20 4.73   2.19     89.41    18
## 137     Kempen   GM  0.14 0.66 0.20 5.27   2.11     82.30    22
## 138     Kempen   PG  0.18 0.62 0.20 4.81   2.72    102.04    13
## 139     Kempen   GM  0.22 0.57 0.21 5.06   2.03     83.58    16
## 140 Leemstreek   SM  0.17 0.24 0.58 5.91   1.82     90.81    16
## 141 Leemstreek   PG  0.20 0.24 0.56 5.96   2.44    123.92    11
## 142 Leemstreek   SM  0.29 0.24 0.48 6.29   1.76     92.04    10
## 143 Leemstreek   PG  0.33 0.27 0.40 5.88   2.44    111.91    16
## 144 Leemstreek   SM  0.34 0.21 0.45 5.84   1.95     95.37    12
## 145     Kempen SMGR  0.15 0.64 0.21   NA   2.26     90.67    23
## 146 Leemstreek   PG  0.22 0.28 0.51 5.84   2.38    117.80    19
## 147 Leemstreek   SM  0.33 0.21 0.46 5.77   1.70     96.13    17
## 148 Leemstreek   PG  0.11 0.27 0.62 6.08   2.42    121.58    18
## 149 Leemstreek   PG  0.19 0.32 0.48 5.83   2.55    116.25    11
## 150     Kempen   GM  0.24 0.57 0.18 5.16   2.07     81.81    21
## 151     Kempen   GM  0.14 0.64 0.22 5.07   1.99     84.01    20
## 152 Leemstreek   SM  0.33 0.23 0.44 5.71   1.87    104.46    12
## 153 Leemstreek SMGR  0.21 0.28 0.51 5.84   1.89    106.09    19
## 154 Leemstreek   PG  0.23 0.26 0.51 6.18   2.36    126.85    20
## 155 Leemstreek   GM  0.18 0.30 0.52 6.18   1.79     98.06    15
## 156 Leemstreek SMGR  0.39 0.21 0.40 6.40   2.02    104.37    20
## 157 Leemstreek SMGR  0.24 0.25 0.51 6.00   1.93  95730.00    19
## 158 Leemstreek   GM  0.25 0.23 0.52 5.70   1.76     91.59    14
```

```
## 159 Leemstreek   GM  0.34 0.24 0.41 5.85   1.70     95.06    24
## 160 Leemstreek   SM  0.21 0.23 0.56 6.08   1.84    101.02    11
## 161 Leemstreek   SM  0.23 0.22 0.55 6.38   1.71     97.15    24
## 162       <NA>   PG  0.09 0.27 0.64 6.02   2.32    122.37    22
## 163     Kempen SMGR  0.23 0.55 0.22 5.33   2.19     86.13    15
## 164 Leemstreek   SM  0.20 0.29 0.50 6.15   1.80     97.29    16
## 165 Leemstreek SMGR  0.28 0.21 0.51 5.77   2.09    105.08    18
## 166       <NA> SMGR  0.24 0.25 0.50 5.97   1.80    102.93    18
## 167     Kempen   GM  0.21 0.63 0.16 5.36   2.06     82.61    11
## 168 Leemstreek   PG  0.33 0.27 0.40 6.18   2.51    117.79    15
## 169 Leemstreek   GM  0.26 0.24 0.50 6.09   1.83     97.74    14
## 170 Leemstreek   PG  0.28 0.25 0.47 6.10   2.45    120.01    11
## 171     Kempen SMGR  0.13 0.65 0.22 5.00   2.14     89.92    25
## 172 Leemstreek   PG  0.42 0.21 0.37 5.74   2.44    112.85    15
## 173     Kempen SMGR  0.26 0.56 0.18 5.36   2.35     87.62    24
## 174     Kempen   GM  0.20 0.62 0.19 4.82   2.08     76.40    22
## 175     Kempen   PG  0.17 0.58 0.25 4.91   2.58     99.98    16
## 176     Kempen SMGR  0.14 0.67 0.19 5.23   2.31     88.27    22
## 177 Leemstreek   GM  0.28 0.27 0.45 5.75   1.79     95.64    14
## 178 Leemstreek   SM  0.24 0.25 0.50 5.89   1.80     99.36    22
## 179     Kempen SMGR  0.27 0.52 0.21 4.86   2.19     94.22    11
## 180 Leemstreek   GM  0.24 0.25 0.51 6.12   1.74     98.01    13
## 181     Kempen   PG  0.13 0.58 0.29 5.19   2.77    109.99    25
## 182 Leemstreek   PG  0.32 0.25 0.43 6.08   2.41    118.32    11
## 183     Kempen   PG  0.25 0.54 0.21 5.19   2.77    104.97    18
## 184 Leemstreek   GM  0.27 0.26 0.47 6.11   1.67    100.45    20
## 185 Leemstreek   PG    NA   NA   NA 6.13   2.47    120.33    14
## 186 Leemstreek   SM  0.30 0.24 0.45 5.61   1.89     91.84    17
## 187 Leemstreek   SM  0.24 0.26 0.50 5.97   1.84     95.08    23
## 188 Leemstreek SMGR  0.23 0.23 0.55 6.02   1.98    108.14    17
## 189     Kempen   PG  0.18 0.63 0.19 5.24   2.66    103.83    12
## 190     Kempen SMGR  0.20 0.53 0.27 4.86   2.20     91.34    15
## 191 Leemstreek   GM  0.29 0.17 0.54 5.90   1.80     96.34    21
## 192 Leemstreek   SM  0.30 0.26 0.44 5.79   1.71     96.37    11
## 193     Kempen   PG  0.12 0.64 0.23 5.31   2.87    104.83    24
## 194     Kempen   SM  0.20 0.59 0.21 5.36   2.01     85.66    20
## 195     Kempen   SM  0.15 0.63 0.23 5.26   2.11     84.95    16
## 196 Leemstreek   PG  0.29 0.22 0.49 5.76   2.46    124.95    12
## 197 Leemstreek   GM  0.19 0.25 0.56 6.09   1.76     95.56    24
## 198     Kempen   GM  0.32 0.50 0.18 5.33   2.03     78.66    13
## 199 Leemstreek   PG  0.25 0.29 0.47 6.03   2.49    120.94    21
## 200 Leemstreek   SM  0.21 0.31 0.49 5.89   1.72    100.96    10
## 201 Leemstreek   SM  0.28 0.28 0.44 5.78   1.69    106.93    14
## 202     Kempen SMGR  0.21 0.60 0.19 4.97   2.14     88.33    25
## 203     Kempen   SM  0.18 0.60 0.22 5.24   2.09     82.91    21
## 204 Leemstreek   GM  0.30 0.20 0.49 5.91   1.85     99.64    20
## 205 Leemstreek   SM  0.38 0.27 0.35 6.05   1.93    103.18    14
## 206     Kempen   PG  0.16 0.59 0.25 5.36   2.77    103.67    11
## 207 Leemstreek   SM  0.25 0.23 0.51 6.12   1.83    100.85    14
## 208 Leemstreek   PG  0.15 0.21 0.64 6.27   2.43    118.21    18
## 209 Leemstreek   GM  0.22 0.24 0.54 5.66   1.75     92.58    22
## 210 Leemstreek   GM  0.30 0.22 0.47 5.94   1.64     98.66    14
## 211 Leemstreek SMGR  0.09 0.24 0.67 6.01   1.92    104.71    15
## 212 Leemstreek   SM  0.12 0.21 0.67 6.12   1.76     99.80    24
```

```
## 213 Leemstreek   SM  0.27 0.30 0.43 5.77   1.82     96.05    17
## 214     Kempen   PG  0.17 0.60 0.23 5.26   2.72    112.98    25
## 215 Leemstreek SMGR  0.23 0.28 0.49 6.06   1.85    106.84    23
## 216     Kempen SMGR  0.30 0.47 0.23 5.15   2.18     88.71    25
## 217 Leemstreek   PG  0.25 0.24 0.51 6.10   2.42    122.99    12
## 218 Leemstreek SMGR  0.35 0.23 0.42 5.98   1.99     99.68    15
## 219     Kempen   PG  0.23 0.54 0.23 5.29   2.76    105.64    24
## 220     Kempen   SM  0.05 0.68 0.27 5.23   2.13     89.48    20
## 221 Leemstreek   PG  0.30 0.21 0.49 5.97   2.35    119.58    25
## 222    Kempeen   PG  0.17 0.62 0.22 5.31   2.87    109.63    25
## 223     Kempen SMGR  0.10 0.64 0.26 5.16   2.21     89.29    21
## 224     Kempen   GM  0.14 0.61 0.25 5.23   1.98     73.42    16
## 225 Leemstreek   SM  0.38 0.22 0.40 5.88   1.82     94.93    20
## 226 Leemstreek   GM  0.21 0.28 0.51 6.32   1.82     97.21    14
## 227 Leemstreek   SM  0.15 0.26 0.60 6.00   1.69    103.80    15
## 228 Leemstreek   PG  0.20 0.22 0.58 6.17   2.58    118.08    23
## 229     Kempen   GM  0.24 0.53 0.23 5.08   2.03     83.17    20
## 230     Kempen   GM  0.05 0.70 0.24 5.04   2.13     78.44    22
## 231     Kempen SMGR  0.25 0.57 0.18 5.26   2.20     86.29    15
## 232 Leemstreek   PG  0.29 0.19 0.52 5.68   2.42    117.29    20
## 233 Leemstreek   GM  0.29 0.27 0.44 6.10   1.60    101.52    19
## 234 Leemstreek SMGR  0.27 0.26 0.47 6.08   1.81    107.89    13
## 235 Leemstreek   SM  0.23 0.21 0.56 5.46   1.79    100.45    14
## 236 Leemstreek   SM  0.38 0.19 0.43 6.07   1.87    102.56    23
## 237 Leemstreek   PG  0.24 0.25 0.52 5.57   2.30    118.60    22
## 238     Kempen SMGR  0.11 0.66 0.23 5.23   2.31     93.04    10
## 239 Leemstreek   PG  0.23 0.27 0.50 5.91   2.56    121.08    21
## 240     Kempen   GM  0.15 0.58 0.27 4.96   1.95     78.11    23
## 241 Leemstreek SMGR  0.34 0.22 0.44 6.00   1.94    101.14    23
## 242       <NA> SMGR  0.15 0.61 0.24 5.37   2.11     90.37    12
## 243 Leemstreek   PG  0.25 0.25 0.49 5.87   2.44    121.06    25
## 244     Kempen   GM  0.16 0.63 0.21 5.21   2.12     80.92    24
## 245 Leemstreek   SM  0.23 0.24 0.53 6.07   1.87     94.53    14
## 246     Kempen SMGR  0.17 0.64 0.19 4.86   2.13     82.13    23
## 247 Leemstreek SMGR  0.35 0.23 0.42 5.65   1.87    105.85    22
## 248     Kempen   SM  0.21 0.58 0.21 4.70   2.02     82.92    17
## 249     Kempen   GM  0.09 0.64 0.27 5.21   2.07     84.23    11
## 250     Kempen SMGR    NA   NA   NA 5.09   2.07     84.25    23
## 251 Leemstreek SMGR    NA   NA   NA 6.37   1.69     99.61    15
## 252 Leemstreek   PG  0.26 0.25 0.49 6.11   2.57    119.90    13
## 253 Leemstreek   GM  0.20 0.27 0.53 6.17   1.93     90.97    22
## 254 Leemstreek SMGR  0.35 0.21 0.44 6.27   1.86     96.82    25
## 255 Leemstreek SMGR  0.17 0.26 0.57 5.90   1.96    107.26    13
## 256     Kempen   PG  0.20 0.56 0.24 5.20   2.75    109.57    11
## 257 Leemstreek   PG  0.27 0.28 0.45 6.17   2.57    116.42    20
## 258 Leemstreek   SM  0.23 0.27 0.50   NA   1.84     97.32    23
## 259 Leemstreek   GM  0.18 0.27 0.54 6.15   1.77     92.02    13
## 260     Kempen   GM  0.16 0.61 0.24 5.18   1.90     83.34    18
## 261     Kempen   PG  0.09 0.64 0.26 4.93   2.83    105.97    15
## 262     Kempen   PG  0.13 0.67 0.20 5.25   2.72    102.85    22
## 263     Kempen   SM  0.05 0.70 0.25 5.20   1.98     81.20    17
## 264     Kempen SMGR  0.15 0.60 0.25 5.23   2.18     86.11    13
## 265 Leemstreek   SM  0.33 0.18 0.49 6.08   1.79     99.98    14
## 266 Leemstreek   SM  0.31 0.25 0.44 5.96   1.88    100.46    22
```

```
## 267 Leemstreek   PG  0.31 0.26 0.43 5.98 234.00    121.73    10
## 268 Leemstreek   SM  0.23 0.25 0.53 6.14   1.75     95.13    16
## 269 Leemstreek SMGR  0.24 0.27 0.50 6.03   2.02     97.52    19
## 270 Leemstreek   SM    NA   NA   NA 5.83   1.91    100.29    18
## 271     Kempen   SM  0.19 0.57 0.24 5.04   2.19     83.44    20
## 272 Leemstreek   GM  0.28 0.24 0.48 5.74   1.88     96.10    18
## 273 Leemstreek   SM  0.30 0.26 0.44 5.81   1.81    102.19    18
## 274 Leemstreek   SM  0.30 0.23 0.47 5.97   1.75     94.90    23
## 275     Kempen   GM  0.17 0.60 0.23 4.74   2.09     86.39    15
## 276     Kempen SMGR  0.29 0.50 0.21 4.76   2.28     88.48    11
## 277     Kempen   PG  0.22 0.54 0.24 5.32   2.94    103.46    22
## 278 Leemstreek   SM  0.31 0.21 0.48 5.88   1.64     97.50    22
## 279 Leemstreek   GM  0.28 0.22 0.49 5.63   1.77     97.71    23
## 280     Kempen   GM  0.24 0.57 0.20 5.08   2.10     81.33    11
## 281     Kempen SMGR  0.10 0.62 0.28 5.36   2.19     86.76    10
## 282 Leemstreek SMGR  0.20 0.28 0.52 6.13   1.77    103.26    19
## 283 Leemstreek   PG  0.26 0.23 0.51 6.09   2.51    110.66    23
## 284 Leemstreek   PG  0.38 0.22 0.40 6.24   2.39    118.42    17
## 285 Leemstreek   PG  0.24 0.22 0.54 5.85   2.49    122.01    11
## 286 Leemstreek SMGR  0.32 0.24 0.44 6.15   1.97    103.69    16
## 287 Leemstreek   GM  0.31 0.24 0.45 5.88   1.85     98.90    15
## 288 Leemstreek   SM  0.18 0.26 0.56 6.00   1.67    100.15    17
## 289 Leemstreek   GM  0.29 0.24 0.47 6.44   1.84     98.84    12
## 290     Kempen   SM  0.34 0.50 0.17 5.29   2.04     86.66    17
## 291 Leemstreek   SM  0.15 0.25 0.61 6.15   1.70     91.73    14
## 292 Leemstreek SMGR  0.36 0.29 0.36 5.78   2.00    104.28    17
## 293 Leemstreek   GM  0.18 0.29 0.53 5.84   1.84    102.47    24
## 294     Kempen   SM  0.19 0.56 0.25 5.56   2.08     82.39    18
## 295     Kempen   SM  0.28 0.52 0.20 4.88   2.15     85.80    21
## 296     Kempen SMGR  0.04 0.72 0.23 5.14   2.13     90.68    11
## 297     Kempen   SM  0.17 0.59 0.23 5.28   2.08     86.28    20
## 298 Leemstreek   SM  0.19 0.25 0.56 6.20   1.79     97.31    22
## 299 Leemstreek   GM  0.21 0.26 0.53 6.22   1.73     93.88    10
## 300     Kempen   GM  0.27 0.52 0.21 5.07   2.10     82.39    11
## 301     Kempen   PG  0.19 0.56 0.25 4.94   2.87    106.88    19
## 302 Leemstreek   SM  0.25 0.20 0.55 5.94   1.76     97.59    22
## 303     Kempen   SM  0.24 0.57 0.19 4.92   2.09     84.20    21
## 304     Kempen   PG  0.22 0.61 0.18 5.23   2.88    107.71    14
## 305     Kempen   GM  0.24 0.53 0.22 5.32   2.02     79.22    22
## 306 Leemstreek SMGR  0.25 0.27 0.48 6.43   1.69    108.71    16
## 307 Leemstreek   GM  0.30 0.26 0.44 6.07   1.79    104.68    12
## 308 Leemstreek   GM  0.27 0.22 0.51 5.83   1.68    101.88    19
## 309 Leemstreek   SM  0.32 0.26 0.42 6.20   1.72     96.61    22
## 310 Leemstreek   GM  0.22 0.27 0.51 6.18   1.86     89.41    22
## 311 Leemstreek SMGR  0.26 0.28 0.46 5.95   2.09    104.50    11
## 312 Leemstreek   GM  0.20 0.24 0.56 5.69   1.66     99.59    11
## 313     Kempen SMGR  0.11 0.60 0.29 5.25   2.28     93.11    24
## 314 Leemstreek SMGR  0.31 0.25 0.45 6.23   1.89     99.67    11
## 315 Leemstreek   GM  0.14 0.29 0.56 5.94   1.76     92.06    13
## 316     Kempen   GM  0.10 0.66 0.24 5.21   2.14     82.65    17
## 317     Kempen   SM  0.14 0.64 0.22 5.04   2.11     86.49    22
## 318 Leemstreek   SM  0.29 0.24 0.47 6.14   1.73    101.65    23
## 319 Leemstreek   PG  0.14 0.26 0.59 5.98   2.41    123.40    19
## 320     Kempen   SM  0.22 0.62 0.16 5.03   2.15     80.43    16
```

```
## 321     Kempen   GM  0.25 0.55 0.20 5.12   1.86     77.14    12
## 322 Leemstreek SMGR  0.39 0.20 0.41 5.94   1.99     96.64    23
## 323 Leemstreek SMGR  0.15 0.27 0.58 5.88   1.77    106.20    23
## 324     Kempen   PG  0.26 0.52 0.21 5.04   2.75    109.26    15
## 325 Leemstreek SMGR  0.28 0.25 0.47 6.00   1.88    105.58    18
## 326 Leemstreek SMGR  0.05 0.26 0.69 6.04   2.00    100.29    17
## 327     Kempen   PG  0.15 0.63 0.22 5.41   2.76    112.14    22
## 328 Leemstreek SMGR  0.27 0.26 0.47 5.73   1.91    102.84    24
## 329 Leemstreek SMGR  0.25 0.23 0.52 5.96   1.95    111.06    19
## 330     Kempen   SM  0.18 0.65 0.17 5.23   2.03     86.24    22
## 331     Kempen   GM  0.08 0.69 0.23 5.45   2.03     86.72    22
## 332 Leemstreek   PG  0.23 0.28 0.49 6.09   2.54    121.58    14
## 333     Kempen SMGR  0.26 0.50 0.23 5.10   2.25     89.88    18
## 334     Kempen   PG  0.14 0.67 0.19 5.04   2.64    102.26    14
## 335  Leemstrek   SM  0.31 0.25 0.44 6.08   1.85     96.20    23
## 336 Leemstreek   GM  0.28 0.27 0.46 6.20   1.64     96.16    21
## 337     Kempen   PG  0.17 0.63 0.20 5.06   2.70    107.19    15
## 338     Kempen   PG  0.15 0.60 0.26 5.13   2.84    106.00    25
## 339     Kempen   GM  0.14 0.60 0.27 5.14   1.87     81.46    21
## 340     Kempen   SM  0.13 0.65 0.22 5.25   1.95     84.85    14
## 341 Leemstreek   PG  0.25 0.27 0.48 6.22   2.55    118.35    20
## 342 Leemstreek   SM  0.24 0.28 0.48 6.06   1.73     96.28    15
## 343 Leemstreek   SM  0.17 0.32 0.50 5.98   1.90     92.77    22
## 344 Leemstreek SMGR  0.22 0.27 0.51 6.28   1.89    107.57    15
## 345 Leemstreek   GM  0.23 0.26 0.52 6.03   1.78     96.49    14
## 346       <NA>   GM  0.36 0.18 0.46 6.32   1.82     96.04    11
## 347     Kempen SMGR  0.07 0.73 0.19 5.09   2.14     87.30    19
## 348 Leemstreek   SM  0.19 0.24 0.57 6.12   1.78     97.38    10
## 349 Leemstreek SMGR  0.21 0.32 0.47 5.69   1.86    100.11    14
## 350     Kempen   PG  0.15 0.62 0.23 5.08   2.97    102.13    14
## 351 Leemstreek   SM  0.21 0.30 0.50 6.00   1.81     91.01    22
## 352     Kempen   SM  0.21 0.59 0.19 4.95   2.13     87.82    12
## 353 Leemstreek   SM  0.20 0.27 0.54 5.79   1.77     98.74    14
## 354 Leemstreek   SM  0.23 0.26 0.51   NA   1.78     94.09    17
## 355     Kempen SMGR  0.15 0.59 0.26 5.28   2.31     94.58    24
## 356     Kempen   GM  0.18 0.59 0.22 4.84   2.02     84.44    15
## 357     Kempen   GM  0.14 0.65 0.21 5.18   2.05     84.96    22
## 358 Leemstreek   PG  0.24 0.24 0.52   NA   2.48    119.40    12
## 359 Leemstreek   SM  0.38 0.19 0.43 5.79   1.76     96.65    17
## 360     Kempen   SM  0.21 0.59 0.19 5.21   2.25     83.77    18
## 361 Leemstreek   GM  0.20 0.27 0.53 5.86   1.63     97.81    23
## 362 Leemstreek   SM  0.25 0.27 0.48 6.20   1.96    101.38    15
## 363     Kempen SMGR  0.15 0.63 0.22 5.11   2.17     96.16    17
## 364 Leemstreek   GM  0.31 0.20 0.49 6.11   1.86     94.39    17
## 365 Leemstreek SMGR  0.28 0.26 0.46 5.90   1.74    100.42    21
## 366     Kempen   SM  0.19 0.65 0.16 4.87   2.17     84.46    11
## 367 Leemstreek   SM  0.19 0.29 0.52 6.18   1.87    109.82    21
## 368 Leemstreek   GM  0.21 0.24 0.55 6.05   1.65     94.68    20
## 369 Leemstreek SMGR  0.34 0.24 0.42 5.95   1.84 108650.00    20
## 370 Leemstreek SMGR  0.12 0.29 0.59 6.01   1.91    103.38    17
## 371 Leemstreek   PG  0.28 0.20 0.52 6.27   2.54    118.53    14
## 372 Leemstreek   PG  0.18 0.24 0.58 5.86   2.39    117.58    16
## 373     Kempen SMGR  0.19 0.62 0.19 4.95   2.18     89.29    12
## 374 Leemstreek   PG  0.29 0.20 0.51 5.88   2.57    126.52    14
```

```
## 375 Leemstreek   SM  0.25 0.26 0.49 5.68   1.71     99.39    11
## 376     Kempen   SM  0.06 0.69 0.24 5.26   2.17     86.01    17
## 377     Kempen   GM  0.20 0.60 0.20 5.14   2.08     85.32    15
## 378 Leemstreek   GM  0.26 0.24 0.50 6.04   1.66    104.18    13
## 379 Leemstreek SMGR  0.34 0.26 0.39 5.99   1.85    107.25    21
## 380     Kempen   PG  0.22 0.59 0.20 5.00   2.81    100.68    14
## 381 Leemstreek   SM  0.22 0.27 0.51 5.84   1.82     94.34    16
## 382     Kempen   PG  0.16 0.65 0.20 5.27   2.86    100.12    21
## 383 Leemstreek   SM  0.25 0.25 0.50 5.71   1.75     87.42    12
## 384     Kempen SMGR  0.25 0.54 0.21 5.19   2.06     95.05    12
## 385 Leemstreek   PG    NA   NA   NA 5.84   2.53    118.40    25
## 386     Kempen   GM  0.17 0.57 0.26 5.16   2.00     83.64    25
## 387 Leemstreek   SM  0.28 0.27 0.45 5.85   1.81    102.76    21
## 388 Leemstreek   SM  0.33 0.19 0.48 5.90   1.94     96.34    10
## 389 Leemstreek   SM  0.22 0.19 0.59 6.35   1.78     97.86    25
## 390 Leemstreek   PG  0.40 0.23 0.37 6.13   2.57    121.66    21
## 391     Kempen   PG  0.17 0.62 0.21 4.67   2.66    106.89    23
## 392 Leemstreek   GM  0.15 0.22 0.62 6.03   1.82     93.36    13
## 393     Kempen   SM  0.07 0.64 0.29 5.01   2.11     86.88    14
## 394     Kempen   GM  0.16 0.62 0.22 4.87   1.99     84.15    19
## 395 Leemstreek   PG  0.31 0.24 0.45 6.04   2.35    119.21    15
## 396 Leemstreek   GM  0.22 0.27 0.51 6.03   1.76     91.66    21
## 397 Leemstreek   SM  0.28 0.26 0.46 5.93   1.82    100.70    11
## 398 Leemstreek   SM  0.18 0.24 0.59 6.39   1.82    103.80    23
## 399 Leemstreek   GM  0.29 0.26 0.45 5.55   1.79    101.46    22
## 400     Kempen   SM  0.14 0.63 0.23 5.27   2.06     87.47    23
```

```
group_number <- 4
rows_to_remove <- c(group_number * 11, group_number^2, 2 * group_number + 21)

print(rows_to_remove)
```

```
## [1] 44 16 29
```

```
removed_rows <- data[-rows_to_remove, ]


print(head(removed_rows))
```

```
##       Region Crop Lutum Sand Silt   pH    C Cstock Years
## 1 Leemstreek   PG  0.23 0.25 0.52 6.12 2.43 118.38    15
## 2     Kempen   SM  0.26 0.54 0.20 4.95 2.15  84.99    16
## 3 Leemstreek   PG  0.22 0.23 0.55 6.17 2.41 118.28    14
## 4     Kempen SMGR  0.15 0.60 0.25 4.95 2.15  94.72    12
## 5     Kempen   GM  0.14 0.63 0.23 5.23 1.91  82.81    14
## 6 Leemstreek   GM  0.29 0.20 0.51 6.22 1.73  94.01    14
```

```
print(removed_rows)
```

```
##         Region Crop Lutum Sand Silt   pH      C    Cstock Years
## 1   Leemstreek   PG  0.23 0.25 0.52 6.12   2.43    118.38    15
## 2       Kempen   SM  0.26 0.54 0.20 4.95   2.15     84.99    16
```

```
## 3   Leemstreek   PG  0.22 0.23 0.55 6.17   2.41    118.28    14
## 4       Kempen SMGR  0.15 0.60 0.25 4.95   2.15     94.72    12
## 5       Kempen   GM  0.14 0.63 0.23 5.23   1.91     82.81    14
## 6   Leemstreek   GM  0.29 0.20 0.51 6.22   1.73     94.01    14
## 7   Leemstreek SMGR  0.27 0.24 0.50 5.80   2.10     97.40    15
## 8       Kempen   SM  0.08 0.64 0.28 5.32   2.18     81.72    10
## 9   Leemstreek   GM  0.25 0.23 0.52 5.90   1.88    104.06    21
## 10  Leemstreek SMGR  0.25 0.26 0.49 6.06   1.97    102.18    17
## 11      Kempen   GM  0.23 0.62 0.14 5.14   2.14     78.95    19
## 12  Leemstreek   PG  0.34 0.26 0.41 5.91 252.00    122.77    12
## 13      Kempen   SM  0.15 0.63 0.22 5.15   2.13     85.81    10
## 14  Leemstreek   GM  0.24 0.25 0.51 5.75   1.70     99.11    13
## 15  Leemstreek   SM  0.25 0.24 0.52 5.97   1.76     97.24    10
## 17  Leemstreek   PG  0.26 0.25 0.49 5.57   2.45    115.76    17
## 18  Leemstreek   GM  0.16 0.26 0.57 5.70   1.82     96.06    22
## 19  Leemstreek   GM  0.29 0.27 0.44 5.77   1.83     93.58    18
## 20      Kempen   SM  0.24 0.57 0.18 4.78   2.10     86.64    12
## 21      Kempen   GM  0.13 0.69 0.18 5.18   2.00     82.63    10
## 22      Kempen SMGR  0.14 0.61 0.24 4.90   2.46     94.30    16
## 23      Kempen SMGR  0.18 0.61 0.22 4.67   2.25     97.85    13
## 24      Kempen   GM  0.09 0.66 0.24 4.97   2.16     82.63    22
## 25      Kempen   PG  0.22 0.56 0.21 5.02   2.72    104.07    18
## 26     Keempen SMGR  0.22 0.58 0.20 5.27   2.19     87.03    18
## 27  Leemstreek   GM  0.20 0.32 0.48 5.78   1.78     95.26    24
## 28  Leemstreek   GM  0.36 0.25 0.39 6.23   1.88     96.08    11
## 30  Leemstreek   PG  0.38 0.21 0.41 6.07   2.52    116.03    18
## 31      Kempen   PG  0.16 0.59 0.25 4.93   2.58    101.93    12
## 32      Kempen SMGR  0.18 0.62 0.20 5.49   2.16     88.65    14
## 33      Kempen   GM  0.14 0.62 0.24 5.07   1.94     84.35    12
## 34      Kempen   PG  0.18 0.55 0.27 4.74   2.83    109.17    12
## 35  Leemstreek   SM  0.26 0.25 0.49 5.96   1.79     94.84    19
## 36  Leemstreek SMGR  0.27 0.22 0.52 5.61   1.87    104.71    11
## 37      Kempen   SM  0.20 0.64 0.17 5.04   2.20     83.95    19
## 38  Leemstreek   GM  0.27 0.28 0.45 5.92   1.84    100.83    21
## 39  Leemstreek   SM  0.31 0.18 0.50 6.14   1.73    102.47    16
## 40  Leemstreek   SM  0.19 0.29 0.52 6.20   1.83     97.15    21
## 41  Leemstreek   SM  0.33 0.21 0.45 5.85   1.70     98.45    11
## 42  Leemstreek SMGR  0.08 0.26 0.66 6.16   1.86    106.81    19
## 43  Leemstreek SMGR  0.26 0.27 0.47 6.28   2.03     98.07    14
## 45  Leemstreek   PG  0.23 0.23 0.54 5.87   2.54    122.46    14
## 46  Leemstreek SMGR  0.25 0.25 0.50 6.13   1.85    102.34    19
## 47  Leemstreek SMGR  0.28 0.26 0.46 6.04   1.75    104.08    16
## 48  Leemstreek   SM  0.16 0.28 0.56 6.11   1.71     98.24    20
## 49  Leemstreek   GM  0.21 0.19 0.60 6.28   1.83     93.61    24
## 50      Kempen SMGR  0.26 0.52 0.22 5.18   2.17     86.36    18
## 51  Leemstreek   GM  0.17 0.29 0.53 6.21   1.58     95.58    14
## 52  Leemstreek   GM  0.22 0.28 0.49 6.12   1.74     95.01    21
## 53      Kempen   SM  0.13 0.62 0.25 5.19   2.21     87.31    18
## 54  Leemstreek   SM  0.30 0.27 0.43 6.08   1.85    104.13    12
## 55  Leemstreek   SM  0.08 0.28 0.64 6.26   1.73     98.31    14
## 56  Leemstreek   PG  0.37 0.17 0.46 5.80   2.32    121.78    12
## 57  Leemstreek   PG  0.30 0.28 0.41 5.78   2.38    119.99    12
## 58      Kempen SMGR  0.21 0.58 0.21 5.22   2.20     92.43    20
## 59      Kempen   SM  0.16 0.61 0.22 5.08   2.17     84.95    21
```

```
## 60  Leemstreek   SM  0.16 0.24 0.60 6.01   1.64     99.20    17
## 61       Kempen   PG  0.11 0.64 0.25 5.25   2.83    110.38    12
## 62  Leemstreek   GM  0.25 0.24 0.51 6.15   1.75     92.58    23
## 63  Leemstreek   GM  0.32 0.27 0.42 5.95   1.79    101.79    25
## 64  Leemstreek   SM  0.31 0.24 0.45 5.94   1.67     99.62    10
## 65       Kempen SMGR  0.24 0.55 0.22 5.11   2.14     90.81    18
## 66  Leemstreek   SM  0.06 0.29 0.65 6.21   1.76     96.21    22
## 67       Kempen   SM  0.20 0.64 0.16 5.19   2.18     83.52    18
## 68       Kempen SMGR  0.06 0.69 0.25 5.39   2.28     91.60    14
## 69       Kempen   SM  0.19 0.60 0.20 5.10   2.13     76.28    17
## 70  Leemstreek   PG  0.12 0.28 0.60 6.23   2.58    118.06    14
## 71       Kempen   SM  0.12 0.70 0.19 5.28   2.05     86.39    25
## 72       Kempen   PG  0.19 0.59 0.22 5.22   2.83    105.03    20
## 73       Kempen   GM  0.28 0.53 0.19 5.18   2.04     79.10    21
## 74  Leemstreek SMGR  0.20 0.24 0.55 5.94   1.85    101.88    10
## 75  Leemstreek SMGR  0.17 0.24 0.59 5.98   1.97    105.19    16
## 76  Leemstreek   GM  0.13 0.25 0.62 6.05   1.86     90.10    17
## 77  Leemstreek SMGR  0.15 0.30 0.55 6.01   1.97    107.62    12
## 78       Kempen   SM  0.14 0.58 0.28 5.29   2.06     75.92    14
## 79  Leemstreek   SM  0.16 0.26 0.57 5.98   1.76     96.39    12
## 80  Leemstreek SMGR  0.26 0.24 0.49 5.99   2.01    101.31    16
## 81  Leemstreek   SM  0.22 0.25 0.53 5.90   1.83     93.54    13
## 82       Kempen   PG  0.19 0.62 0.19 5.07   2.81    102.02    18
## 83  Leemstreek   GM  0.22 0.29 0.49 5.99   1.72     97.25    23
## 84       Kempen SMGR  0.14 0.61 0.26 4.64   2.29     89.57    18
## 85   Lemstreek   PG  0.20 0.27 0.53 5.73   2.38    119.11    16
## 86  Leemstreek   SM  0.20 0.27 0.53 5.99   1.85     89.48    14
## 87       Kempen   SM  0.17 0.65 0.19 5.05   2.03     79.08    13
## 88       Kempen SMGR  0.23 0.57 0.20 4.86   2.00     89.57    11
## 89       Kempen   GM  0.12 0.68 0.20 4.70   1.94     78.70    22
## 90  Leemstreek   PG  0.24 0.24 0.52 5.93   2.46    117.26    24
## 91  Leemstreek   PG  0.31 0.25 0.44 6.13   2.52    124.32    24
## 92       Kempen   SM  0.09 0.67 0.24 5.45   1.96     84.86    14
## 93         <NA>   GM  0.28 0.29 0.43 5.99   1.65     92.33    14
## 94       Kempen SMGR  0.18 0.55 0.27 5.40   2.34     88.40    15
## 95  Leemstreek   SM  0.25 0.22 0.53 6.02   1.76     97.22    15
## 96  Leemstreek   PG  0.33 0.21 0.46   NA   2.49    117.72    19
## 97       Kempen   SM  0.16 0.61 0.23 5.16   2.05     80.47    20
## 98  Leemstreek   SM  0.27 0.25 0.48 5.83   1.77     98.07    12
## 99  Leemstreek SMGR  0.22 0.26 0.52 5.99   1.91    107.12    17
## 100 Leemstreek SMGR  0.22 0.27 0.51 6.18   1.90    105.20    11
## 101 Leemstreek   GM  0.33 0.23 0.44 5.85   1.73     95.51    20
## 102 Leemstreek   GM  0.33 0.22 0.45 5.94   1.71     97.76    24
## 103 Leemstreek   SM  0.20 0.28 0.52 5.77   1.55    101.27    24
## 104      Kempen   SM  0.09 0.64 0.27 5.17   2.24     83.60    24
## 105 Leemstreek   GM  0.23 0.20 0.57 6.08   1.89     98.05    17
## 106      Kempen   SM  0.20 0.60 0.20 5.23   2.01     79.80    16
## 107      Kempen   GM  0.20 0.56 0.24 4.86   2.03     81.12    16
## 108      Kempen   GM  0.28 0.50 0.22 5.12   2.16     80.53    24
## 109 Leemstreek   PG  0.23 0.25 0.52 5.81   2.53    121.01    20
## 110 Leemstreek   GM  0.24 0.25 0.52 6.00   1.85     97.66    16
## 111      Kempen   GM  0.11 0.60 0.30 5.37   1.96     81.71    17
## 112 Leemstreek   SM  0.31 0.26 0.43 5.90   2.00     95.37    21
## 113 Leemstreek   PG  0.22 0.28 0.51 6.14   2.36    116.80    21
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## 114     Kempen   GM  0.22 0.61 0.17 4.95   2.10     83.15    17
## 115     Kempen SMGR  0.19 0.57 0.24 5.00   2.21     95.26    16
## 116 Leemstreek   SM  0.21 0.23 0.57 6.09   1.77     95.33    16
## 117 Leemstreek   GM  0.17 0.25 0.59 5.86   1.72     94.81    20
## 118     Kempen SMGR  0.22 0.62 0.16 4.76   2.11     91.31    20
## 119 Leemstreek   SM  0.26 0.22 0.52 6.25   1.93     96.24    11
## 120 Leemstreek   PG  0.16 0.24 0.60 6.07   2.46    120.78    17
## 121     Kempen   GM  0.14 0.65 0.21 4.98   2.07     81.92    18
## 122 Leemstreek   GM  0.27 0.25 0.49 5.60   1.76     93.27    12
## 123 Leemstreek   GM  0.27 0.23 0.50 6.38   1.72     99.53    15
## 124 Leemstreek   SM  0.24 0.24 0.52 6.68   1.71    101.65    13
## 125 Leemstreek   GM  0.18 0.28 0.54 6.04   1.77    100.20    12
## 126     Kempen SMGR  0.11 0.63 0.26 5.27   2.03     94.98    23
## 127 Leemstreek   PG  0.22 0.29 0.49 6.24   2.34    118.42    15
## 128 Leemstreek   SM  0.23 0.25 0.51 5.86   1.71     96.51    11
## 129 Leemstreek   PG  0.09 0.26 0.65 5.93   2.36    121.28    20
## 130     Kempen   GM  0.19 0.57 0.23 4.75   2.05     84.44    21
## 131     Kempen   PG  0.14 0.63 0.23 4.95   2.87    104.80    18
## 132     Kempen SMGR  0.25 0.57 0.18 5.22   2.07     88.54    18
## 133     Kempen   GM  0.25 0.53 0.22 5.04 208.00     79.76    21
## 134     Kempen   PG  0.18 0.61 0.21 5.06   2.82    111.75    23
## 135 Leemstreek   SM  0.19 0.31 0.50 5.98   1.72     88.77    22
## 136     Kempen SMGR  0.16 0.64 0.20 4.73   2.19     89.41    18
## 137     Kempen   GM  0.14 0.66 0.20 5.27   2.11     82.30    22
## 138     Kempen   PG  0.18 0.62 0.20 4.81   2.72    102.04    13
## 139     Kempen   GM  0.22 0.57 0.21 5.06   2.03     83.58    16
## 140 Leemstreek   SM  0.17 0.24 0.58 5.91   1.82     90.81    16
## 141 Leemstreek   PG  0.20 0.24 0.56 5.96   2.44    123.92    11
## 142 Leemstreek   SM  0.29 0.24 0.48 6.29   1.76     92.04    10
## 143 Leemstreek   PG  0.33 0.27 0.40 5.88   2.44    111.91    16
## 144 Leemstreek   SM  0.34 0.21 0.45 5.84   1.95     95.37    12
## 145     Kempen SMGR  0.15 0.64 0.21   NA   2.26     90.67    23
## 146 Leemstreek   PG  0.22 0.28 0.51 5.84   2.38    117.80    19
## 147 Leemstreek   SM  0.33 0.21 0.46 5.77   1.70     96.13    17
## 148 Leemstreek   PG  0.11 0.27 0.62 6.08   2.42    121.58    18
## 149 Leemstreek   PG  0.19 0.32 0.48 5.83   2.55    116.25    11
## 150     Kempen   GM  0.24 0.57 0.18 5.16   2.07     81.81    21
## 151     Kempen   GM  0.14 0.64 0.22 5.07   1.99     84.01    20
## 152 Leemstreek   SM  0.33 0.23 0.44 5.71   1.87    104.46    12
## 153 Leemstreek SMGR  0.21 0.28 0.51 5.84   1.89    106.09    19
## 154 Leemstreek   PG  0.23 0.26 0.51 6.18   2.36    126.85    20
## 155 Leemstreek   GM  0.18 0.30 0.52 6.18   1.79     98.06    15
## 156 Leemstreek SMGR  0.39 0.21 0.40 6.40   2.02    104.37    20
## 157 Leemstreek SMGR  0.24 0.25 0.51 6.00   1.93  95730.00    19
## 158 Leemstreek   GM  0.25 0.23 0.52 5.70   1.76     91.59    14
## 159 Leemstreek   GM  0.34 0.24 0.41 5.85   1.70     95.06    24
## 160 Leemstreek   SM  0.21 0.23 0.56 6.08   1.84    101.02    11
## 161 Leemstreek   SM  0.23 0.22 0.55 6.38   1.71     97.15    24
## 162       <NA>   PG  0.09 0.27 0.64 6.02   2.32    122.37    22
## 163     Kempen SMGR  0.23 0.55 0.22 5.33   2.19     86.13    15
## 164 Leemstreek   SM  0.20 0.29 0.50 6.15   1.80     97.29    16
## 165 Leemstreek SMGR  0.28 0.21 0.51 5.77   2.09    105.08    18
## 166       <NA> SMGR  0.24 0.25 0.50 5.97   1.80    102.93    18
## 167     Kempen   GM  0.21 0.63 0.16 5.36   2.06     82.61    11
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## 168 Leemstreek   PG  0.33 0.27 0.40 6.18   2.51    117.79    15
## 169 Leemstreek   GM  0.26 0.24 0.50 6.09   1.83     97.74    14
## 170 Leemstreek   PG  0.28 0.25 0.47 6.10   2.45    120.01    11
## 171     Kempen SMGR  0.13 0.65 0.22 5.00   2.14     89.92    25
## 172 Leemstreek   PG  0.42 0.21 0.37 5.74   2.44    112.85    15
## 173     Kempen SMGR  0.26 0.56 0.18 5.36   2.35     87.62    24
## 174     Kempen   GM  0.20 0.62 0.19 4.82   2.08     76.40    22
## 175     Kempen   PG  0.17 0.58 0.25 4.91   2.58     99.98    16
## 176     Kempen SMGR  0.14 0.67 0.19 5.23   2.31     88.27    22
## 177 Leemstreek   GM  0.28 0.27 0.45 5.75   1.79     95.64    14
## 178 Leemstreek   SM  0.24 0.25 0.50 5.89   1.80     99.36    22
## 179     Kempen SMGR  0.27 0.52 0.21 4.86   2.19     94.22    11
## 180 Leemstreek   GM  0.24 0.25 0.51 6.12   1.74     98.01    13
## 181     Kempen   PG  0.13 0.58 0.29 5.19   2.77    109.99    25
## 182 Leemstreek   PG  0.32 0.25 0.43 6.08   2.41    118.32    11
## 183     Kempen   PG  0.25 0.54 0.21 5.19   2.77    104.97    18
## 184 Leemstreek   GM  0.27 0.26 0.47 6.11   1.67    100.45    20
## 185 Leemstreek   PG    NA   NA   NA 6.13   2.47    120.33    14
## 186 Leemstreek   SM  0.30 0.24 0.45 5.61   1.89     91.84    17
## 187 Leemstreek   SM  0.24 0.26 0.50 5.97   1.84     95.08    23
## 188 Leemstreek SMGR  0.23 0.23 0.55 6.02   1.98    108.14    17
## 189     Kempen   PG  0.18 0.63 0.19 5.24   2.66    103.83    12
## 190     Kempen SMGR  0.20 0.53 0.27 4.86   2.20     91.34    15
## 191 Leemstreek   GM  0.29 0.17 0.54 5.90   1.80     96.34    21
## 192 Leemstreek   SM  0.30 0.26 0.44 5.79   1.71     96.37    11
## 193     Kempen   PG  0.12 0.64 0.23 5.31   2.87    104.83    24
## 194     Kempen   SM  0.20 0.59 0.21 5.36   2.01     85.66    20
## 195     Kempen   SM  0.15 0.63 0.23 5.26   2.11     84.95    16
## 196 Leemstreek   PG  0.29 0.22 0.49 5.76   2.46    124.95    12
## 197 Leemstreek   GM  0.19 0.25 0.56 6.09   1.76     95.56    24
## 198     Kempen   GM  0.32 0.50 0.18 5.33   2.03     78.66    13
## 199 Leemstreek   PG  0.25 0.29 0.47 6.03   2.49    120.94    21
## 200 Leemstreek   SM  0.21 0.31 0.49 5.89   1.72    100.96    10
## 201 Leemstreek   SM  0.28 0.28 0.44 5.78   1.69    106.93    14
## 202     Kempen SMGR  0.21 0.60 0.19 4.97   2.14     88.33    25
## 203     Kempen   SM  0.18 0.60 0.22 5.24   2.09     82.91    21
## 204 Leemstreek   GM  0.30 0.20 0.49 5.91   1.85     99.64    20
## 205 Leemstreek   SM  0.38 0.27 0.35 6.05   1.93    103.18    14
## 206     Kempen   PG  0.16 0.59 0.25 5.36   2.77    103.67    11
## 207 Leemstreek   SM  0.25 0.23 0.51 6.12   1.83    100.85    14
## 208 Leemstreek   PG  0.15 0.21 0.64 6.27   2.43    118.21    18
## 209 Leemstreek   GM  0.22 0.24 0.54 5.66   1.75     92.58    22
## 210 Leemstreek   GM  0.30 0.22 0.47 5.94   1.64     98.66    14
## 211 Leemstreek SMGR  0.09 0.24 0.67 6.01   1.92    104.71    15
## 212 Leemstreek   SM  0.12 0.21 0.67 6.12   1.76     99.80    24
## 213 Leemstreek   SM  0.27 0.30 0.43 5.77   1.82     96.05    17
## 214     Kempen   PG  0.17 0.60 0.23 5.26   2.72    112.98    25
## 215 Leemstreek SMGR  0.23 0.28 0.49 6.06   1.85    106.84    23
## 216     Kempen SMGR  0.30 0.47 0.23 5.15   2.18     88.71    25
## 217 Leemstreek   PG  0.25 0.24 0.51 6.10   2.42    122.99    12
## 218 Leemstreek SMGR  0.35 0.23 0.42 5.98   1.99     99.68    15
## 219     Kempen   PG  0.23 0.54 0.23 5.29   2.76    105.64    24
## 220     Kempen   SM  0.05 0.68 0.27 5.23   2.13     89.48    20
## 221 Leemstreek   PG  0.30 0.21 0.49 5.97   2.35    119.58    25
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## 222    Kempeen   PG  0.17 0.62 0.22 5.31   2.87    109.63    25
## 223     Kempen SMGR  0.10 0.64 0.26 5.16   2.21     89.29    21
## 224     Kempen   GM  0.14 0.61 0.25 5.23   1.98     73.42    16
## 225 Leemstreek   SM  0.38 0.22 0.40 5.88   1.82     94.93    20
## 226 Leemstreek   GM  0.21 0.28 0.51 6.32   1.82     97.21    14
## 227 Leemstreek   SM  0.15 0.26 0.60 6.00   1.69    103.80    15
## 228 Leemstreek   PG  0.20 0.22 0.58 6.17   2.58    118.08    23
## 229     Kempen   GM  0.24 0.53 0.23 5.08   2.03     83.17    20
## 230     Kempen   GM  0.05 0.70 0.24 5.04   2.13     78.44    22
## 231     Kempen SMGR  0.25 0.57 0.18 5.26   2.20     86.29    15
## 232 Leemstreek   PG  0.29 0.19 0.52 5.68   2.42    117.29    20
## 233 Leemstreek   GM  0.29 0.27 0.44 6.10   1.60    101.52    19
## 234 Leemstreek SMGR  0.27 0.26 0.47 6.08   1.81    107.89    13
## 235 Leemstreek   SM  0.23 0.21 0.56 5.46   1.79    100.45    14
## 236 Leemstreek   SM  0.38 0.19 0.43 6.07   1.87    102.56    23
## 237 Leemstreek   PG  0.24 0.25 0.52 5.57   2.30    118.60    22
## 238     Kempen SMGR  0.11 0.66 0.23 5.23   2.31     93.04    10
## 239 Leemstreek   PG  0.23 0.27 0.50 5.91   2.56    121.08    21
## 240     Kempen   GM  0.15 0.58 0.27 4.96   1.95     78.11    23
## 241 Leemstreek SMGR  0.34 0.22 0.44 6.00   1.94    101.14    23
## 242       <NA> SMGR  0.15 0.61 0.24 5.37   2.11     90.37    12
## 243 Leemstreek   PG  0.25 0.25 0.49 5.87   2.44    121.06    25
## 244     Kempen   GM  0.16 0.63 0.21 5.21   2.12     80.92    24
## 245 Leemstreek   SM  0.23 0.24 0.53 6.07   1.87     94.53    14
## 246     Kempen SMGR  0.17 0.64 0.19 4.86   2.13     82.13    23
## 247 Leemstreek SMGR  0.35 0.23 0.42 5.65   1.87    105.85    22
## 248     Kempen   SM  0.21 0.58 0.21 4.70   2.02     82.92    17
## 249     Kempen   GM  0.09 0.64 0.27 5.21   2.07     84.23    11
## 250     Kempen SMGR    NA   NA   NA 5.09   2.07     84.25    23
## 251 Leemstreek SMGR    NA   NA   NA 6.37   1.69     99.61    15
## 252 Leemstreek   PG  0.26 0.25 0.49 6.11   2.57    119.90    13
## 253 Leemstreek   GM  0.20 0.27 0.53 6.17   1.93     90.97    22
## 254 Leemstreek SMGR  0.35 0.21 0.44 6.27   1.86     96.82    25
## 255 Leemstreek SMGR  0.17 0.26 0.57 5.90   1.96    107.26    13
## 256     Kempen   PG  0.20 0.56 0.24 5.20   2.75    109.57    11
## 257 Leemstreek   PG  0.27 0.28 0.45 6.17   2.57    116.42    20
## 258 Leemstreek   SM  0.23 0.27 0.50   NA   1.84     97.32    23
## 259 Leemstreek   GM  0.18 0.27 0.54 6.15   1.77     92.02    13
## 260     Kempen   GM  0.16 0.61 0.24 5.18   1.90     83.34    18
## 261     Kempen   PG  0.09 0.64 0.26 4.93   2.83    105.97    15
## 262     Kempen   PG  0.13 0.67 0.20 5.25   2.72    102.85    22
## 263     Kempen   SM  0.05 0.70 0.25 5.20   1.98     81.20    17
## 264     Kempen SMGR  0.15 0.60 0.25 5.23   2.18     86.11    13
## 265 Leemstreek   SM  0.33 0.18 0.49 6.08   1.79     99.98    14
## 266 Leemstreek   SM  0.31 0.25 0.44 5.96   1.88    100.46    22
## 267 Leemstreek   PG  0.31 0.26 0.43 5.98 234.00    121.73    10
## 268 Leemstreek   SM  0.23 0.25 0.53 6.14   1.75     95.13    16
## 269 Leemstreek SMGR  0.24 0.27 0.50 6.03   2.02     97.52    19
## 270 Leemstreek   SM    NA   NA   NA 5.83   1.91    100.29    18
## 271     Kempen   SM  0.19 0.57 0.24 5.04   2.19     83.44    20
## 272 Leemstreek   GM  0.28 0.24 0.48 5.74   1.88     96.10    18
## 273 Leemstreek   SM  0.30 0.26 0.44 5.81   1.81    102.19    18
## 274 Leemstreek   SM  0.30 0.23 0.47 5.97   1.75     94.90    23
## 275     Kempen   GM  0.17 0.60 0.23 4.74   2.09     86.39    15
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## 276     Kempen SMGR  0.29 0.50 0.21 4.76   2.28     88.48    11
## 277     Kempen   PG  0.22 0.54 0.24 5.32   2.94    103.46    22
## 278 Leemstreek   SM  0.31 0.21 0.48 5.88   1.64     97.50    22
## 279 Leemstreek   GM  0.28 0.22 0.49 5.63   1.77     97.71    23
## 280     Kempen   GM  0.24 0.57 0.20 5.08   2.10     81.33    11
## 281     Kempen SMGR  0.10 0.62 0.28 5.36   2.19     86.76    10
## 282 Leemstreek SMGR  0.20 0.28 0.52 6.13   1.77    103.26    19
## 283 Leemstreek   PG  0.26 0.23 0.51 6.09   2.51    110.66    23
## 284 Leemstreek   PG  0.38 0.22 0.40 6.24   2.39    118.42    17
## 285 Leemstreek   PG  0.24 0.22 0.54 5.85   2.49    122.01    11
## 286 Leemstreek SMGR  0.32 0.24 0.44 6.15   1.97    103.69    16
## 287 Leemstreek   GM  0.31 0.24 0.45 5.88   1.85     98.90    15
## 288 Leemstreek   SM  0.18 0.26 0.56 6.00   1.67    100.15    17
## 289 Leemstreek   GM  0.29 0.24 0.47 6.44   1.84     98.84    12
## 290     Kempen   SM  0.34 0.50 0.17 5.29   2.04     86.66    17
## 291 Leemstreek   SM  0.15 0.25 0.61 6.15   1.70     91.73    14
## 292 Leemstreek SMGR  0.36 0.29 0.36 5.78   2.00    104.28    17
## 293 Leemstreek   GM  0.18 0.29 0.53 5.84   1.84    102.47    24
## 294     Kempen   SM  0.19 0.56 0.25 5.56   2.08     82.39    18
## 295     Kempen   SM  0.28 0.52 0.20 4.88   2.15     85.80    21
## 296     Kempen SMGR  0.04 0.72 0.23 5.14   2.13     90.68    11
## 297     Kempen   SM  0.17 0.59 0.23 5.28   2.08     86.28    20
## 298 Leemstreek   SM  0.19 0.25 0.56 6.20   1.79     97.31    22
## 299 Leemstreek   GM  0.21 0.26 0.53 6.22   1.73     93.88    10
## 300     Kempen   GM  0.27 0.52 0.21 5.07   2.10     82.39    11
## 301     Kempen   PG  0.19 0.56 0.25 4.94   2.87    106.88    19
## 302 Leemstreek   SM  0.25 0.20 0.55 5.94   1.76     97.59    22
## 303     Kempen   SM  0.24 0.57 0.19 4.92   2.09     84.20    21
## 304     Kempen   PG  0.22 0.61 0.18 5.23   2.88    107.71    14
## 305     Kempen   GM  0.24 0.53 0.22 5.32   2.02     79.22    22
## 306 Leemstreek SMGR  0.25 0.27 0.48 6.43   1.69    108.71    16
## 307 Leemstreek   GM  0.30 0.26 0.44 6.07   1.79    104.68    12
## 308 Leemstreek   GM  0.27 0.22 0.51 5.83   1.68    101.88    19
## 309 Leemstreek   SM  0.32 0.26 0.42 6.20   1.72     96.61    22
## 310 Leemstreek   GM  0.22 0.27 0.51 6.18   1.86     89.41    22
## 311 Leemstreek SMGR  0.26 0.28 0.46 5.95   2.09    104.50    11
## 312 Leemstreek   GM  0.20 0.24 0.56 5.69   1.66     99.59    11
## 313     Kempen SMGR  0.11 0.60 0.29 5.25   2.28     93.11    24
## 314 Leemstreek SMGR  0.31 0.25 0.45 6.23   1.89     99.67    11
## 315 Leemstreek   GM  0.14 0.29 0.56 5.94   1.76     92.06    13
## 316     Kempen   GM  0.10 0.66 0.24 5.21   2.14     82.65    17
## 317     Kempen   SM  0.14 0.64 0.22 5.04   2.11     86.49    22
## 318 Leemstreek   SM  0.29 0.24 0.47 6.14   1.73    101.65    23
## 319 Leemstreek   PG  0.14 0.26 0.59 5.98   2.41    123.40    19
## 320     Kempen   SM  0.22 0.62 0.16 5.03   2.15     80.43    16
## 321     Kempen   GM  0.25 0.55 0.20 5.12   1.86     77.14    12
## 322 Leemstreek SMGR  0.39 0.20 0.41 5.94   1.99     96.64    23
## 323 Leemstreek SMGR  0.15 0.27 0.58 5.88   1.77    106.20    23
## 324     Kempen   PG  0.26 0.52 0.21 5.04   2.75    109.26    15
## 325 Leemstreek SMGR  0.28 0.25 0.47 6.00   1.88    105.58    18
## 326 Leemstreek SMGR  0.05 0.26 0.69 6.04   2.00    100.29    17
## 327     Kempen   PG  0.15 0.63 0.22 5.41   2.76    112.14    22
## 328 Leemstreek SMGR  0.27 0.26 0.47 5.73   1.91    102.84    24
## 329 Leemstreek SMGR  0.25 0.23 0.52 5.96   1.95    111.06    19
```

```
## 330     Kempen   SM  0.18 0.65 0.17 5.23   2.03     86.24    22
## 331     Kempen   GM  0.08 0.69 0.23 5.45   2.03     86.72    22
## 332 Leemstreek   PG  0.23 0.28 0.49 6.09   2.54    121.58    14
## 333     Kempen SMGR  0.26 0.50 0.23 5.10   2.25     89.88    18
## 334     Kempen   PG  0.14 0.67 0.19 5.04   2.64    102.26    14
## 335  Leemstrek   SM  0.31 0.25 0.44 6.08   1.85     96.20    23
## 336 Leemstreek   GM  0.28 0.27 0.46 6.20   1.64     96.16    21
## 337     Kempen   PG  0.17 0.63 0.20 5.06   2.70    107.19    15
## 338     Kempen   PG  0.15 0.60 0.26 5.13   2.84    106.00    25
## 339     Kempen   GM  0.14 0.60 0.27 5.14   1.87     81.46    21
## 340     Kempen   SM  0.13 0.65 0.22 5.25   1.95     84.85    14
## 341 Leemstreek   PG  0.25 0.27 0.48 6.22   2.55    118.35    20
## 342 Leemstreek   SM  0.24 0.28 0.48 6.06   1.73     96.28    15
## 343 Leemstreek   SM  0.17 0.32 0.50 5.98   1.90     92.77    22
## 344 Leemstreek SMGR  0.22 0.27 0.51 6.28   1.89    107.57    15
## 345 Leemstreek   GM  0.23 0.26 0.52 6.03   1.78     96.49    14
## 346       <NA>   GM  0.36 0.18 0.46 6.32   1.82     96.04    11
## 347     Kempen SMGR  0.07 0.73 0.19 5.09   2.14     87.30    19
## 348 Leemstreek   SM  0.19 0.24 0.57 6.12   1.78     97.38    10
## 349 Leemstreek SMGR  0.21 0.32 0.47 5.69   1.86    100.11    14
## 350     Kempen   PG  0.15 0.62 0.23 5.08   2.97    102.13    14
## 351 Leemstreek   SM  0.21 0.30 0.50 6.00   1.81     91.01    22
## 352     Kempen   SM  0.21 0.59 0.19 4.95   2.13     87.82    12
## 353 Leemstreek   SM  0.20 0.27 0.54 5.79   1.77     98.74    14
## 354 Leemstreek   SM  0.23 0.26 0.51   NA   1.78     94.09    17
## 355     Kempen SMGR  0.15 0.59 0.26 5.28   2.31     94.58    24
## 356     Kempen   GM  0.18 0.59 0.22 4.84   2.02     84.44    15
## 357     Kempen   GM  0.14 0.65 0.21 5.18   2.05     84.96    22
## 358 Leemstreek   PG  0.24 0.24 0.52   NA   2.48    119.40    12
## 359 Leemstreek   SM  0.38 0.19 0.43 5.79   1.76     96.65    17
## 360     Kempen   SM  0.21 0.59 0.19 5.21   2.25     83.77    18
## 361 Leemstreek   GM  0.20 0.27 0.53 5.86   1.63     97.81    23
## 362 Leemstreek   SM  0.25 0.27 0.48 6.20   1.96    101.38    15
## 363     Kempen SMGR  0.15 0.63 0.22 5.11   2.17     96.16    17
## 364 Leemstreek   GM  0.31 0.20 0.49 6.11   1.86     94.39    17
## 365 Leemstreek SMGR  0.28 0.26 0.46 5.90   1.74    100.42    21
## 366     Kempen   SM  0.19 0.65 0.16 4.87   2.17     84.46    11
## 367 Leemstreek   SM  0.19 0.29 0.52 6.18   1.87    109.82    21
## 368 Leemstreek   GM  0.21 0.24 0.55 6.05   1.65     94.68    20
## 369 Leemstreek SMGR  0.34 0.24 0.42 5.95   1.84 108650.00    20
## 370 Leemstreek SMGR  0.12 0.29 0.59 6.01   1.91    103.38    17
## 371 Leemstreek   PG  0.28 0.20 0.52 6.27   2.54    118.53    14
## 372 Leemstreek   PG  0.18 0.24 0.58 5.86   2.39    117.58    16
## 373     Kempen SMGR  0.19 0.62 0.19 4.95   2.18     89.29    12
## 374 Leemstreek   PG  0.29 0.20 0.51 5.88   2.57    126.52    14
## 375 Leemstreek   SM  0.25 0.26 0.49 5.68   1.71     99.39    11
## 376     Kempen   SM  0.06 0.69 0.24 5.26   2.17     86.01    17
## 377     Kempen   GM  0.20 0.60 0.20 5.14   2.08     85.32    15
## 378 Leemstreek   GM  0.26 0.24 0.50 6.04   1.66    104.18    13
## 379 Leemstreek SMGR  0.34 0.26 0.39 5.99   1.85    107.25    21
## 380     Kempen   PG  0.22 0.59 0.20 5.00   2.81    100.68    14
## 381 Leemstreek   SM  0.22 0.27 0.51 5.84   1.82     94.34    16
## 382     Kempen   PG  0.16 0.65 0.20 5.27   2.86    100.12    21
## 383 Leemstreek   SM  0.25 0.25 0.50 5.71   1.75     87.42    12
```

```
## 384     Kempen SMGR  0.25 0.54 0.21 5.19   2.06     95.05    12
## 385 Leemstreek   PG    NA   NA   NA 5.84   2.53    118.40    25
## 386     Kempen   GM  0.17 0.57 0.26 5.16   2.00     83.64    25
## 387 Leemstreek   SM  0.28 0.27 0.45 5.85   1.81    102.76    21
## 388 Leemstreek   SM  0.33 0.19 0.48 5.90   1.94     96.34    10
## 389 Leemstreek   SM  0.22 0.19 0.59 6.35   1.78     97.86    25
## 390 Leemstreek   PG  0.40 0.23 0.37 6.13   2.57    121.66    21
## 391     Kempen   PG  0.17 0.62 0.21 4.67   2.66    106.89    23
## 392 Leemstreek   GM  0.15 0.22 0.62 6.03   1.82     93.36    13
## 393     Kempen   SM  0.07 0.64 0.29 5.01   2.11     86.88    14
## 394     Kempen   GM  0.16 0.62 0.22 4.87   1.99     84.15    19
## 395 Leemstreek   PG  0.31 0.24 0.45 6.04   2.35    119.21    15
## 396 Leemstreek   GM  0.22 0.27 0.51 6.03   1.76     91.66    21
## 397 Leemstreek   SM  0.28 0.26 0.46 5.93   1.82    100.70    11
## 398 Leemstreek   SM  0.18 0.24 0.59 6.39   1.82    103.80    23
## 399 Leemstreek   GM  0.29 0.26 0.45 5.55   1.79    101.46    22
## 400     Kempen   SM  0.14 0.63 0.23 5.27   2.06     87.47    23
```

```
summary(removed_rows)
```

```
##     Region              Crop               Lutum             Sand
##  Length:397         Length:397         Min.   :0.0400   Min.   :0.1700
##  Class :character   Class :character   1st Qu.:0.1700   1st Qu.:0.2400
##  Mode  :character   Mode  :character   Median :0.2200   Median :0.2750
##                                        Mean   :0.2198   Mean   :0.3885
##                                        3rd Qu.:0.2700   3rd Qu.:0.5900
##                                        Max.   :0.4200   Max.   :0.7300
##                                        NA's   :5        NA's   :5
##       Silt              pH              C               Cstock
##  Min.   :0.1400   Min.   :4.64   Min.   :  1.550   Min.   :    73.42
##  1st Qu.:0.2300   1st Qu.:5.21   1st Qu.:  1.820   1st Qu.:    89.41
##  Median :0.4400   Median :5.83   Median :  2.010   Median :    97.38
##  Mean   :0.3916   Mean   :5.65   Mean   :  3.806   Mean   :   612.54
##  3rd Qu.:0.5100   3rd Qu.:6.04   3rd Qu.:  2.280   3rd Qu.:   104.71
##  Max.   :0.6900   Max.   :6.68   Max.   :252.000   Max.   :108650.00
##  NA's   :5        NA's   :5
##      Years
##  Min.   :10.00
##  1st Qu.:14.00
##  Median :17.00
##  Mean   :17.34
##  3rd Qu.:21.00
##  Max.   :25.00
##
```

```
na_positions <- is.na(removed_rows)

# Print the row and column indices where NAs are present
which(na_positions, arr.ind = TRUE)
```

```
##     row col
## 93   90   1
## 162 159   1
```

```
## 166 163   1
## 242 239   1
## 346 343   1
## 185 182   3
## 250 247   3
## 251 248   3
## 270 267   3
## 385 382   3
## 185 182   4
## 250 247   4
## 251 248   4
## 270 267   4
## 385 382   4
## 185 182   5
## 250 247   5
## 251 248   5
## 270 267   5
## 385 382   5
## 96   93   6
## 145 142   6
## 258 255   6
## 354 351   6
## 358 355   6
```

```
sum(rowSums(is.na(removed_rows)) > 0)
```

```
## [1] 15
```

```
data_clean <- na.omit(removed_rows)
print(data_clean)
```

```
##         Region Crop Lutum Sand Silt   pH      C    Cstock Years
## 1   Leemstreek   PG  0.23 0.25 0.52 6.12   2.43    118.38    15
## 2       Kempen   SM  0.26 0.54 0.20 4.95   2.15     84.99    16
## 3   Leemstreek   PG  0.22 0.23 0.55 6.17   2.41    118.28    14
## 4       Kempen SMGR  0.15 0.60 0.25 4.95   2.15     94.72    12
## 5       Kempen   GM  0.14 0.63 0.23 5.23   1.91     82.81    14
## 6   Leemstreek   GM  0.29 0.20 0.51 6.22   1.73     94.01    14
## 7   Leemstreek SMGR  0.27 0.24 0.50 5.80   2.10     97.40    15
## 8       Kempen   SM  0.08 0.64 0.28 5.32   2.18     81.72    10
## 9   Leemstreek   GM  0.25 0.23 0.52 5.90   1.88    104.06    21
## 10  Leemstreek SMGR  0.25 0.26 0.49 6.06   1.97    102.18    17
## 11      Kempen   GM  0.23 0.62 0.14 5.14   2.14     78.95    19
## 12  Leemstreek   PG  0.34 0.26 0.41 5.91 252.00    122.77    12
## 13      Kempen   SM  0.15 0.63 0.22 5.15   2.13     85.81    10
## 14  Leemstreek   GM  0.24 0.25 0.51 5.75   1.70     99.11    13
## 15  Leemstreek   SM  0.25 0.24 0.52 5.97   1.76     97.24    10
## 17  Leemstreek   PG  0.26 0.25 0.49 5.57   2.45    115.76    17
## 18  Leemstreek   GM  0.16 0.26 0.57 5.70   1.82     96.06    22
## 19  Leemstreek   GM  0.29 0.27 0.44 5.77   1.83     93.58    18
## 20      Kempen   SM  0.24 0.57 0.18 4.78   2.10     86.64    12
## 21      Kempen   GM  0.13 0.69 0.18 5.18   2.00     82.63    10
## 22      Kempen SMGR  0.14 0.61 0.24 4.90   2.46     94.30    16
```

```
## 23      Kempen SMGR  0.18 0.61 0.22 4.67   2.25     97.85    13
## 24      Kempen   GM  0.09 0.66 0.24 4.97   2.16     82.63    22
## 25      Kempen   PG  0.22 0.56 0.21 5.02   2.72    104.07    18
## 26     Keempen SMGR  0.22 0.58 0.20 5.27   2.19     87.03    18
## 27  Leemstreek   GM  0.20 0.32 0.48 5.78   1.78     95.26    24
## 28  Leemstreek   GM  0.36 0.25 0.39 6.23   1.88     96.08    11
## 30  Leemstreek   PG  0.38 0.21 0.41 6.07   2.52    116.03    18
## 31      Kempen   PG  0.16 0.59 0.25 4.93   2.58    101.93    12
## 32      Kempen SMGR  0.18 0.62 0.20 5.49   2.16     88.65    14
## 33      Kempen   GM  0.14 0.62 0.24 5.07   1.94     84.35    12
## 34      Kempen   PG  0.18 0.55 0.27 4.74   2.83    109.17    12
## 35  Leemstreek   SM  0.26 0.25 0.49 5.96   1.79     94.84    19
## 36  Leemstreek SMGR  0.27 0.22 0.52 5.61   1.87    104.71    11
## 37      Kempen   SM  0.20 0.64 0.17 5.04   2.20     83.95    19
## 38  Leemstreek   GM  0.27 0.28 0.45 5.92   1.84    100.83    21
## 39  Leemstreek   SM  0.31 0.18 0.50 6.14   1.73    102.47    16
## 40  Leemstreek   SM  0.19 0.29 0.52 6.20   1.83     97.15    21
## 41  Leemstreek   SM  0.33 0.21 0.45 5.85   1.70     98.45    11
## 42  Leemstreek SMGR  0.08 0.26 0.66 6.16   1.86    106.81    19
## 43  Leemstreek SMGR  0.26 0.27 0.47 6.28   2.03     98.07    14
## 45  Leemstreek   PG  0.23 0.23 0.54 5.87   2.54    122.46    14
## 46  Leemstreek SMGR  0.25 0.25 0.50 6.13   1.85    102.34    19
## 47  Leemstreek SMGR  0.28 0.26 0.46 6.04   1.75    104.08    16
## 48  Leemstreek   SM  0.16 0.28 0.56 6.11   1.71     98.24    20
## 49  Leemstreek   GM  0.21 0.19 0.60 6.28   1.83     93.61    24
## 50      Kempen SMGR  0.26 0.52 0.22 5.18   2.17     86.36    18
## 51  Leemstreek   GM  0.17 0.29 0.53 6.21   1.58     95.58    14
## 52  Leemstreek   GM  0.22 0.28 0.49 6.12   1.74     95.01    21
## 53      Kempen   SM  0.13 0.62 0.25 5.19   2.21     87.31    18
## 54  Leemstreek   SM  0.30 0.27 0.43 6.08   1.85    104.13    12
## 55  Leemstreek   SM  0.08 0.28 0.64 6.26   1.73     98.31    14
## 56  Leemstreek   PG  0.37 0.17 0.46 5.80   2.32    121.78    12
## 57  Leemstreek   PG  0.30 0.28 0.41 5.78   2.38    119.99    12
## 58      Kempen SMGR  0.21 0.58 0.21 5.22   2.20     92.43    20
## 59      Kempen   SM  0.16 0.61 0.22 5.08   2.17     84.95    21
## 60  Leemstreek   SM  0.16 0.24 0.60 6.01   1.64     99.20    17
## 61      Kempen   PG  0.11 0.64 0.25 5.25   2.83    110.38    12
## 62  Leemstreek   GM  0.25 0.24 0.51 6.15   1.75     92.58    23
## 63  Leemstreek   GM  0.32 0.27 0.42 5.95   1.79    101.79    25
## 64  Leemstreek   SM  0.31 0.24 0.45 5.94   1.67     99.62    10
## 65      Kempen SMGR  0.24 0.55 0.22 5.11   2.14     90.81    18
## 66  Leemstreek   SM  0.06 0.29 0.65 6.21   1.76     96.21    22
## 67      Kempen   SM  0.20 0.64 0.16 5.19   2.18     83.52    18
## 68      Kempen SMGR  0.06 0.69 0.25 5.39   2.28     91.60    14
## 69      Kempen   SM  0.19 0.60 0.20 5.10   2.13     76.28    17
## 70  Leemstreek   PG  0.12 0.28 0.60 6.23   2.58    118.06    14
## 71      Kempen   SM  0.12 0.70 0.19 5.28   2.05     86.39    25
## 72      Kempen   PG  0.19 0.59 0.22 5.22   2.83    105.03    20
## 73      Kempen   GM  0.28 0.53 0.19 5.18   2.04     79.10    21
## 74  Leemstreek SMGR  0.20 0.24 0.55 5.94   1.85    101.88    10
## 75  Leemstreek SMGR  0.17 0.24 0.59 5.98   1.97    105.19    16
## 76  Leemstreek   GM  0.13 0.25 0.62 6.05   1.86     90.10    17
## 77  Leemstreek SMGR  0.15 0.30 0.55 6.01   1.97    107.62    12
## 78      Kempen   SM  0.14 0.58 0.28 5.29   2.06     75.92    14
```

```
## 79  Leemstreek   SM  0.16 0.26 0.57 5.98   1.76     96.39    12
## 80  Leemstreek SMGR  0.26 0.24 0.49 5.99   2.01    101.31    16
## 81  Leemstreek   SM  0.22 0.25 0.53 5.90   1.83     93.54    13
## 82      Kempen   PG  0.19 0.62 0.19 5.07   2.81    102.02    18
## 83  Leemstreek   GM  0.22 0.29 0.49 5.99   1.72     97.25    23
## 84      Kempen SMGR  0.14 0.61 0.26 4.64   2.29     89.57    18
## 85   Lemstreek   PG  0.20 0.27 0.53 5.73   2.38    119.11    16
## 86  Leemstreek   SM  0.20 0.27 0.53 5.99   1.85     89.48    14
## 87      Kempen   SM  0.17 0.65 0.19 5.05   2.03     79.08    13
## 88      Kempen SMGR  0.23 0.57 0.20 4.86   2.00     89.57    11
## 89      Kempen   GM  0.12 0.68 0.20 4.70   1.94     78.70    22
## 90  Leemstreek   PG  0.24 0.24 0.52 5.93   2.46    117.26    24
## 91  Leemstreek   PG  0.31 0.25 0.44 6.13   2.52    124.32    24
## 92      Kempen   SM  0.09 0.67 0.24 5.45   1.96     84.86    14
## 94      Kempen SMGR  0.18 0.55 0.27 5.40   2.34     88.40    15
## 95  Leemstreek   SM  0.25 0.22 0.53 6.02   1.76     97.22    15
## 97      Kempen   SM  0.16 0.61 0.23 5.16   2.05     80.47    20
## 98  Leemstreek   SM  0.27 0.25 0.48 5.83   1.77     98.07    12
## 99  Leemstreek SMGR  0.22 0.26 0.52 5.99   1.91    107.12    17
## 100 Leemstreek SMGR  0.22 0.27 0.51 6.18   1.90    105.20    11
## 101 Leemstreek   GM  0.33 0.23 0.44 5.85   1.73     95.51    20
## 102 Leemstreek   GM  0.33 0.22 0.45 5.94   1.71     97.76    24
## 103 Leemstreek   SM  0.20 0.28 0.52 5.77   1.55    101.27    24
## 104     Kempen   SM  0.09 0.64 0.27 5.17   2.24     83.60    24
## 105 Leemstreek   GM  0.23 0.20 0.57 6.08   1.89     98.05    17
## 106     Kempen   SM  0.20 0.60 0.20 5.23   2.01     79.80    16
## 107     Kempen   GM  0.20 0.56 0.24 4.86   2.03     81.12    16
## 108     Kempen   GM  0.28 0.50 0.22 5.12   2.16     80.53    24
## 109 Leemstreek   PG  0.23 0.25 0.52 5.81   2.53    121.01    20
## 110 Leemstreek   GM  0.24 0.25 0.52 6.00   1.85     97.66    16
## 111     Kempen   GM  0.11 0.60 0.30 5.37   1.96     81.71    17
## 112 Leemstreek   SM  0.31 0.26 0.43 5.90   2.00     95.37    21
## 113 Leemstreek   PG  0.22 0.28 0.51 6.14   2.36    116.80    21
## 114     Kempen   GM  0.22 0.61 0.17 4.95   2.10     83.15    17
## 115     Kempen SMGR  0.19 0.57 0.24 5.00   2.21     95.26    16
## 116 Leemstreek   SM  0.21 0.23 0.57 6.09   1.77     95.33    16
## 117 Leemstreek   GM  0.17 0.25 0.59 5.86   1.72     94.81    20
## 118     Kempen SMGR  0.22 0.62 0.16 4.76   2.11     91.31    20
## 119 Leemstreek   SM  0.26 0.22 0.52 6.25   1.93     96.24    11
## 120 Leemstreek   PG  0.16 0.24 0.60 6.07   2.46    120.78    17
## 121     Kempen   GM  0.14 0.65 0.21 4.98   2.07     81.92    18
## 122 Leemstreek   GM  0.27 0.25 0.49 5.60   1.76     93.27    12
## 123 Leemstreek   GM  0.27 0.23 0.50 6.38   1.72     99.53    15
## 124 Leemstreek   SM  0.24 0.24 0.52 6.68   1.71    101.65    13
## 125 Leemstreek   GM  0.18 0.28 0.54 6.04   1.77    100.20    12
## 126     Kempen SMGR  0.11 0.63 0.26 5.27   2.03     94.98    23
## 127 Leemstreek   PG  0.22 0.29 0.49 6.24   2.34    118.42    15
## 128 Leemstreek   SM  0.23 0.25 0.51 5.86   1.71     96.51    11
## 129 Leemstreek   PG  0.09 0.26 0.65 5.93   2.36    121.28    20
## 130     Kempen   GM  0.19 0.57 0.23 4.75   2.05     84.44    21
## 131     Kempen   PG  0.14 0.63 0.23 4.95   2.87    104.80    18
## 132     Kempen SMGR  0.25 0.57 0.18 5.22   2.07     88.54    18
## 133     Kempen   GM  0.25 0.53 0.22 5.04 208.00     79.76    21
## 134     Kempen   PG  0.18 0.61 0.21 5.06   2.82    111.75    23
```

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## 135 Leemstreek   SM  0.19 0.31 0.50 5.98   1.72     88.77    22
## 136     Kempen SMGR  0.16 0.64 0.20 4.73   2.19     89.41    18
## 137     Kempen   GM  0.14 0.66 0.20 5.27   2.11     82.30    22
## 138     Kempen   PG  0.18 0.62 0.20 4.81   2.72    102.04    13
## 139     Kempen   GM  0.22 0.57 0.21 5.06   2.03     83.58    16
## 140 Leemstreek   SM  0.17 0.24 0.58 5.91   1.82     90.81    16
## 141 Leemstreek   PG  0.20 0.24 0.56 5.96   2.44    123.92    11
## 142 Leemstreek   SM  0.29 0.24 0.48 6.29   1.76     92.04    10
## 143 Leemstreek   PG  0.33 0.27 0.40 5.88   2.44    111.91    16
## 144 Leemstreek   SM  0.34 0.21 0.45 5.84   1.95     95.37    12
## 146 Leemstreek   PG  0.22 0.28 0.51 5.84   2.38    117.80    19
## 147 Leemstreek   SM  0.33 0.21 0.46 5.77   1.70     96.13    17
## 148 Leemstreek   PG  0.11 0.27 0.62 6.08   2.42    121.58    18
## 149 Leemstreek   PG  0.19 0.32 0.48 5.83   2.55    116.25    11
## 150     Kempen   GM  0.24 0.57 0.18 5.16   2.07     81.81    21
## 151     Kempen   GM  0.14 0.64 0.22 5.07   1.99     84.01    20
## 152 Leemstreek   SM  0.33 0.23 0.44 5.71   1.87    104.46    12
## 153 Leemstreek SMGR  0.21 0.28 0.51 5.84   1.89    106.09    19
## 154 Leemstreek   PG  0.23 0.26 0.51 6.18   2.36    126.85    20
## 155 Leemstreek   GM  0.18 0.30 0.52 6.18   1.79     98.06    15
## 156 Leemstreek SMGR  0.39 0.21 0.40 6.40   2.02    104.37    20
## 157 Leemstreek SMGR  0.24 0.25 0.51 6.00   1.93  95730.00    19
## 158 Leemstreek   GM  0.25 0.23 0.52 5.70   1.76     91.59    14
## 159 Leemstreek   GM  0.34 0.24 0.41 5.85   1.70     95.06    24
## 160 Leemstreek   SM  0.21 0.23 0.56 6.08   1.84    101.02    11
## 161 Leemstreek   SM  0.23 0.22 0.55 6.38   1.71     97.15    24
## 163     Kempen SMGR  0.23 0.55 0.22 5.33   2.19     86.13    15
## 164 Leemstreek   SM  0.20 0.29 0.50 6.15   1.80     97.29    16
## 165 Leemstreek SMGR  0.28 0.21 0.51 5.77   2.09    105.08    18
## 167     Kempen   GM  0.21 0.63 0.16 5.36   2.06     82.61    11
## 168 Leemstreek   PG  0.33 0.27 0.40 6.18   2.51    117.79    15
## 169 Leemstreek   GM  0.26 0.24 0.50 6.09   1.83     97.74    14
## 170 Leemstreek   PG  0.28 0.25 0.47 6.10   2.45    120.01    11
## 171     Kempen SMGR  0.13 0.65 0.22 5.00   2.14     89.92    25
## 172 Leemstreek   PG  0.42 0.21 0.37 5.74   2.44    112.85    15
## 173     Kempen SMGR  0.26 0.56 0.18 5.36   2.35     87.62    24
## 174     Kempen   GM  0.20 0.62 0.19 4.82   2.08     76.40    22
## 175     Kempen   PG  0.17 0.58 0.25 4.91   2.58     99.98    16
## 176     Kempen SMGR  0.14 0.67 0.19 5.23   2.31     88.27    22
## 177 Leemstreek   GM  0.28 0.27 0.45 5.75   1.79     95.64    14
## 178 Leemstreek   SM  0.24 0.25 0.50 5.89   1.80     99.36    22
## 179     Kempen SMGR  0.27 0.52 0.21 4.86   2.19     94.22    11
## 180 Leemstreek   GM  0.24 0.25 0.51 6.12   1.74     98.01    13
## 181     Kempen   PG  0.13 0.58 0.29 5.19   2.77    109.99    25
## 182 Leemstreek   PG  0.32 0.25 0.43 6.08   2.41    118.32    11
## 183     Kempen   PG  0.25 0.54 0.21 5.19   2.77    104.97    18
## 184 Leemstreek   GM  0.27 0.26 0.47 6.11   1.67    100.45    20
## 186 Leemstreek   SM  0.30 0.24 0.45 5.61   1.89     91.84    17
## 187 Leemstreek   SM  0.24 0.26 0.50 5.97   1.84     95.08    23
## 188 Leemstreek SMGR  0.23 0.23 0.55 6.02   1.98    108.14    17
## 189     Kempen   PG  0.18 0.63 0.19 5.24   2.66    103.83    12
## 190     Kempen SMGR  0.20 0.53 0.27 4.86   2.20     91.34    15
## 191 Leemstreek   GM  0.29 0.17 0.54 5.90   1.80     96.34    21
## 192 Leemstreek   SM  0.30 0.26 0.44 5.79   1.71     96.37    11
```

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## 193     Kempen   PG  0.12 0.64 0.23 5.31   2.87    104.83    24
## 194     Kempen   SM  0.20 0.59 0.21 5.36   2.01     85.66    20
## 195     Kempen   SM  0.15 0.63 0.23 5.26   2.11     84.95    16
## 196 Leemstreek   PG  0.29 0.22 0.49 5.76   2.46    124.95    12
## 197 Leemstreek   GM  0.19 0.25 0.56 6.09   1.76     95.56    24
## 198     Kempen   GM  0.32 0.50 0.18 5.33   2.03     78.66    13
## 199 Leemstreek   PG  0.25 0.29 0.47 6.03   2.49    120.94    21
## 200 Leemstreek   SM  0.21 0.31 0.49 5.89   1.72    100.96    10
## 201 Leemstreek   SM  0.28 0.28 0.44 5.78   1.69    106.93    14
## 202     Kempen SMGR  0.21 0.60 0.19 4.97   2.14     88.33    25
## 203     Kempen   SM  0.18 0.60 0.22 5.24   2.09     82.91    21
## 204 Leemstreek   GM  0.30 0.20 0.49 5.91   1.85     99.64    20
## 205 Leemstreek   SM  0.38 0.27 0.35 6.05   1.93    103.18    14
## 206     Kempen   PG  0.16 0.59 0.25 5.36   2.77    103.67    11
## 207 Leemstreek   SM  0.25 0.23 0.51 6.12   1.83    100.85    14
## 208 Leemstreek   PG  0.15 0.21 0.64 6.27   2.43    118.21    18
## 209 Leemstreek   GM  0.22 0.24 0.54 5.66   1.75     92.58    22
## 210 Leemstreek   GM  0.30 0.22 0.47 5.94   1.64     98.66    14
## 211 Leemstreek SMGR  0.09 0.24 0.67 6.01   1.92    104.71    15
## 212 Leemstreek   SM  0.12 0.21 0.67 6.12   1.76     99.80    24
## 213 Leemstreek   SM  0.27 0.30 0.43 5.77   1.82     96.05    17
## 214     Kempen   PG  0.17 0.60 0.23 5.26   2.72    112.98    25
## 215 Leemstreek SMGR  0.23 0.28 0.49 6.06   1.85    106.84    23
## 216     Kempen SMGR  0.30 0.47 0.23 5.15   2.18     88.71    25
## 217 Leemstreek   PG  0.25 0.24 0.51 6.10   2.42    122.99    12
## 218 Leemstreek SMGR  0.35 0.23 0.42 5.98   1.99     99.68    15
## 219     Kempen   PG  0.23 0.54 0.23 5.29   2.76    105.64    24
## 220     Kempen   SM  0.05 0.68 0.27 5.23   2.13     89.48    20
## 221 Leemstreek   PG  0.30 0.21 0.49 5.97   2.35    119.58    25
## 222    Kempeen   PG  0.17 0.62 0.22 5.31   2.87    109.63    25
## 223     Kempen SMGR  0.10 0.64 0.26 5.16   2.21     89.29    21
## 224     Kempen   GM  0.14 0.61 0.25 5.23   1.98     73.42    16
## 225 Leemstreek   SM  0.38 0.22 0.40 5.88   1.82     94.93    20
## 226 Leemstreek   GM  0.21 0.28 0.51 6.32   1.82     97.21    14
## 227 Leemstreek   SM  0.15 0.26 0.60 6.00   1.69    103.80    15
## 228 Leemstreek   PG  0.20 0.22 0.58 6.17   2.58    118.08    23
## 229     Kempen   GM  0.24 0.53 0.23 5.08   2.03     83.17    20
## 230     Kempen   GM  0.05 0.70 0.24 5.04   2.13     78.44    22
## 231     Kempen SMGR  0.25 0.57 0.18 5.26   2.20     86.29    15
## 232 Leemstreek   PG  0.29 0.19 0.52 5.68   2.42    117.29    20
## 233 Leemstreek   GM  0.29 0.27 0.44 6.10   1.60    101.52    19
## 234 Leemstreek SMGR  0.27 0.26 0.47 6.08   1.81    107.89    13
## 235 Leemstreek   SM  0.23 0.21 0.56 5.46   1.79    100.45    14
## 236 Leemstreek   SM  0.38 0.19 0.43 6.07   1.87    102.56    23
## 237 Leemstreek   PG  0.24 0.25 0.52 5.57   2.30    118.60    22
## 238     Kempen SMGR  0.11 0.66 0.23 5.23   2.31     93.04    10
## 239 Leemstreek   PG  0.23 0.27 0.50 5.91   2.56    121.08    21
## 240     Kempen   GM  0.15 0.58 0.27 4.96   1.95     78.11    23
## 241 Leemstreek SMGR  0.34 0.22 0.44 6.00   1.94    101.14    23
## 243 Leemstreek   PG  0.25 0.25 0.49 5.87   2.44    121.06    25
## 244     Kempen   GM  0.16 0.63 0.21 5.21   2.12     80.92    24
## 245 Leemstreek   SM  0.23 0.24 0.53 6.07   1.87     94.53    14
## 246     Kempen SMGR  0.17 0.64 0.19 4.86   2.13     82.13    23
## 247 Leemstreek SMGR  0.35 0.23 0.42 5.65   1.87    105.85    22
```

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## 248     Kempen   SM  0.21 0.58 0.21 4.70   2.02     82.92    17
## 249     Kempen   GM  0.09 0.64 0.27 5.21   2.07     84.23    11
## 252 Leemstreek   PG  0.26 0.25 0.49 6.11   2.57    119.90    13
## 253 Leemstreek   GM  0.20 0.27 0.53 6.17   1.93     90.97    22
## 254 Leemstreek SMGR  0.35 0.21 0.44 6.27   1.86     96.82    25
## 255 Leemstreek SMGR  0.17 0.26 0.57 5.90   1.96    107.26    13
## 256     Kempen   PG  0.20 0.56 0.24 5.20   2.75    109.57    11
## 257 Leemstreek   PG  0.27 0.28 0.45 6.17   2.57    116.42    20
## 259 Leemstreek   GM  0.18 0.27 0.54 6.15   1.77     92.02    13
## 260     Kempen   GM  0.16 0.61 0.24 5.18   1.90     83.34    18
## 261     Kempen   PG  0.09 0.64 0.26 4.93   2.83    105.97    15
## 262     Kempen   PG  0.13 0.67 0.20 5.25   2.72    102.85    22
## 263     Kempen   SM  0.05 0.70 0.25 5.20   1.98     81.20    17
## 264     Kempen SMGR  0.15 0.60 0.25 5.23   2.18     86.11    13
## 265 Leemstreek   SM  0.33 0.18 0.49 6.08   1.79     99.98    14
## 266 Leemstreek   SM  0.31 0.25 0.44 5.96   1.88    100.46    22
## 267 Leemstreek   PG  0.31 0.26 0.43 5.98 234.00    121.73    10
## 268 Leemstreek   SM  0.23 0.25 0.53 6.14   1.75     95.13    16
## 269 Leemstreek SMGR  0.24 0.27 0.50 6.03   2.02     97.52    19
## 271     Kempen   SM  0.19 0.57 0.24 5.04   2.19     83.44    20
## 272 Leemstreek   GM  0.28 0.24 0.48 5.74   1.88     96.10    18
## 273 Leemstreek   SM  0.30 0.26 0.44 5.81   1.81    102.19    18
## 274 Leemstreek   SM  0.30 0.23 0.47 5.97   1.75     94.90    23
## 275     Kempen   GM  0.17 0.60 0.23 4.74   2.09     86.39    15
## 276     Kempen SMGR  0.29 0.50 0.21 4.76   2.28     88.48    11
## 277     Kempen   PG  0.22 0.54 0.24 5.32   2.94    103.46    22
## 278 Leemstreek   SM  0.31 0.21 0.48 5.88   1.64     97.50    22
## 279 Leemstreek   GM  0.28 0.22 0.49 5.63   1.77     97.71    23
## 280     Kempen   GM  0.24 0.57 0.20 5.08   2.10     81.33    11
## 281     Kempen SMGR  0.10 0.62 0.28 5.36   2.19     86.76    10
## 282 Leemstreek SMGR  0.20 0.28 0.52 6.13   1.77    103.26    19
## 283 Leemstreek   PG  0.26 0.23 0.51 6.09   2.51    110.66    23
## 284 Leemstreek   PG  0.38 0.22 0.40 6.24   2.39    118.42    17
## 285 Leemstreek   PG  0.24 0.22 0.54 5.85   2.49    122.01    11
## 286 Leemstreek SMGR  0.32 0.24 0.44 6.15   1.97    103.69    16
## 287 Leemstreek   GM  0.31 0.24 0.45 5.88   1.85     98.90    15
## 288 Leemstreek   SM  0.18 0.26 0.56 6.00   1.67    100.15    17
## 289 Leemstreek   GM  0.29 0.24 0.47 6.44   1.84     98.84    12
## 290     Kempen   SM  0.34 0.50 0.17 5.29   2.04     86.66    17
## 291 Leemstreek   SM  0.15 0.25 0.61 6.15   1.70     91.73    14
## 292 Leemstreek SMGR  0.36 0.29 0.36 5.78   2.00    104.28    17
## 293 Leemstreek   GM  0.18 0.29 0.53 5.84   1.84    102.47    24
## 294     Kempen   SM  0.19 0.56 0.25 5.56   2.08     82.39    18
## 295     Kempen   SM  0.28 0.52 0.20 4.88   2.15     85.80    21
## 296     Kempen SMGR  0.04 0.72 0.23 5.14   2.13     90.68    11
## 297     Kempen   SM  0.17 0.59 0.23 5.28   2.08     86.28    20
## 298 Leemstreek   SM  0.19 0.25 0.56 6.20   1.79     97.31    22
## 299 Leemstreek   GM  0.21 0.26 0.53 6.22   1.73     93.88    10
## 300     Kempen   GM  0.27 0.52 0.21 5.07   2.10     82.39    11
## 301     Kempen   PG  0.19 0.56 0.25 4.94   2.87    106.88    19
## 302 Leemstreek   SM  0.25 0.20 0.55 5.94   1.76     97.59    22
## 303     Kempen   SM  0.24 0.57 0.19 4.92   2.09     84.20    21
## 304     Kempen   PG  0.22 0.61 0.18 5.23   2.88    107.71    14
## 305     Kempen   GM  0.24 0.53 0.22 5.32   2.02     79.22    22
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## 306 Leemstreek SMGR  0.25 0.27 0.48 6.43   1.69    108.71    16
## 307 Leemstreek   GM  0.30 0.26 0.44 6.07   1.79    104.68    12
## 308 Leemstreek   GM  0.27 0.22 0.51 5.83   1.68    101.88    19
## 309 Leemstreek   SM  0.32 0.26 0.42 6.20   1.72     96.61    22
## 310 Leemstreek   GM  0.22 0.27 0.51 6.18   1.86     89.41    22
## 311 Leemstreek SMGR  0.26 0.28 0.46 5.95   2.09    104.50    11
## 312 Leemstreek   GM  0.20 0.24 0.56 5.69   1.66     99.59    11
## 313     Kempen SMGR  0.11 0.60 0.29 5.25   2.28     93.11    24
## 314 Leemstreek SMGR  0.31 0.25 0.45 6.23   1.89     99.67    11
## 315 Leemstreek   GM  0.14 0.29 0.56 5.94   1.76     92.06    13
## 316     Kempen   GM  0.10 0.66 0.24 5.21   2.14     82.65    17
## 317     Kempen   SM  0.14 0.64 0.22 5.04   2.11     86.49    22
## 318 Leemstreek   SM  0.29 0.24 0.47 6.14   1.73    101.65    23
## 319 Leemstreek   PG  0.14 0.26 0.59 5.98   2.41    123.40    19
## 320     Kempen   SM  0.22 0.62 0.16 5.03   2.15     80.43    16
## 321     Kempen   GM  0.25 0.55 0.20 5.12   1.86     77.14    12
## 322 Leemstreek SMGR  0.39 0.20 0.41 5.94   1.99     96.64    23
## 323 Leemstreek SMGR  0.15 0.27 0.58 5.88   1.77    106.20    23
## 324     Kempen   PG  0.26 0.52 0.21 5.04   2.75    109.26    15
## 325 Leemstreek SMGR  0.28 0.25 0.47 6.00   1.88    105.58    18
## 326 Leemstreek SMGR  0.05 0.26 0.69 6.04   2.00    100.29    17
## 327     Kempen   PG  0.15 0.63 0.22 5.41   2.76    112.14    22
## 328 Leemstreek SMGR  0.27 0.26 0.47 5.73   1.91    102.84    24
## 329 Leemstreek SMGR  0.25 0.23 0.52 5.96   1.95    111.06    19
## 330     Kempen   SM  0.18 0.65 0.17 5.23   2.03     86.24    22
## 331     Kempen   GM  0.08 0.69 0.23 5.45   2.03     86.72    22
## 332 Leemstreek   PG  0.23 0.28 0.49 6.09   2.54    121.58    14
## 333     Kempen SMGR  0.26 0.50 0.23 5.10   2.25     89.88    18
## 334     Kempen   PG  0.14 0.67 0.19 5.04   2.64    102.26    14
## 335  Leemstrek   SM  0.31 0.25 0.44 6.08   1.85     96.20    23
## 336 Leemstreek   GM  0.28 0.27 0.46 6.20   1.64     96.16    21
## 337     Kempen   PG  0.17 0.63 0.20 5.06   2.70    107.19    15
## 338     Kempen   PG  0.15 0.60 0.26 5.13   2.84    106.00    25
## 339     Kempen   GM  0.14 0.60 0.27 5.14   1.87     81.46    21
## 340     Kempen   SM  0.13 0.65 0.22 5.25   1.95     84.85    14
## 341 Leemstreek   PG  0.25 0.27 0.48 6.22   2.55    118.35    20
## 342 Leemstreek   SM  0.24 0.28 0.48 6.06   1.73     96.28    15
## 343 Leemstreek   SM  0.17 0.32 0.50 5.98   1.90     92.77    22
## 344 Leemstreek SMGR  0.22 0.27 0.51 6.28   1.89    107.57    15
## 345 Leemstreek   GM  0.23 0.26 0.52 6.03   1.78     96.49    14
## 347     Kempen SMGR  0.07 0.73 0.19 5.09   2.14     87.30    19
## 348 Leemstreek   SM  0.19 0.24 0.57 6.12   1.78     97.38    10
## 349 Leemstreek SMGR  0.21 0.32 0.47 5.69   1.86    100.11    14
## 350     Kempen   PG  0.15 0.62 0.23 5.08   2.97    102.13    14
## 351 Leemstreek   SM  0.21 0.30 0.50 6.00   1.81     91.01    22
## 352     Kempen   SM  0.21 0.59 0.19 4.95   2.13     87.82    12
## 353 Leemstreek   SM  0.20 0.27 0.54 5.79   1.77     98.74    14
## 355     Kempen SMGR  0.15 0.59 0.26 5.28   2.31     94.58    24
## 356     Kempen   GM  0.18 0.59 0.22 4.84   2.02     84.44    15
## 357     Kempen   GM  0.14 0.65 0.21 5.18   2.05     84.96    22
## 359 Leemstreek   SM  0.38 0.19 0.43 5.79   1.76     96.65    17
## 360     Kempen   SM  0.21 0.59 0.19 5.21   2.25     83.77    18
## 361 Leemstreek   GM  0.20 0.27 0.53 5.86   1.63     97.81    23
## 362 Leemstreek   SM  0.25 0.27 0.48 6.20   1.96    101.38    15
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## 363     Kempen SMGR  0.15 0.63 0.22 5.11   2.17     96.16    17
## 364 Leemstreek   GM  0.31 0.20 0.49 6.11   1.86     94.39    17
## 365 Leemstreek SMGR  0.28 0.26 0.46 5.90   1.74    100.42    21
## 366     Kempen   SM  0.19 0.65 0.16 4.87   2.17     84.46    11
## 367 Leemstreek   SM  0.19 0.29 0.52 6.18   1.87    109.82    21
## 368 Leemstreek   GM  0.21 0.24 0.55 6.05   1.65     94.68    20
## 369 Leemstreek SMGR  0.34 0.24 0.42 5.95   1.84 108650.00    20
## 370 Leemstreek SMGR  0.12 0.29 0.59 6.01   1.91    103.38    17
## 371 Leemstreek   PG  0.28 0.20 0.52 6.27   2.54    118.53    14
## 372 Leemstreek   PG  0.18 0.24 0.58 5.86   2.39    117.58    16
## 373     Kempen SMGR  0.19 0.62 0.19 4.95   2.18     89.29    12
## 374 Leemstreek   PG  0.29 0.20 0.51 5.88   2.57    126.52    14
## 375 Leemstreek   SM  0.25 0.26 0.49 5.68   1.71     99.39    11
## 376     Kempen   SM  0.06 0.69 0.24 5.26   2.17     86.01    17
## 377     Kempen   GM  0.20 0.60 0.20 5.14   2.08     85.32    15
## 378 Leemstreek   GM  0.26 0.24 0.50 6.04   1.66    104.18    13
## 379 Leemstreek SMGR  0.34 0.26 0.39 5.99   1.85    107.25    21
## 380     Kempen   PG  0.22 0.59 0.20 5.00   2.81    100.68    14
## 381 Leemstreek   SM  0.22 0.27 0.51 5.84   1.82     94.34    16
## 382     Kempen   PG  0.16 0.65 0.20 5.27   2.86    100.12    21
## 383 Leemstreek   SM  0.25 0.25 0.50 5.71   1.75     87.42    12
## 384     Kempen SMGR  0.25 0.54 0.21 5.19   2.06     95.05    12
## 386     Kempen   GM  0.17 0.57 0.26 5.16   2.00     83.64    25
## 387 Leemstreek   SM  0.28 0.27 0.45 5.85   1.81    102.76    21
## 388 Leemstreek   SM  0.33 0.19 0.48 5.90   1.94     96.34    10
## 389 Leemstreek   SM  0.22 0.19 0.59 6.35   1.78     97.86    25
## 390 Leemstreek   PG  0.40 0.23 0.37 6.13   2.57    121.66    21
## 391     Kempen   PG  0.17 0.62 0.21 4.67   2.66    106.89    23
## 392 Leemstreek   GM  0.15 0.22 0.62 6.03   1.82     93.36    13
## 393     Kempen   SM  0.07 0.64 0.29 5.01   2.11     86.88    14
## 394     Kempen   GM  0.16 0.62 0.22 4.87   1.99     84.15    19
## 395 Leemstreek   PG  0.31 0.24 0.45 6.04   2.35    119.21    15
## 396 Leemstreek   GM  0.22 0.27 0.51 6.03   1.76     91.66    21
## 397 Leemstreek   SM  0.28 0.26 0.46 5.93   1.82    100.70    11
## 398 Leemstreek   SM  0.18 0.24 0.59 6.39   1.82    103.80    23
## 399 Leemstreek   GM  0.29 0.26 0.45 5.55   1.79    101.46    22
## 400     Kempen   SM  0.14 0.63 0.23 5.27   2.06     87.47    23
```

```
summary(data_clean)
```

```
##     Region              Crop               Lutum             Sand
##  Length:382         Length:382         Min.   :0.0400   Min.   :0.1700
##  Class :character   Class :character   1st Qu.:0.1700   1st Qu.:0.2400
##  Mode  :character   Mode  :character   Median :0.2200   Median :0.2800
##                                        Mean   :0.2196   Mean   :0.3902
##                                        3rd Qu.:0.2700   3rd Qu.:0.5900
##                                        Max.   :0.4200   Max.   :0.7300
##       Silt              pH               C               Cstock
##  Min.   :0.1400   Min.   :4.640   Min.   :  1.550   Min.   :    73.42
##  1st Qu.:0.2300   1st Qu.:5.210   1st Qu.:  1.820   1st Qu.:    88.90
##  Median :0.4400   Median :5.805   Median :  2.010   Median :    97.34
##  Mean   :0.3901   Mean   :5.643   Mean   :  3.874   Mean   :   632.55
##  3rd Qu.:0.5100   3rd Qu.:6.040   3rd Qu.:  2.250   3rd Qu.:   104.64
##  Max.   :0.6900   Max.   :6.680   Max.   :252.000   Max.   :108650.00
```

```
##      Years
##  Min.   :10.00
##  1st Qu.:14.00
##  Median :17.00
##  Mean   :17.32
##  3rd Qu.:21.00
##  Max.   :25.00
```

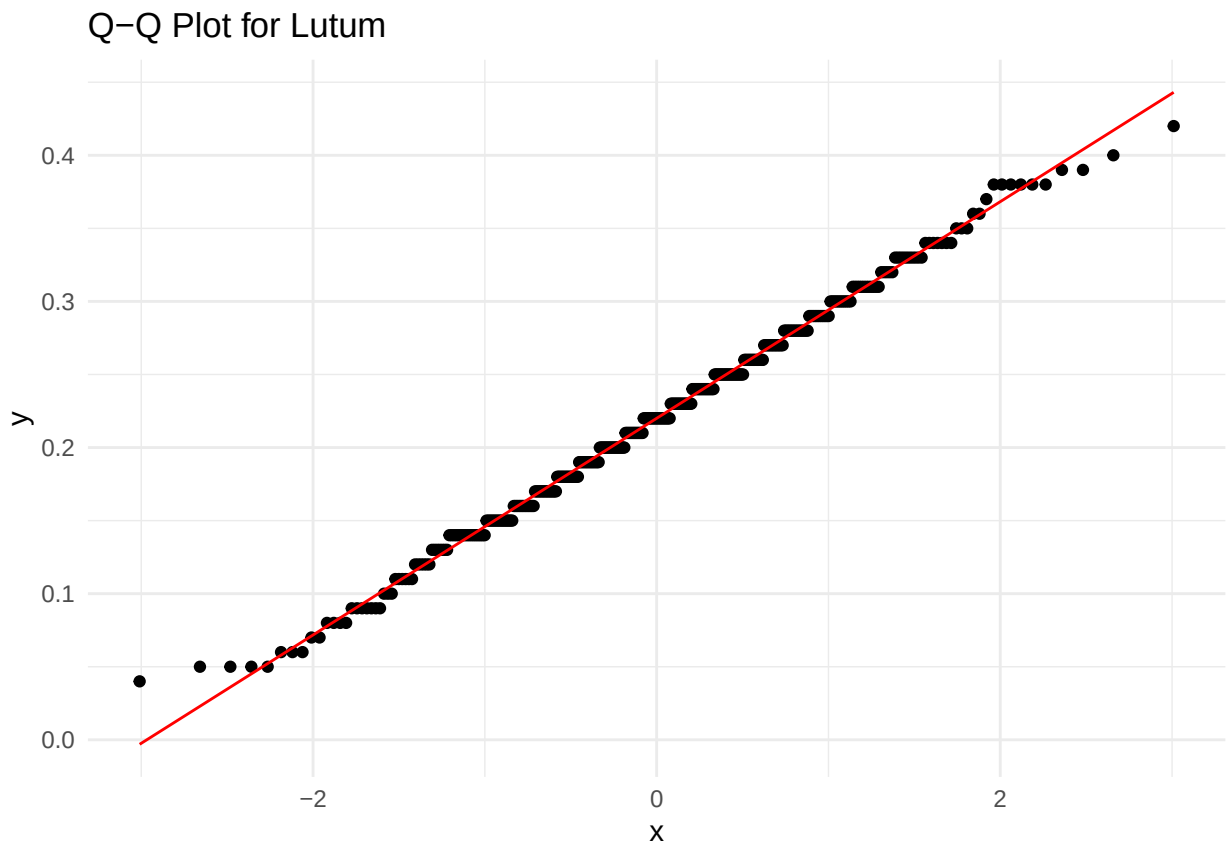
```r
na_positions <- is.na(data_clean)

# Print the row and column indices where NAs are present
which(na_positions, arr.ind = TRUE)
```

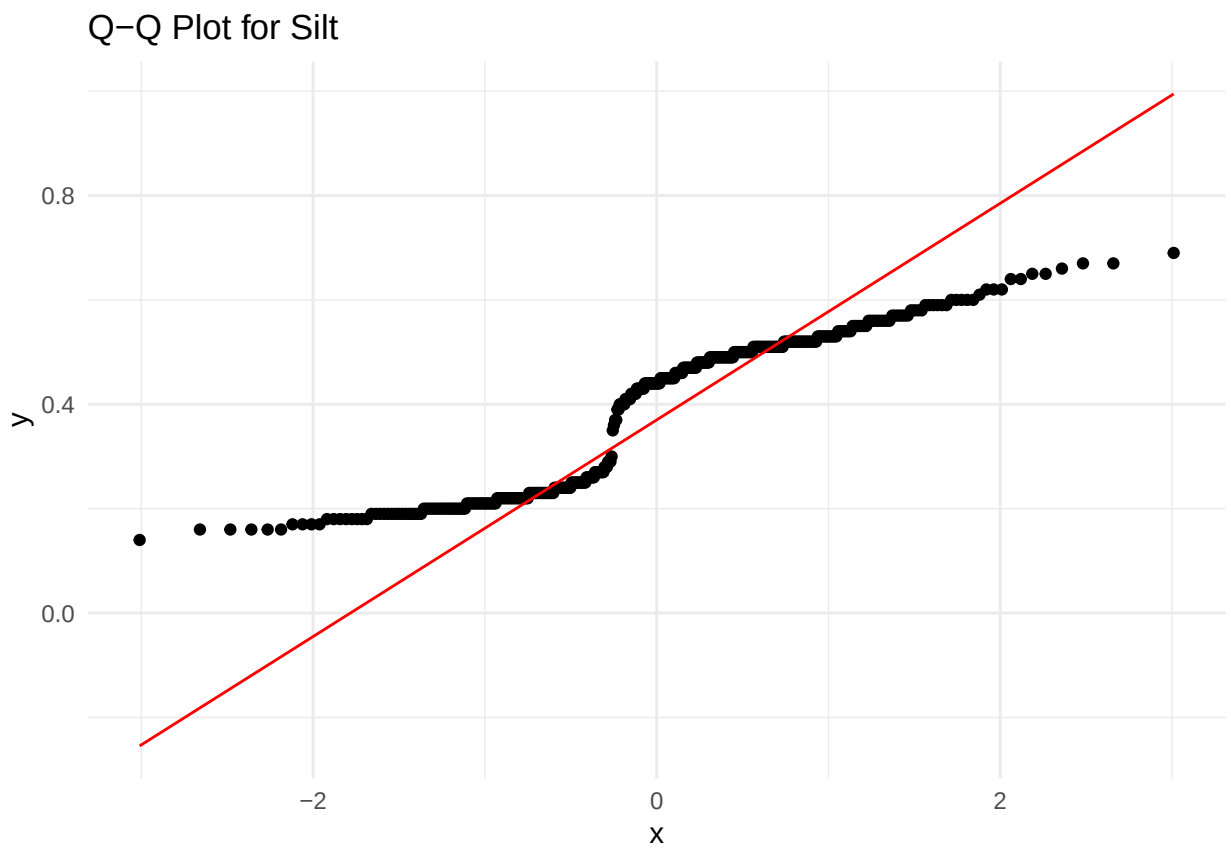
```
##      row col
```

```r
create_qq_plot <- function(var, var_name) {
  ggplot(data_clean, aes(sample = var)) +
    stat_qq() +
    stat_qq_line(color = "red") +
    ggtitle(paste("Q-Q Plot for", var_name)) +
    theme_minimal()
}

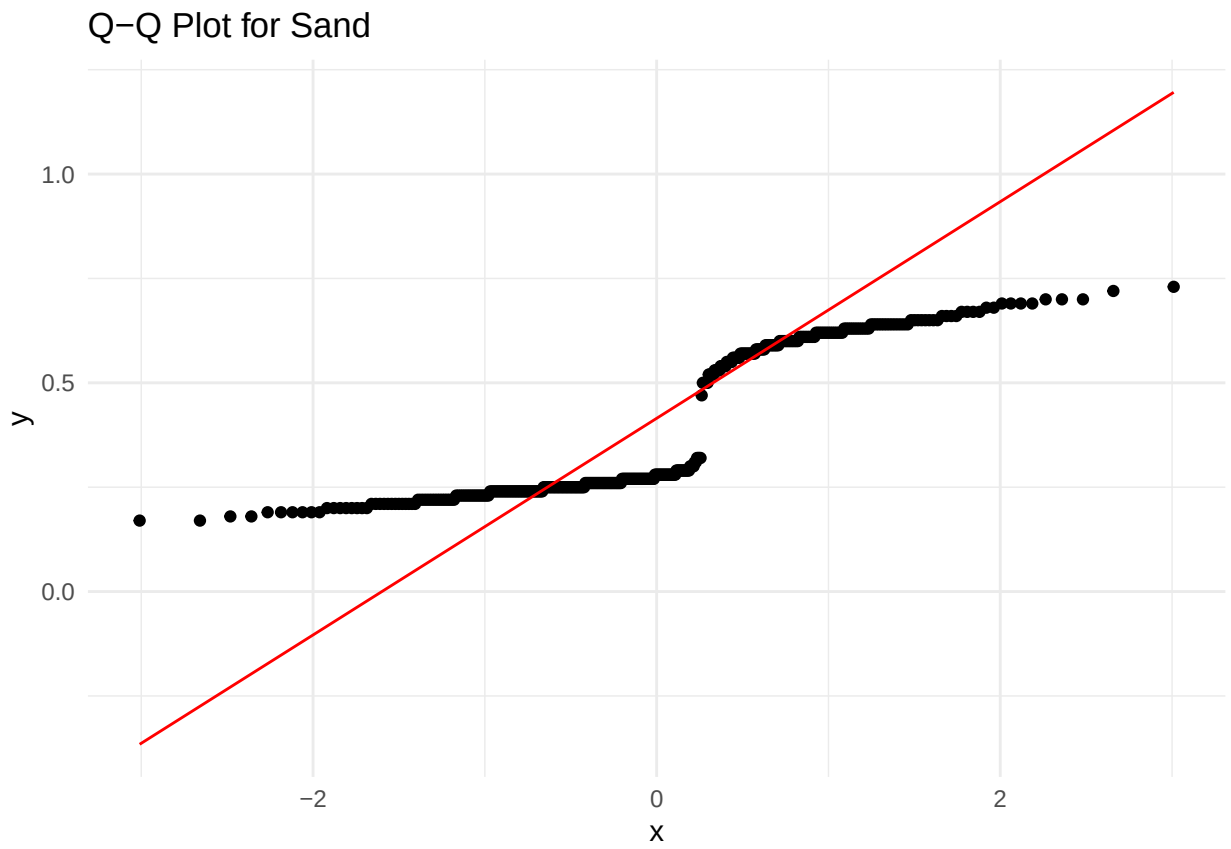
# Create Q-Q plots for each independent variable
create_qq_plot(data_clean$Lutum, "Lutum")
```



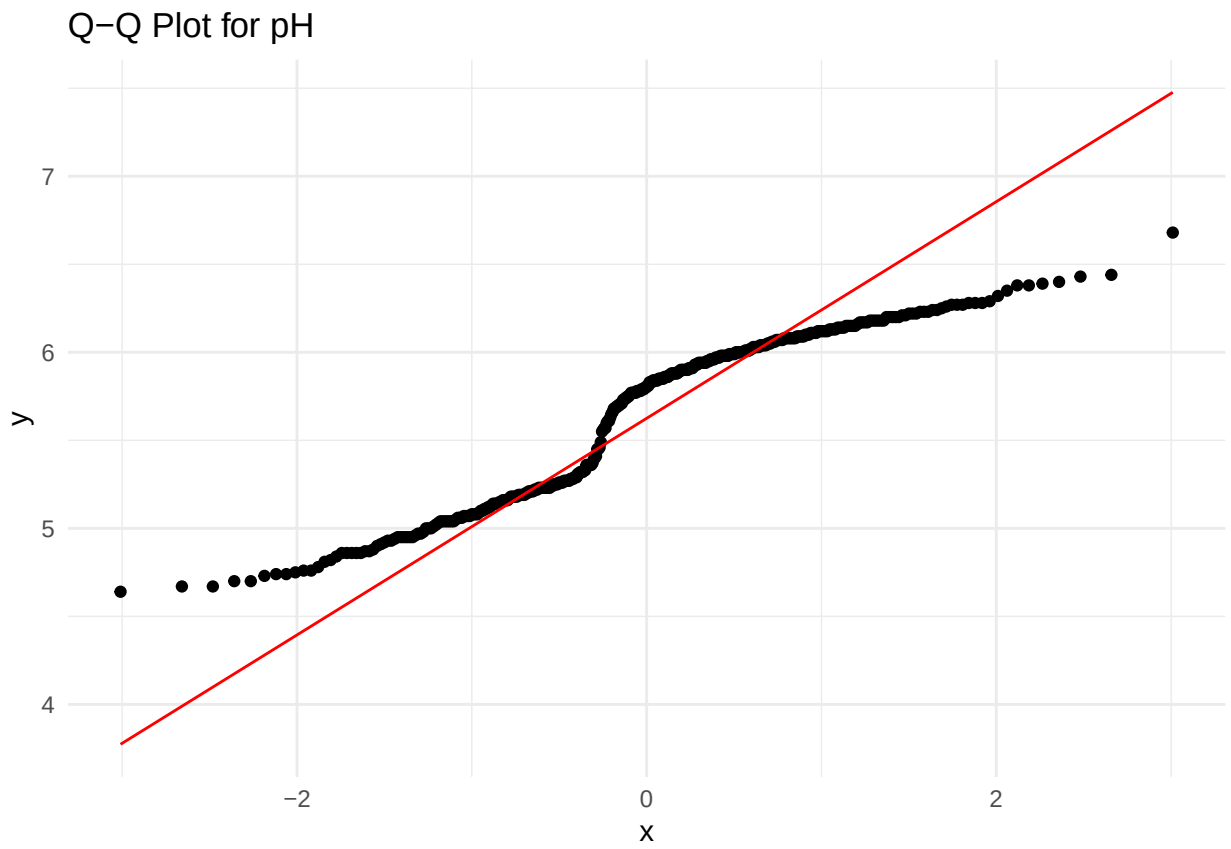
```
create_qq_plot(data_clean$Silt, "Silt")
```




```
create_qq_plot(data_clean$Sand, "Sand")
```

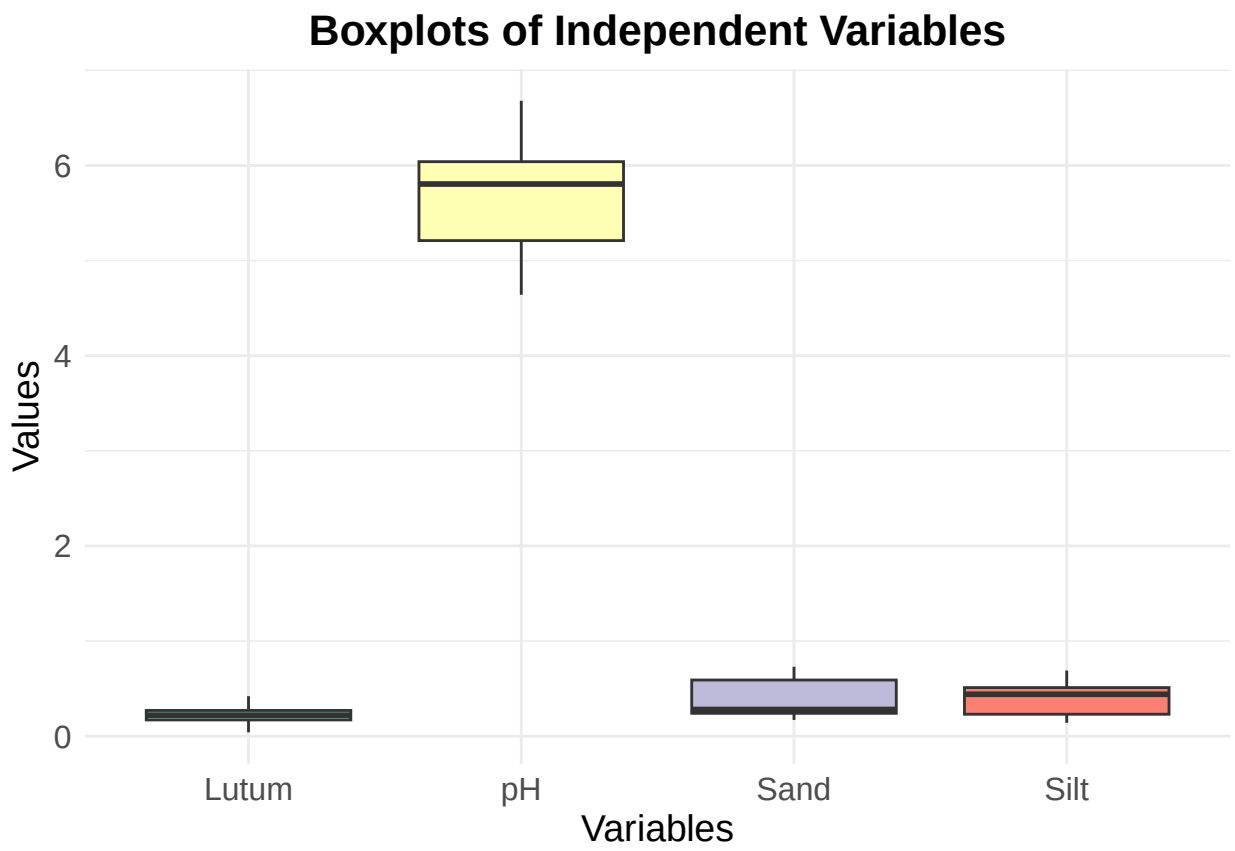



```
create_qq_plot(data_clean$pH, "pH")
```

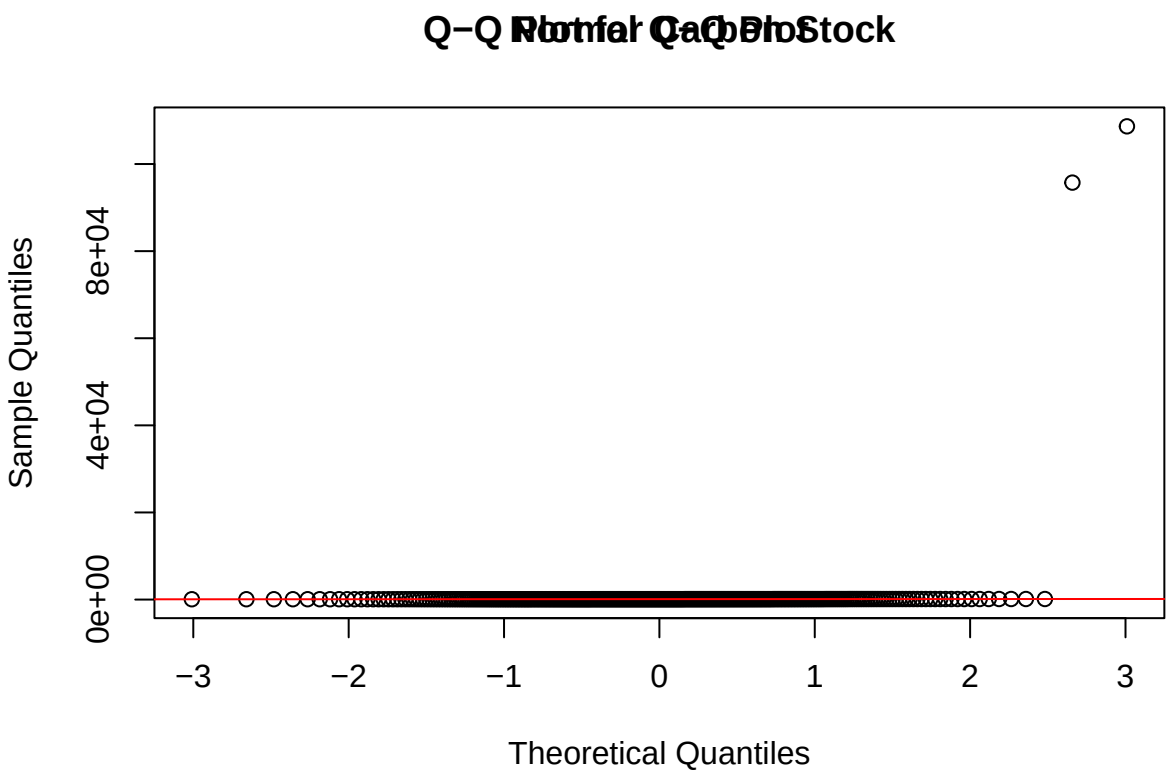



```r
# Reshape data to long format
data_long <- data_clean %>%
  pivot_longer(cols = c(Lutum, Silt, Sand, pH), names_to = "Variable", values_to = "Value")

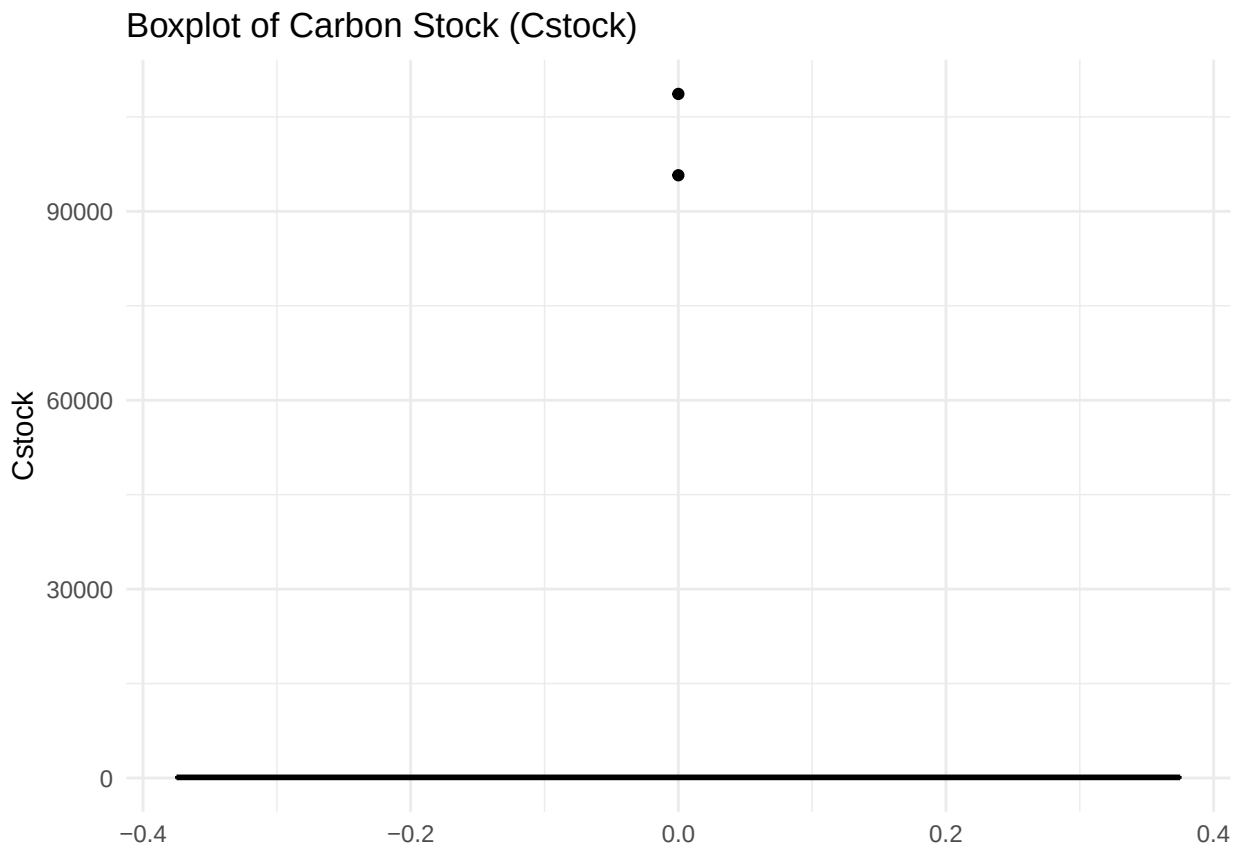
# Create a boxplot for all variables
ggplot(data_long, aes(x = Variable, y = Value, fill = Variable)) +
  geom_boxplot(outlier.shape = 21, outlier.fill = "white", outlier.color = "black") +
  ggtitle("Boxplots of Independent Variables") +
  ylab("Values") +
  xlab("Variables") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
    axis.title = element_text(size = 14),
    axis.text = element_text(size = 12),
    legend.position = "none"
  ) +
  scale_fill_brewer(palette = "Set3")
```




```r
# Q-Q Plot for Carbon Stock
qqnorm(data_clean$Cstock)
qqline(data_clean$Cstock, col = "red")
title(main = "Q-Q Plot for Carbon Stock")
```




```
ggplot(data_clean, aes(y = Cstock)) +
  geom_boxplot(fill = "lightgreen", color = "black") +
  labs(title = "Boxplot of Carbon Stock (Cstock)", y = "Cstock") +
  theme_minimal()
```




```
Q1 <- quantile(data_clean$Cstock, 0.25, na.rm = TRUE)
Q3 <- quantile(data_clean$Cstock, 0.75, na.rm = TRUE)
IQR_value <- Q3 - Q1

# Calculate lower and upper bounds for outliers
lower_bound <- Q1 - 1.5 * IQR_value
upper_bound <- Q3 + 1.5 * IQR_value

# Filter out outliers
data_clean <- subset(data_clean, Cstock >= lower_bound & Cstock <= upper_bound)

# Verify the data after outlier removal
print(head(data_clean))
```

```
##       Region Crop Lutum Sand Silt   pH    C Cstock Years
## 1 Leemstreek   PG  0.23 0.25 0.52 6.12 2.43 118.38    15
## 2     Kempen   SM  0.26 0.54 0.20 4.95 2.15  84.99    16
## 3 Leemstreek   PG  0.22 0.23 0.55 6.17 2.41 118.28    14
## 4     Kempen SMGR  0.15 0.60 0.25 4.95 2.15  94.72    12
## 5     Kempen   GM  0.14 0.63 0.23 5.23 1.91  82.81    14
## 6 Leemstreek   GM  0.29 0.20 0.51 6.22 1.73  94.01    14
```

```
print(data_clean)
```

```
##         Region Crop Lutum Sand Silt   pH      C Cstock Years
```

```
## 1   Leemstreek   PG  0.23 0.25 0.52 6.12   2.43 118.38    15
## 2       Kempen   SM  0.26 0.54 0.20 4.95   2.15  84.99    16
## 3   Leemstreek   PG  0.22 0.23 0.55 6.17   2.41 118.28    14
## 4       Kempen SMGR  0.15 0.60 0.25 4.95   2.15  94.72    12
## 5       Kempen   GM  0.14 0.63 0.23 5.23   1.91  82.81    14
## 6   Leemstreek   GM  0.29 0.20 0.51 6.22   1.73  94.01    14
## 7   Leemstreek SMGR  0.27 0.24 0.50 5.80   2.10  97.40    15
## 8       Kempen   SM  0.08 0.64 0.28 5.32   2.18  81.72    10
## 9   Leemstreek   GM  0.25 0.23 0.52 5.90   1.88 104.06    21
## 10  Leemstreek SMGR  0.25 0.26 0.49 6.06   1.97 102.18    17
## 11      Kempen   GM  0.23 0.62 0.14 5.14   2.14  78.95    19
## 12  Leemstreek   PG  0.34 0.26 0.41 5.91 252.00 122.77    12
## 13      Kempen   SM  0.15 0.63 0.22 5.15   2.13  85.81    10
## 14  Leemstreek   GM  0.24 0.25 0.51 5.75   1.70  99.11    13
## 15  Leemstreek   SM  0.25 0.24 0.52 5.97   1.76  97.24    10
## 17  Leemstreek   PG  0.26 0.25 0.49 5.57   2.45 115.76    17
## 18  Leemstreek   GM  0.16 0.26 0.57 5.70   1.82  96.06    22
## 19  Leemstreek   GM  0.29 0.27 0.44 5.77   1.83  93.58    18
## 20      Kempen   SM  0.24 0.57 0.18 4.78   2.10  86.64    12
## 21      Kempen   GM  0.13 0.69 0.18 5.18   2.00  82.63    10
## 22      Kempen SMGR  0.14 0.61 0.24 4.90   2.46  94.30    16
## 23      Kempen SMGR  0.18 0.61 0.22 4.67   2.25  97.85    13
## 24      Kempen   GM  0.09 0.66 0.24 4.97   2.16  82.63    22
## 25      Kempen   PG  0.22 0.56 0.21 5.02   2.72 104.07    18
## 26     Keempen SMGR  0.22 0.58 0.20 5.27   2.19  87.03    18
## 27  Leemstreek   GM  0.20 0.32 0.48 5.78   1.78  95.26    24
## 28  Leemstreek   GM  0.36 0.25 0.39 6.23   1.88  96.08    11
## 30  Leemstreek   PG  0.38 0.21 0.41 6.07   2.52 116.03    18
## 31      Kempen   PG  0.16 0.59 0.25 4.93   2.58 101.93    12
## 32      Kempen SMGR  0.18 0.62 0.20 5.49   2.16  88.65    14
## 33      Kempen   GM  0.14 0.62 0.24 5.07   1.94  84.35    12
## 34      Kempen   PG  0.18 0.55 0.27 4.74   2.83 109.17    12
## 35  Leemstreek   SM  0.26 0.25 0.49 5.96   1.79  94.84    19
## 36  Leemstreek SMGR  0.27 0.22 0.52 5.61   1.87 104.71    11
## 37      Kempen   SM  0.20 0.64 0.17 5.04   2.20  83.95    19
## 38  Leemstreek   GM  0.27 0.28 0.45 5.92   1.84 100.83    21
## 39  Leemstreek   SM  0.31 0.18 0.50 6.14   1.73 102.47    16
## 40  Leemstreek   SM  0.19 0.29 0.52 6.20   1.83  97.15    21
## 41  Leemstreek   SM  0.33 0.21 0.45 5.85   1.70  98.45    11
## 42  Leemstreek SMGR  0.08 0.26 0.66 6.16   1.86 106.81    19
## 43  Leemstreek SMGR  0.26 0.27 0.47 6.28   2.03  98.07    14
## 45  Leemstreek   PG  0.23 0.23 0.54 5.87   2.54 122.46    14
## 46  Leemstreek SMGR  0.25 0.25 0.50 6.13   1.85 102.34    19
## 47  Leemstreek SMGR  0.28 0.26 0.46 6.04   1.75 104.08    16
## 48  Leemstreek   SM  0.16 0.28 0.56 6.11   1.71  98.24    20
## 49  Leemstreek   GM  0.21 0.19 0.60 6.28   1.83  93.61    24
## 50      Kempen SMGR  0.26 0.52 0.22 5.18   2.17  86.36    18
## 51  Leemstreek   GM  0.17 0.29 0.53 6.21   1.58  95.58    14
## 52  Leemstreek   GM  0.22 0.28 0.49 6.12   1.74  95.01    21
## 53      Kempen   SM  0.13 0.62 0.25 5.19   2.21  87.31    18
## 54  Leemstreek   SM  0.30 0.27 0.43 6.08   1.85 104.13    12
## 55  Leemstreek   SM  0.08 0.28 0.64 6.26   1.73  98.31    14
## 56  Leemstreek   PG  0.37 0.17 0.46 5.80   2.32 121.78    12
## 57  Leemstreek   PG  0.30 0.28 0.41 5.78   2.38 119.99    12
```

```
## 58      Kempen SMGR  0.21 0.58 0.21 5.22   2.20  92.43    20
## 59      Kempen   SM  0.16 0.61 0.22 5.08   2.17  84.95    21
## 60  Leemstreek   SM  0.16 0.24 0.60 6.01   1.64  99.20    17
## 61      Kempen   PG  0.11 0.64 0.25 5.25   2.83 110.38    12
## 62  Leemstreek   GM  0.25 0.24 0.51 6.15   1.75  92.58    23
## 63  Leemstreek   GM  0.32 0.27 0.42 5.95   1.79 101.79    25
## 64  Leemstreek   SM  0.31 0.24 0.45 5.94   1.67  99.62    10
## 65      Kempen SMGR  0.24 0.55 0.22 5.11   2.14  90.81    18
## 66  Leemstreek   SM  0.06 0.29 0.65 6.21   1.76  96.21    22
## 67      Kempen   SM  0.20 0.64 0.16 5.19   2.18  83.52    18
## 68      Kempen SMGR  0.06 0.69 0.25 5.39   2.28  91.60    14
## 69      Kempen   SM  0.19 0.60 0.20 5.10   2.13  76.28    17
## 70  Leemstreek   PG  0.12 0.28 0.60 6.23   2.58 118.06    14
## 71      Kempen   SM  0.12 0.70 0.19 5.28   2.05  86.39    25
## 72      Kempen   PG  0.19 0.59 0.22 5.22   2.83 105.03    20
## 73      Kempen   GM  0.28 0.53 0.19 5.18   2.04  79.10    21
## 74  Leemstreek SMGR  0.20 0.24 0.55 5.94   1.85 101.88    10
## 75  Leemstreek SMGR  0.17 0.24 0.59 5.98   1.97 105.19    16
## 76  Leemstreek   GM  0.13 0.25 0.62 6.05   1.86  90.10    17
## 77  Leemstreek SMGR  0.15 0.30 0.55 6.01   1.97 107.62    12
## 78      Kempen   SM  0.14 0.58 0.28 5.29   2.06  75.92    14
## 79  Leemstreek   SM  0.16 0.26 0.57 5.98   1.76  96.39    12
## 80  Leemstreek SMGR  0.26 0.24 0.49 5.99   2.01 101.31    16
## 81  Leemstreek   SM  0.22 0.25 0.53 5.90   1.83  93.54    13
## 82      Kempen   PG  0.19 0.62 0.19 5.07   2.81 102.02    18
## 83  Leemstreek   GM  0.22 0.29 0.49 5.99   1.72  97.25    23
## 84      Kempen SMGR  0.14 0.61 0.26 4.64   2.29  89.57    18
## 85   Lemstreek   PG  0.20 0.27 0.53 5.73   2.38 119.11    16
## 86  Leemstreek   SM  0.20 0.27 0.53 5.99   1.85  89.48    14
## 87      Kempen   SM  0.17 0.65 0.19 5.05   2.03  79.08    13
## 88      Kempen SMGR  0.23 0.57 0.20 4.86   2.00  89.57    11
## 89      Kempen   GM  0.12 0.68 0.20 4.70   1.94  78.70    22
## 90  Leemstreek   PG  0.24 0.24 0.52 5.93   2.46 117.26    24
## 91  Leemstreek   PG  0.31 0.25 0.44 6.13   2.52 124.32    24
## 92      Kempen   SM  0.09 0.67 0.24 5.45   1.96  84.86    14
## 94      Kempen SMGR  0.18 0.55 0.27 5.40   2.34  88.40    15
## 95  Leemstreek   SM  0.25 0.22 0.53 6.02   1.76  97.22    15
## 97      Kempen   SM  0.16 0.61 0.23 5.16   2.05  80.47    20
## 98  Leemstreek   SM  0.27 0.25 0.48 5.83   1.77  98.07    12
## 99  Leemstreek SMGR  0.22 0.26 0.52 5.99   1.91 107.12    17
## 100 Leemstreek SMGR  0.22 0.27 0.51 6.18   1.90 105.20    11
## 101 Leemstreek   GM  0.33 0.23 0.44 5.85   1.73  95.51    20
## 102 Leemstreek   GM  0.33 0.22 0.45 5.94   1.71  97.76    24
## 103 Leemstreek   SM  0.20 0.28 0.52 5.77   1.55 101.27    24
## 104     Kempen   SM  0.09 0.64 0.27 5.17   2.24  83.60    24
## 105 Leemstreek   GM  0.23 0.20 0.57 6.08   1.89  98.05    17
## 106     Kempen   SM  0.20 0.60 0.20 5.23   2.01  79.80    16
## 107     Kempen   GM  0.20 0.56 0.24 4.86   2.03  81.12    16
## 108     Kempen   GM  0.28 0.50 0.22 5.12   2.16  80.53    24
## 109 Leemstreek   PG  0.23 0.25 0.52 5.81   2.53 121.01    20
## 110 Leemstreek   GM  0.24 0.25 0.52 6.00   1.85  97.66    16
## 111     Kempen   GM  0.11 0.60 0.30 5.37   1.96  81.71    17
## 112 Leemstreek   SM  0.31 0.26 0.43 5.90   2.00  95.37    21
## 113 Leemstreek   PG  0.22 0.28 0.51 6.14   2.36 116.80    21
```

```
## 114     Kempen   GM  0.22 0.61 0.17 4.95   2.10  83.15    17
## 115     Kempen SMGR  0.19 0.57 0.24 5.00   2.21  95.26    16
## 116 Leemstreek   SM  0.21 0.23 0.57 6.09   1.77  95.33    16
## 117 Leemstreek   GM  0.17 0.25 0.59 5.86   1.72  94.81    20
## 118     Kempen SMGR  0.22 0.62 0.16 4.76   2.11  91.31    20
## 119 Leemstreek   SM  0.26 0.22 0.52 6.25   1.93  96.24    11
## 120 Leemstreek   PG  0.16 0.24 0.60 6.07   2.46 120.78    17
## 121     Kempen   GM  0.14 0.65 0.21 4.98   2.07  81.92    18
## 122 Leemstreek   GM  0.27 0.25 0.49 5.60   1.76  93.27    12
## 123 Leemstreek   GM  0.27 0.23 0.50 6.38   1.72  99.53    15
## 124 Leemstreek   SM  0.24 0.24 0.52 6.68   1.71 101.65    13
## 125 Leemstreek   GM  0.18 0.28 0.54 6.04   1.77 100.20    12
## 126     Kempen SMGR  0.11 0.63 0.26 5.27   2.03  94.98    23
## 127 Leemstreek   PG  0.22 0.29 0.49 6.24   2.34 118.42    15
## 128 Leemstreek   SM  0.23 0.25 0.51 5.86   1.71  96.51    11
## 129 Leemstreek   PG  0.09 0.26 0.65 5.93   2.36 121.28    20
## 130     Kempen   GM  0.19 0.57 0.23 4.75   2.05  84.44    21
## 131     Kempen   PG  0.14 0.63 0.23 4.95   2.87 104.80    18
## 132     Kempen SMGR  0.25 0.57 0.18 5.22   2.07  88.54    18
## 133     Kempen   GM  0.25 0.53 0.22 5.04 208.00  79.76    21
## 134     Kempen   PG  0.18 0.61 0.21 5.06   2.82 111.75    23
## 135 Leemstreek   SM  0.19 0.31 0.50 5.98   1.72  88.77    22
## 136     Kempen SMGR  0.16 0.64 0.20 4.73   2.19  89.41    18
## 137     Kempen   GM  0.14 0.66 0.20 5.27   2.11  82.30    22
## 138     Kempen   PG  0.18 0.62 0.20 4.81   2.72 102.04    13
## 139     Kempen   GM  0.22 0.57 0.21 5.06   2.03  83.58    16
## 140 Leemstreek   SM  0.17 0.24 0.58 5.91   1.82  90.81    16
## 141 Leemstreek   PG  0.20 0.24 0.56 5.96   2.44 123.92    11
## 142 Leemstreek   SM  0.29 0.24 0.48 6.29   1.76  92.04    10
## 143 Leemstreek   PG  0.33 0.27 0.40 5.88   2.44 111.91    16
## 144 Leemstreek   SM  0.34 0.21 0.45 5.84   1.95  95.37    12
## 146 Leemstreek   PG  0.22 0.28 0.51 5.84   2.38 117.80    19
## 147 Leemstreek   SM  0.33 0.21 0.46 5.77   1.70  96.13    17
## 148 Leemstreek   PG  0.11 0.27 0.62 6.08   2.42 121.58    18
## 149 Leemstreek   PG  0.19 0.32 0.48 5.83   2.55 116.25    11
## 150     Kempen   GM  0.24 0.57 0.18 5.16   2.07  81.81    21
## 151     Kempen   GM  0.14 0.64 0.22 5.07   1.99  84.01    20
## 152 Leemstreek   SM  0.33 0.23 0.44 5.71   1.87 104.46    12
## 153 Leemstreek SMGR  0.21 0.28 0.51 5.84   1.89 106.09    19
## 154 Leemstreek   PG  0.23 0.26 0.51 6.18   2.36 126.85    20
## 155 Leemstreek   GM  0.18 0.30 0.52 6.18   1.79  98.06    15
## 156 Leemstreek SMGR  0.39 0.21 0.40 6.40   2.02 104.37    20
## 158 Leemstreek   GM  0.25 0.23 0.52 5.70   1.76  91.59    14
## 159 Leemstreek   GM  0.34 0.24 0.41 5.85   1.70  95.06    24
## 160 Leemstreek   SM  0.21 0.23 0.56 6.08   1.84 101.02    11
## 161 Leemstreek   SM  0.23 0.22 0.55 6.38   1.71  97.15    24
## 163     Kempen SMGR  0.23 0.55 0.22 5.33   2.19  86.13    15
## 164 Leemstreek   SM  0.20 0.29 0.50 6.15   1.80  97.29    16
## 165 Leemstreek SMGR  0.28 0.21 0.51 5.77   2.09 105.08    18
## 167     Kempen   GM  0.21 0.63 0.16 5.36   2.06  82.61    11
## 168 Leemstreek   PG  0.33 0.27 0.40 6.18   2.51 117.79    15
## 169 Leemstreek   GM  0.26 0.24 0.50 6.09   1.83  97.74    14
## 170 Leemstreek   PG  0.28 0.25 0.47 6.10   2.45 120.01    11
## 171     Kempen SMGR  0.13 0.65 0.22 5.00   2.14  89.92    25
```

```
## 172 Leemstreek   PG  0.42 0.21 0.37 5.74   2.44 112.85    15
## 173     Kempen SMGR  0.26 0.56 0.18 5.36   2.35  87.62    24
## 174     Kempen   GM  0.20 0.62 0.19 4.82   2.08  76.40    22
## 175     Kempen   PG  0.17 0.58 0.25 4.91   2.58  99.98    16
## 176     Kempen SMGR  0.14 0.67 0.19 5.23   2.31  88.27    22
## 177 Leemstreek   GM  0.28 0.27 0.45 5.75   1.79  95.64    14
## 178 Leemstreek   SM  0.24 0.25 0.50 5.89   1.80  99.36    22
## 179     Kempen SMGR  0.27 0.52 0.21 4.86   2.19  94.22    11
## 180 Leemstreek   GM  0.24 0.25 0.51 6.12   1.74  98.01    13
## 181     Kempen   PG  0.13 0.58 0.29 5.19   2.77 109.99    25
## 182 Leemstreek   PG  0.32 0.25 0.43 6.08   2.41 118.32    11
## 183     Kempen   PG  0.25 0.54 0.21 5.19   2.77 104.97    18
## 184 Leemstreek   GM  0.27 0.26 0.47 6.11   1.67 100.45    20
## 186 Leemstreek   SM  0.30 0.24 0.45 5.61   1.89  91.84    17
## 187 Leemstreek   SM  0.24 0.26 0.50 5.97   1.84  95.08    23
## 188 Leemstreek SMGR  0.23 0.23 0.55 6.02   1.98 108.14    17
## 189     Kempen   PG  0.18 0.63 0.19 5.24   2.66 103.83    12
## 190     Kempen SMGR  0.20 0.53 0.27 4.86   2.20  91.34    15
## 191 Leemstreek   GM  0.29 0.17 0.54 5.90   1.80  96.34    21
## 192 Leemstreek   SM  0.30 0.26 0.44 5.79   1.71  96.37    11
## 193     Kempen   PG  0.12 0.64 0.23 5.31   2.87 104.83    24
## 194     Kempen   SM  0.20 0.59 0.21 5.36   2.01  85.66    20
## 195     Kempen   SM  0.15 0.63 0.23 5.26   2.11  84.95    16
## 196 Leemstreek   PG  0.29 0.22 0.49 5.76   2.46 124.95    12
## 197 Leemstreek   GM  0.19 0.25 0.56 6.09   1.76  95.56    24
## 198     Kempen   GM  0.32 0.50 0.18 5.33   2.03  78.66    13
## 199 Leemstreek   PG  0.25 0.29 0.47 6.03   2.49 120.94    21
## 200 Leemstreek   SM  0.21 0.31 0.49 5.89   1.72 100.96    10
## 201 Leemstreek   SM  0.28 0.28 0.44 5.78   1.69 106.93    14
## 202     Kempen SMGR  0.21 0.60 0.19 4.97   2.14  88.33    25
## 203     Kempen   SM  0.18 0.60 0.22 5.24   2.09  82.91    21
## 204 Leemstreek   GM  0.30 0.20 0.49 5.91   1.85  99.64    20
## 205 Leemstreek   SM  0.38 0.27 0.35 6.05   1.93 103.18    14
## 206     Kempen   PG  0.16 0.59 0.25 5.36   2.77 103.67    11
## 207 Leemstreek   SM  0.25 0.23 0.51 6.12   1.83 100.85    14
## 208 Leemstreek   PG  0.15 0.21 0.64 6.27   2.43 118.21    18
## 209 Leemstreek   GM  0.22 0.24 0.54 5.66   1.75  92.58    22
## 210 Leemstreek   GM  0.30 0.22 0.47 5.94   1.64  98.66    14
## 211 Leemstreek SMGR  0.09 0.24 0.67 6.01   1.92 104.71    15
## 212 Leemstreek   SM  0.12 0.21 0.67 6.12   1.76  99.80    24
## 213 Leemstreek   SM  0.27 0.30 0.43 5.77   1.82  96.05    17
## 214     Kempen   PG  0.17 0.60 0.23 5.26   2.72 112.98    25
## 215 Leemstreek SMGR  0.23 0.28 0.49 6.06   1.85 106.84    23
## 216     Kempen SMGR  0.30 0.47 0.23 5.15   2.18  88.71    25
## 217 Leemstreek   PG  0.25 0.24 0.51 6.10   2.42 122.99    12
## 218 Leemstreek SMGR  0.35 0.23 0.42 5.98   1.99  99.68    15
## 219     Kempen   PG  0.23 0.54 0.23 5.29   2.76 105.64    24
## 220     Kempen   SM  0.05 0.68 0.27 5.23   2.13  89.48    20
## 221 Leemstreek   PG  0.30 0.21 0.49 5.97   2.35 119.58    25
## 222    Kempeen   PG  0.17 0.62 0.22 5.31   2.87 109.63    25
## 223     Kempen SMGR  0.10 0.64 0.26 5.16   2.21  89.29    21
## 224     Kempen   GM  0.14 0.61 0.25 5.23   1.98  73.42    16
## 225 Leemstreek   SM  0.38 0.22 0.40 5.88   1.82  94.93    20
## 226 Leemstreek   GM  0.21 0.28 0.51 6.32   1.82  97.21    14
```

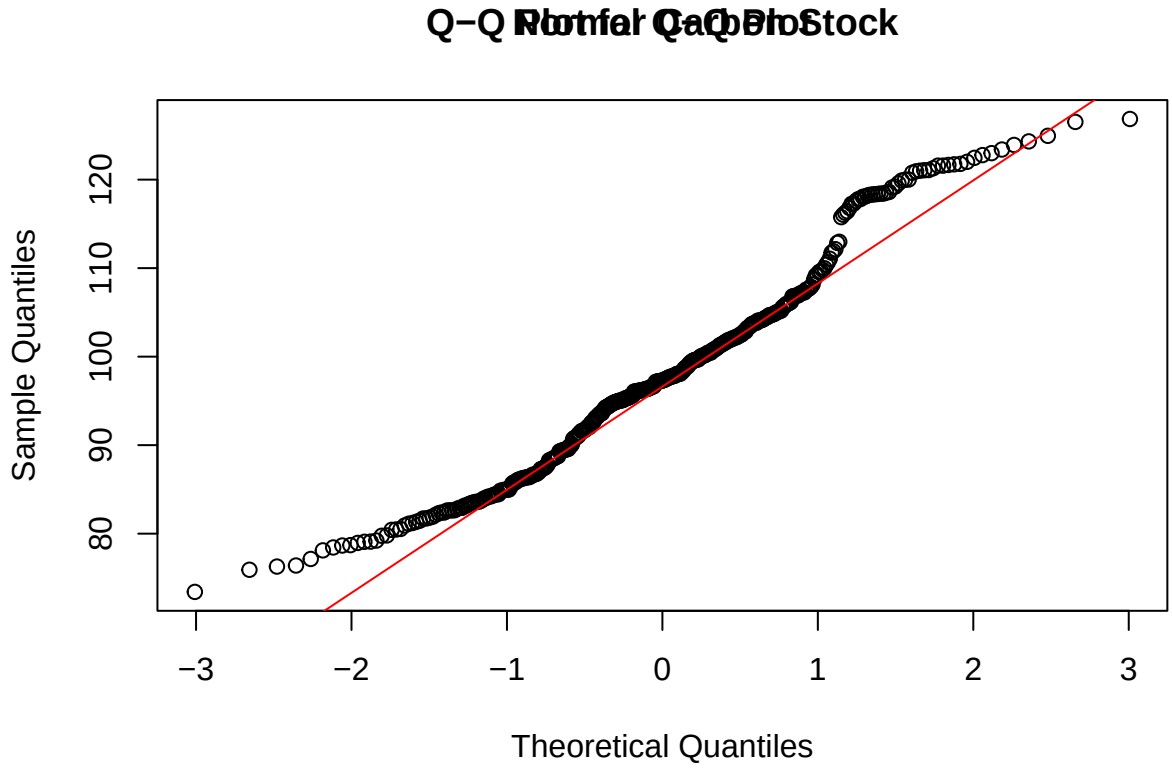
```
## 227 Leemstreek   SM  0.15 0.26 0.60 6.00   1.69 103.80    15
## 228 Leemstreek   PG  0.20 0.22 0.58 6.17   2.58 118.08    23
## 229     Kempen   GM  0.24 0.53 0.23 5.08   2.03  83.17    20
## 230     Kempen   GM  0.05 0.70 0.24 5.04   2.13  78.44    22
## 231     Kempen SMGR  0.25 0.57 0.18 5.26   2.20  86.29    15
## 232 Leemstreek   PG  0.29 0.19 0.52 5.68   2.42 117.29    20
## 233 Leemstreek   GM  0.29 0.27 0.44 6.10   1.60 101.52    19
## 234 Leemstreek SMGR  0.27 0.26 0.47 6.08   1.81 107.89    13
## 235 Leemstreek   SM  0.23 0.21 0.56 5.46   1.79 100.45    14
## 236 Leemstreek   SM  0.38 0.19 0.43 6.07   1.87 102.56    23
## 237 Leemstreek   PG  0.24 0.25 0.52 5.57   2.30 118.60    22
## 238     Kempen SMGR  0.11 0.66 0.23 5.23   2.31  93.04    10
## 239 Leemstreek   PG  0.23 0.27 0.50 5.91   2.56 121.08    21
## 240     Kempen   GM  0.15 0.58 0.27 4.96   1.95  78.11    23
## 241 Leemstreek SMGR  0.34 0.22 0.44 6.00   1.94 101.14    23
## 243 Leemstreek   PG  0.25 0.25 0.49 5.87   2.44 121.06    25
## 244     Kempen   GM  0.16 0.63 0.21 5.21   2.12  80.92    24
## 245 Leemstreek   SM  0.23 0.24 0.53 6.07   1.87  94.53    14
## 246     Kempen SMGR  0.17 0.64 0.19 4.86   2.13  82.13    23
## 247 Leemstreek SMGR  0.35 0.23 0.42 5.65   1.87 105.85    22
## 248     Kempen   SM  0.21 0.58 0.21 4.70   2.02  82.92    17
## 249     Kempen   GM  0.09 0.64 0.27 5.21   2.07  84.23    11
## 252 Leemstreek   PG  0.26 0.25 0.49 6.11   2.57 119.90    13
## 253 Leemstreek   GM  0.20 0.27 0.53 6.17   1.93  90.97    22
## 254 Leemstreek SMGR  0.35 0.21 0.44 6.27   1.86  96.82    25
## 255 Leemstreek SMGR  0.17 0.26 0.57 5.90   1.96 107.26    13
## 256     Kempen   PG  0.20 0.56 0.24 5.20   2.75 109.57    11
## 257 Leemstreek   PG  0.27 0.28 0.45 6.17   2.57 116.42    20
## 259 Leemstreek   GM  0.18 0.27 0.54 6.15   1.77  92.02    13
## 260     Kempen   GM  0.16 0.61 0.24 5.18   1.90  83.34    18
## 261     Kempen   PG  0.09 0.64 0.26 4.93   2.83 105.97    15
## 262     Kempen   PG  0.13 0.67 0.20 5.25   2.72 102.85    22
## 263     Kempen   SM  0.05 0.70 0.25 5.20   1.98  81.20    17
## 264     Kempen SMGR  0.15 0.60 0.25 5.23   2.18  86.11    13
## 265 Leemstreek   SM  0.33 0.18 0.49 6.08   1.79  99.98    14
## 266 Leemstreek   SM  0.31 0.25 0.44 5.96   1.88 100.46    22
## 267 Leemstreek   PG  0.31 0.26 0.43 5.98 234.00 121.73    10
## 268 Leemstreek   SM  0.23 0.25 0.53 6.14   1.75  95.13    16
## 269 Leemstreek SMGR  0.24 0.27 0.50 6.03   2.02  97.52    19
## 271     Kempen   SM  0.19 0.57 0.24 5.04   2.19  83.44    20
## 272 Leemstreek   GM  0.28 0.24 0.48 5.74   1.88  96.10    18
## 273 Leemstreek   SM  0.30 0.26 0.44 5.81   1.81 102.19    18
## 274 Leemstreek   SM  0.30 0.23 0.47 5.97   1.75  94.90    23
## 275     Kempen   GM  0.17 0.60 0.23 4.74   2.09  86.39    15
## 276     Kempen SMGR  0.29 0.50 0.21 4.76   2.28  88.48    11
## 277     Kempen   PG  0.22 0.54 0.24 5.32   2.94 103.46    22
## 278 Leemstreek   SM  0.31 0.21 0.48 5.88   1.64  97.50    22
## 279 Leemstreek   GM  0.28 0.22 0.49 5.63   1.77  97.71    23
## 280     Kempen   GM  0.24 0.57 0.20 5.08   2.10  81.33    11
## 281     Kempen SMGR  0.10 0.62 0.28 5.36   2.19  86.76    10
## 282 Leemstreek SMGR  0.20 0.28 0.52 6.13   1.77 103.26    19
## 283 Leemstreek   PG  0.26 0.23 0.51 6.09   2.51 110.66    23
## 284 Leemstreek   PG  0.38 0.22 0.40 6.24   2.39 118.42    17
## 285 Leemstreek   PG  0.24 0.22 0.54 5.85   2.49 122.01    11
```

```
## 286 Leemstreek SMGR  0.32 0.24 0.44 6.15   1.97 103.69    16
## 287 Leemstreek   GM  0.31 0.24 0.45 5.88   1.85  98.90    15
## 288 Leemstreek   SM  0.18 0.26 0.56 6.00   1.67 100.15    17
## 289 Leemstreek   GM  0.29 0.24 0.47 6.44   1.84  98.84    12
## 290     Kempen   SM  0.34 0.50 0.17 5.29   2.04  86.66    17
## 291 Leemstreek   SM  0.15 0.25 0.61 6.15   1.70  91.73    14
## 292 Leemstreek SMGR  0.36 0.29 0.36 5.78   2.00 104.28    17
## 293 Leemstreek   GM  0.18 0.29 0.53 5.84   1.84 102.47    24
## 294     Kempen   SM  0.19 0.56 0.25 5.56   2.08  82.39    18
## 295     Kempen   SM  0.28 0.52 0.20 4.88   2.15  85.80    21
## 296     Kempen SMGR  0.04 0.72 0.23 5.14   2.13  90.68    11
## 297     Kempen   SM  0.17 0.59 0.23 5.28   2.08  86.28    20
## 298 Leemstreek   SM  0.19 0.25 0.56 6.20   1.79  97.31    22
## 299 Leemstreek   GM  0.21 0.26 0.53 6.22   1.73  93.88    10
## 300     Kempen   GM  0.27 0.52 0.21 5.07   2.10  82.39    11
## 301     Kempen   PG  0.19 0.56 0.25 4.94   2.87 106.88    19
## 302 Leemstreek   SM  0.25 0.20 0.55 5.94   1.76  97.59    22
## 303     Kempen   SM  0.24 0.57 0.19 4.92   2.09  84.20    21
## 304     Kempen   PG  0.22 0.61 0.18 5.23   2.88 107.71    14
## 305     Kempen   GM  0.24 0.53 0.22 5.32   2.02  79.22    22
## 306 Leemstreek SMGR  0.25 0.27 0.48 6.43   1.69 108.71    16
## 307 Leemstreek   GM  0.30 0.26 0.44 6.07   1.79 104.68    12
## 308 Leemstreek   GM  0.27 0.22 0.51 5.83   1.68 101.88    19
## 309 Leemstreek   SM  0.32 0.26 0.42 6.20   1.72  96.61    22
## 310 Leemstreek   GM  0.22 0.27 0.51 6.18   1.86  89.41    22
## 311 Leemstreek SMGR  0.26 0.28 0.46 5.95   2.09 104.50    11
## 312 Leemstreek   GM  0.20 0.24 0.56 5.69   1.66  99.59    11
## 313     Kempen SMGR  0.11 0.60 0.29 5.25   2.28  93.11    24
## 314 Leemstreek SMGR  0.31 0.25 0.45 6.23   1.89  99.67    11
## 315 Leemstreek   GM  0.14 0.29 0.56 5.94   1.76  92.06    13
## 316     Kempen   GM  0.10 0.66 0.24 5.21   2.14  82.65    17
## 317     Kempen   SM  0.14 0.64 0.22 5.04   2.11  86.49    22
## 318 Leemstreek   SM  0.29 0.24 0.47 6.14   1.73 101.65    23
## 319 Leemstreek   PG  0.14 0.26 0.59 5.98   2.41 123.40    19
## 320     Kempen   SM  0.22 0.62 0.16 5.03   2.15  80.43    16
## 321     Kempen   GM  0.25 0.55 0.20 5.12   1.86  77.14    12
## 322 Leemstreek SMGR  0.39 0.20 0.41 5.94   1.99  96.64    23
## 323 Leemstreek SMGR  0.15 0.27 0.58 5.88   1.77 106.20    23
## 324     Kempen   PG  0.26 0.52 0.21 5.04   2.75 109.26    15
## 325 Leemstreek SMGR  0.28 0.25 0.47 6.00   1.88 105.58    18
## 326 Leemstreek SMGR  0.05 0.26 0.69 6.04   2.00 100.29    17
## 327     Kempen   PG  0.15 0.63 0.22 5.41   2.76 112.14    22
## 328 Leemstreek SMGR  0.27 0.26 0.47 5.73   1.91 102.84    24
## 329 Leemstreek SMGR  0.25 0.23 0.52 5.96   1.95 111.06    19
## 330     Kempen   SM  0.18 0.65 0.17 5.23   2.03  86.24    22
## 331     Kempen   GM  0.08 0.69 0.23 5.45   2.03  86.72    22
## 332 Leemstreek   PG  0.23 0.28 0.49 6.09   2.54 121.58    14
## 333     Kempen SMGR  0.26 0.50 0.23 5.10   2.25  89.88    18
## 334     Kempen   PG  0.14 0.67 0.19 5.04   2.64 102.26    14
## 335  Leemstrek   SM  0.31 0.25 0.44 6.08   1.85  96.20    23
## 336 Leemstreek   GM  0.28 0.27 0.46 6.20   1.64  96.16    21
## 337     Kempen   PG  0.17 0.63 0.20 5.06   2.70 107.19    15
## 338     Kempen   PG  0.15 0.60 0.26 5.13   2.84 106.00    25
## 339     Kempen   GM  0.14 0.60 0.27 5.14   1.87  81.46    21
```

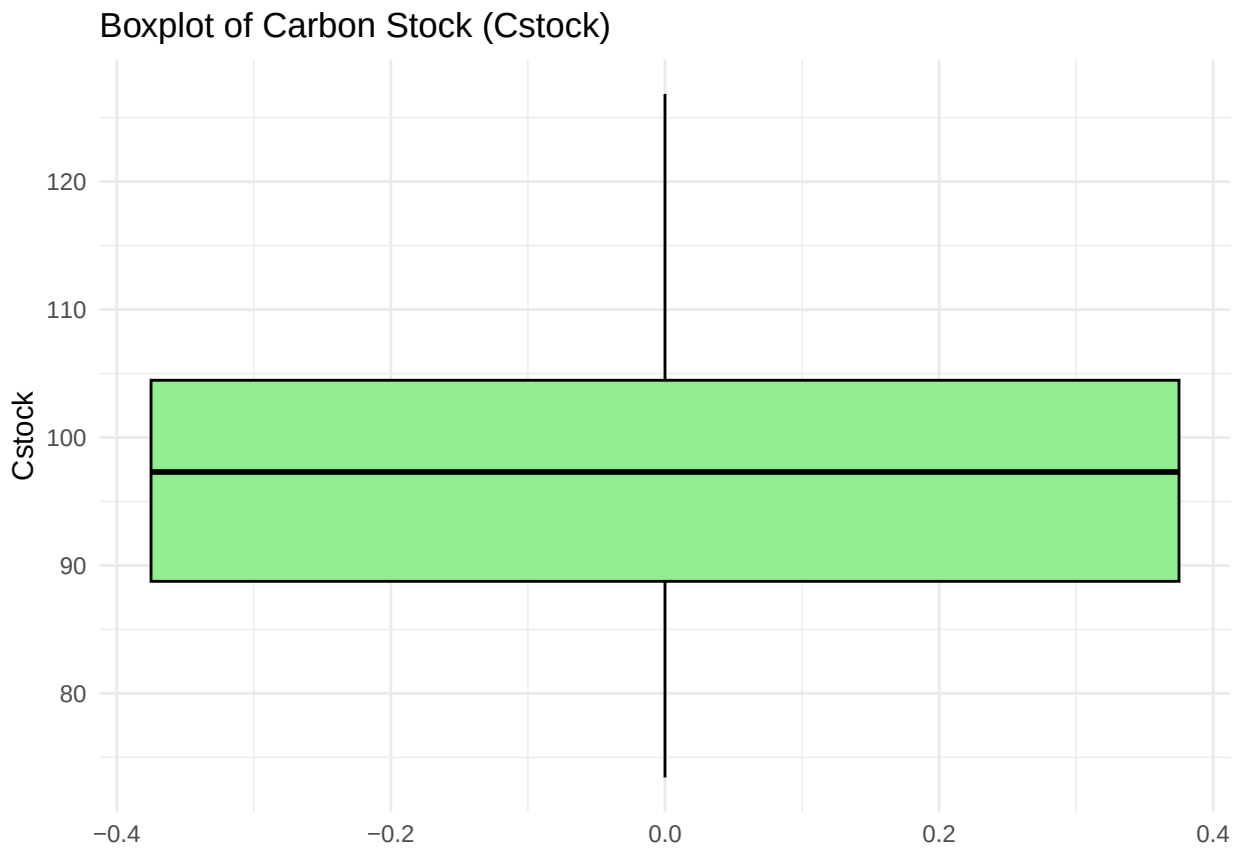
```
## 340     Kempen   SM  0.13 0.65 0.22 5.25   1.95  84.85    14
## 341 Leemstreek   PG  0.25 0.27 0.48 6.22   2.55 118.35    20
## 342 Leemstreek   SM  0.24 0.28 0.48 6.06   1.73  96.28    15
## 343 Leemstreek   SM  0.17 0.32 0.50 5.98   1.90  92.77    22
## 344 Leemstreek SMGR  0.22 0.27 0.51 6.28   1.89 107.57    15
## 345 Leemstreek   GM  0.23 0.26 0.52 6.03   1.78  96.49    14
## 347     Kempen SMGR  0.07 0.73 0.19 5.09   2.14  87.30    19
## 348 Leemstreek   SM  0.19 0.24 0.57 6.12   1.78  97.38    10
## 349 Leemstreek SMGR  0.21 0.32 0.47 5.69   1.86 100.11    14
## 350     Kempen   PG  0.15 0.62 0.23 5.08   2.97 102.13    14
## 351 Leemstreek   SM  0.21 0.30 0.50 6.00   1.81  91.01    22
## 352     Kempen   SM  0.21 0.59 0.19 4.95   2.13  87.82    12
## 353 Leemstreek   SM  0.20 0.27 0.54 5.79   1.77  98.74    14
## 355     Kempen SMGR  0.15 0.59 0.26 5.28   2.31  94.58    24
## 356     Kempen   GM  0.18 0.59 0.22 4.84   2.02  84.44    15
## 357     Kempen   GM  0.14 0.65 0.21 5.18   2.05  84.96    22
## 359 Leemstreek   SM  0.38 0.19 0.43 5.79   1.76  96.65    17
## 360     Kempen   SM  0.21 0.59 0.19 5.21   2.25  83.77    18
## 361 Leemstreek   GM  0.20 0.27 0.53 5.86   1.63  97.81    23
## 362 Leemstreek   SM  0.25 0.27 0.48 6.20   1.96 101.38    15
## 363     Kempen SMGR  0.15 0.63 0.22 5.11   2.17  96.16    17
## 364 Leemstreek   GM  0.31 0.20 0.49 6.11   1.86  94.39    17
## 365 Leemstreek SMGR  0.28 0.26 0.46 5.90   1.74 100.42    21
## 366     Kempen   SM  0.19 0.65 0.16 4.87   2.17  84.46    11
## 367 Leemstreek   SM  0.19 0.29 0.52 6.18   1.87 109.82    21
## 368 Leemstreek   GM  0.21 0.24 0.55 6.05   1.65  94.68    20
## 370 Leemstreek SMGR  0.12 0.29 0.59 6.01   1.91 103.38    17
## 371 Leemstreek   PG  0.28 0.20 0.52 6.27   2.54 118.53    14
## 372 Leemstreek   PG  0.18 0.24 0.58 5.86   2.39 117.58    16
## 373     Kempen SMGR  0.19 0.62 0.19 4.95   2.18  89.29    12
## 374 Leemstreek   PG  0.29 0.20 0.51 5.88   2.57 126.52    14
## 375 Leemstreek   SM  0.25 0.26 0.49 5.68   1.71  99.39    11
## 376     Kempen   SM  0.06 0.69 0.24 5.26   2.17  86.01    17
## 377     Kempen   GM  0.20 0.60 0.20 5.14   2.08  85.32    15
## 378 Leemstreek   GM  0.26 0.24 0.50 6.04   1.66 104.18    13
## 379 Leemstreek SMGR  0.34 0.26 0.39 5.99   1.85 107.25    21
## 380     Kempen   PG  0.22 0.59 0.20 5.00   2.81 100.68    14
## 381 Leemstreek   SM  0.22 0.27 0.51 5.84   1.82  94.34    16
## 382     Kempen   PG  0.16 0.65 0.20 5.27   2.86 100.12    21
## 383 Leemstreek   SM  0.25 0.25 0.50 5.71   1.75  87.42    12
## 384     Kempen SMGR  0.25 0.54 0.21 5.19   2.06  95.05    12
## 386     Kempen   GM  0.17 0.57 0.26 5.16   2.00  83.64    25
## 387 Leemstreek   SM  0.28 0.27 0.45 5.85   1.81 102.76    21
## 388 Leemstreek   SM  0.33 0.19 0.48 5.90   1.94  96.34    10
## 389 Leemstreek   SM  0.22 0.19 0.59 6.35   1.78  97.86    25
## 390 Leemstreek   PG  0.40 0.23 0.37 6.13   2.57 121.66    21
## 391     Kempen   PG  0.17 0.62 0.21 4.67   2.66 106.89    23
## 392 Leemstreek   GM  0.15 0.22 0.62 6.03   1.82  93.36    13
## 393     Kempen   SM  0.07 0.64 0.29 5.01   2.11  86.88    14
## 394     Kempen   GM  0.16 0.62 0.22 4.87   1.99  84.15    19
## 395 Leemstreek   PG  0.31 0.24 0.45 6.04   2.35 119.21    15
## 396 Leemstreek   GM  0.22 0.27 0.51 6.03   1.76  91.66    21
## 397 Leemstreek   SM  0.28 0.26 0.46 5.93   1.82 100.70    11
## 398 Leemstreek   SM  0.18 0.24 0.59 6.39   1.82 103.80    23
```

```
## 399 Leemstreek   GM  0.29 0.26 0.45 5.55   1.79 101.46    22
## 400     Kempen   SM  0.14 0.63 0.23 5.27   2.06  87.47    23
```

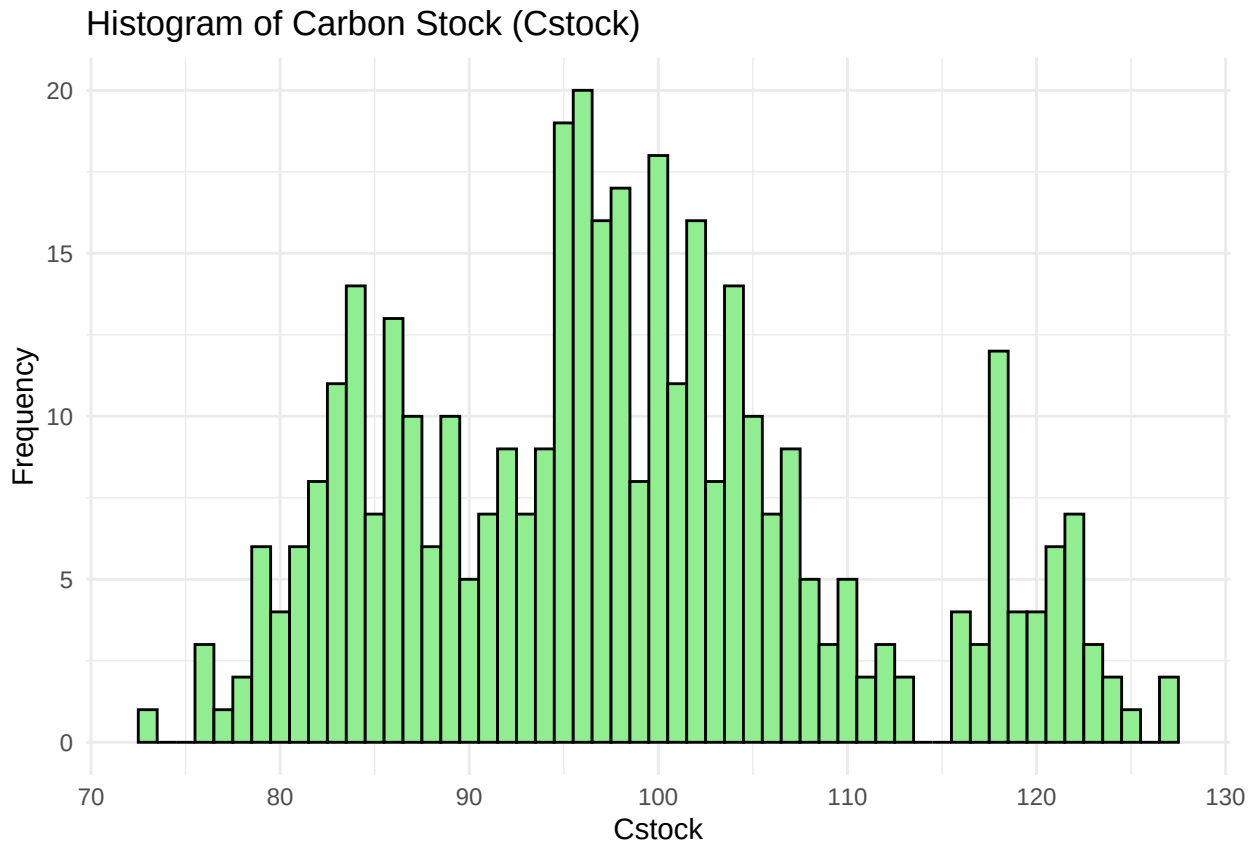
```r
qqnorm(data_clean$Cstock)
qqline(data_clean$Cstock, col = "red")
title(main = "Q-Q Plot for Carbon Stock")
```




```r
ggplot(data_clean, aes(y = Cstock)) +
  geom_boxplot(fill = "lightgreen", color = "black") +
  labs(title = "Boxplot of Carbon Stock (Cstock)", y = "Cstock") +
  theme_minimal()
```




```r
#Histogram
ggplot(data_clean, aes(x = Cstock)) +
  geom_histogram(binwidth = 1, color = "black", fill = "lightgreen") +
  labs(title = "Histogram of Carbon Stock (Cstock)",
       x = "Cstock",
       y = "Frequency") +
  theme_minimal()
```

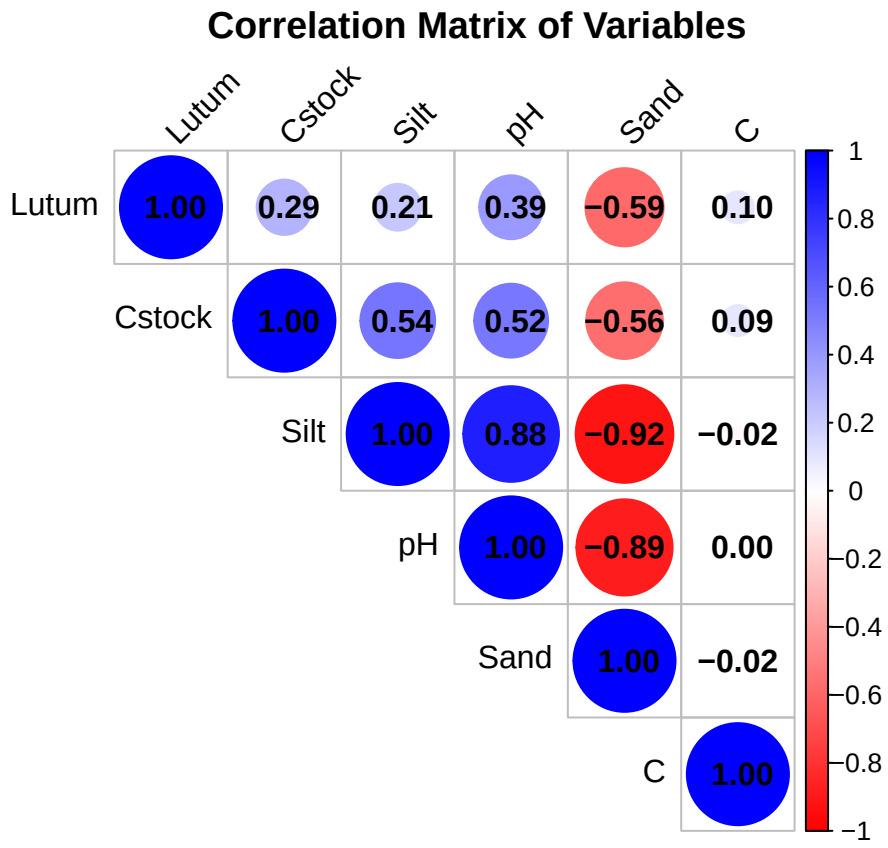



```
summary(data_clean)
```

```
##     Region              Crop              Lutum             Sand        
##  Length:380         Length:380         Min.   :0.0400   Min.   :0.170  
##  Class :character   Class :character   1st Qu.:0.1700   1st Qu.:0.240  
##  Mode  :character   Mode  :character   Median :0.2200   Median :0.280  
##                                        Mean   :0.2192   Mean   :0.391  
##                                        3rd Qu.:0.2700   3rd Qu.:0.590  
##                                        Max.   :0.4200   Max.   :0.730  
##       Silt              pH               C               Cstock      
##  Min.   :0.1400   Min.   :4.640   Min.   :  1.550   Min.   : 73.42  
##  1st Qu.:0.2300   1st Qu.:5.207   1st Qu.:  1.820   1st Qu.: 88.75  
##  Median :0.4400   Median :5.800   Median :  2.015   Median : 97.30  
##  Mean   :0.3897   Mean   :5.641   Mean   :  3.884   Mean   : 98.04  
##  3rd Qu.:0.5100   3rd Qu.:6.040   3rd Qu.:  2.257   3rd Qu.:104.47  
##  Max.   :0.6900   Max.   :6.680   Max.   :252.000   Max.   :126.85  
##      Years      
##  Min.   :10.00  
##  1st Qu.:14.00  
##  Median :17.00  
##  Mean   :17.31  
##  3rd Qu.:21.00  
##  Max.   :25.00
```

```r
# Compute the correlation matrix
cor_matrix <- cor(data_clean[, c("Lutum", "Silt", "Sand", "pH", "C", "Cstock")], use = "complete.obs")

corrplot(
  cor_matrix,
  method = "circle",
  type = "upper",
  order = "hclust",
  addCoef.col = "black",
  tl.col = "black",
  tl.srt = 45,
  col = colorRampPalette(c("red", "white", "blue"))(200),
  title = "Correlation Matrix of Variables",
  mar = c(0, 0, 2, 0)
)
```

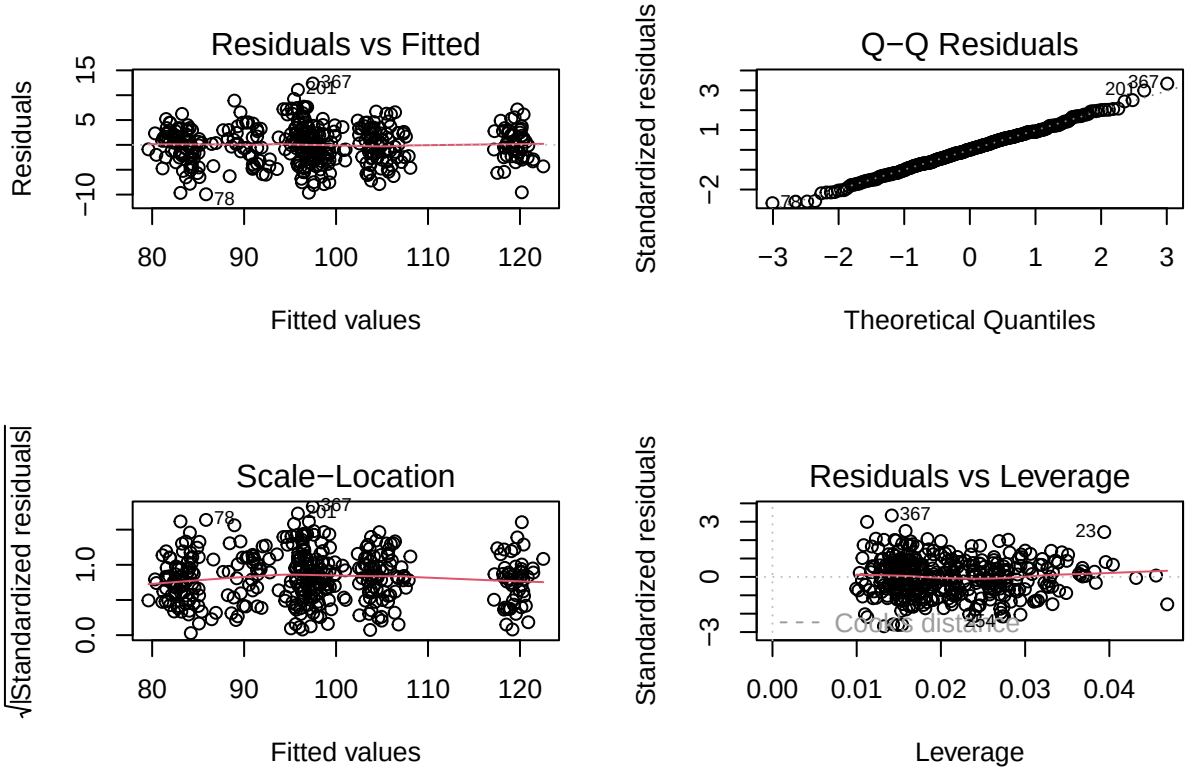



```r
#We can see that Sand has negative correlation with other variables
model <- lm(Cstock ~ Crop + Lutum + Sand + Silt + pH, data = data_clean)
summary(model)
```

```
##
## Call:
## lm(formula = Cstock ~ Crop + Lutum + Sand + Silt + pH, data = data_clean)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -9.9364 -2.4825  0.0521  2.4328 12.3513
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  87.5575    38.6906   2.263  0.02421 *
## CropPG       23.2470     0.5526  42.072  < 2e-16 ***
## CropSM        1.5954     0.5133   3.108  0.00203 **
## CropSMGR      7.6125     0.5549  13.718  < 2e-16 ***
## Lutum        -5.6290    38.6035  -0.146  0.88414
## Sand        -29.1203    38.6021  -0.754  0.45110
## Silt          5.3363    38.7196   0.138  0.89046
## pH            2.4361     0.9420   2.586  0.01008 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.726 on 372 degrees of freedom
## Multiple R-squared:  0.9013, Adjusted R-squared:  0.8995
## F-statistic: 485.4 on 7 and 372 DF,  p-value: < 2.2e-16
```

```r
par(mfrow = c(2, 2))
plot(model)
```



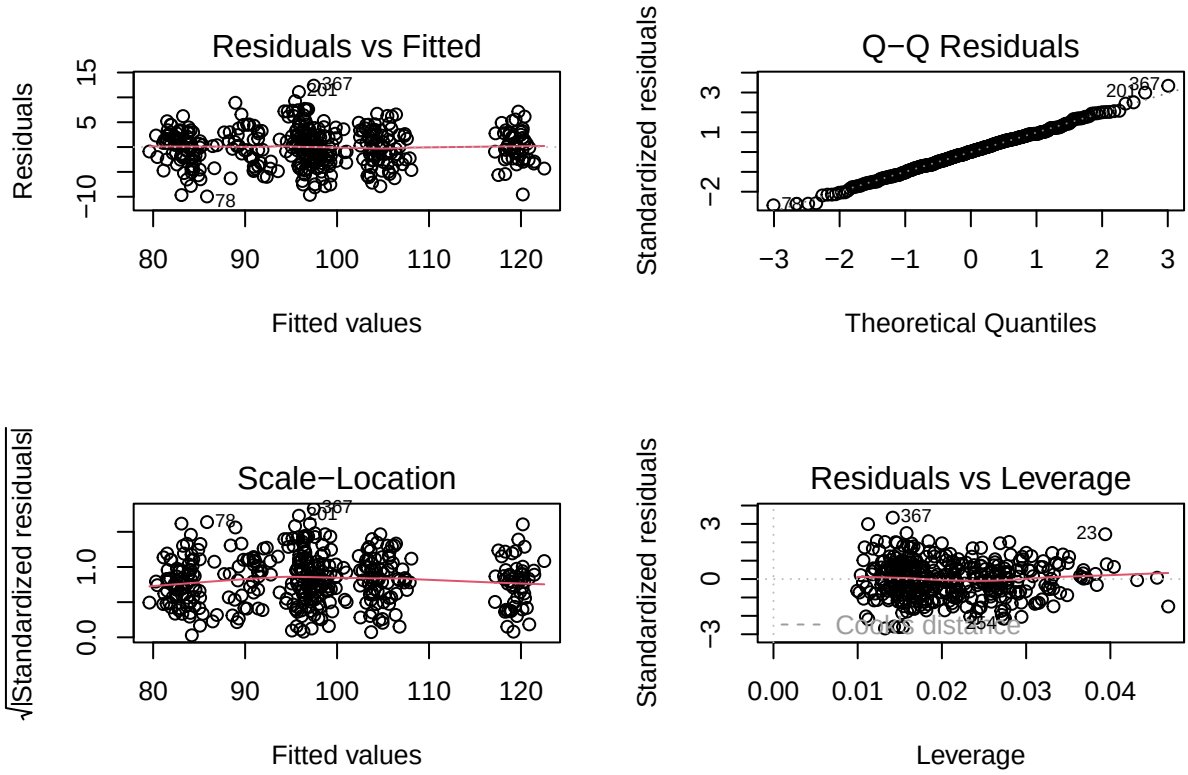

```r
#Remove Sand for multicolinearity
model2 <- lm(Cstock ~ Crop + Lutum + Silt + pH, data = data_clean)
summary(model2)
```

```
##
## Call:
## lm(formula = Cstock ~ Crop + Lutum + Silt + pH, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.9521 -2.4267  0.0279  2.4762 12.3480
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  58.5356     4.1076  14.250  < 2e-16 ***
## CropPG       23.2234     0.5513  42.121  < 2e-16 ***
## CropSM        1.5842     0.5128   3.089  0.00216 **
## CropSMGR      7.5874     0.5536  13.705  < 2e-16 ***
## Lutum        23.4061     2.9662   7.891 3.37e-14 ***
## Silt         34.4664     2.8422  12.127  < 2e-16 ***
## pH            2.4244     0.9413   2.575  0.01039 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.723 on 373 degrees of freedom
## Multiple R-squared:  0.9012, Adjusted R-squared:  0.8996
## F-statistic: 566.8 on 6 and 373 DF,  p-value: < 2.2e-16
```

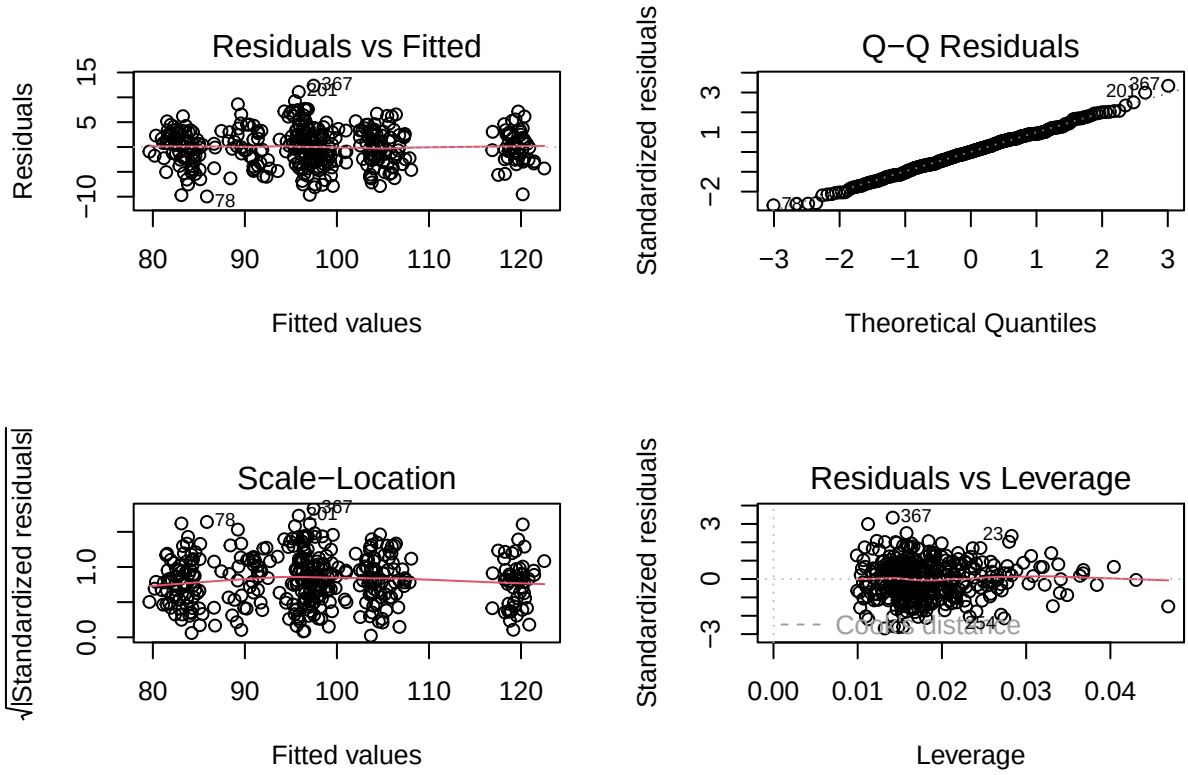
```
#We remove crop type to check is there any effect on CStock
model3 = lm(Cstock ~ Lutum + Silt + pH, data = data_clean)
summary(model3)
```

```
##
## Call:
## lm(formula = Cstock ~ Lutum + Silt + pH, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -16.846  -6.900  -3.235   4.745  23.357
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   67.918     10.686   6.355 6.02e-10 ***
## Lutum         27.593      7.730   3.570 0.000404 ***
## Silt          34.421      7.407   4.647 4.67e-06 ***
## pH             1.889      2.450   0.771 0.441158
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.711 on 376 degrees of freedom
## Multiple R-squared:  0.3223, Adjusted R-squared:  0.3169
## F-statistic:  59.6 on 3 and 376 DF,  p-value: < 2.2e-16
```

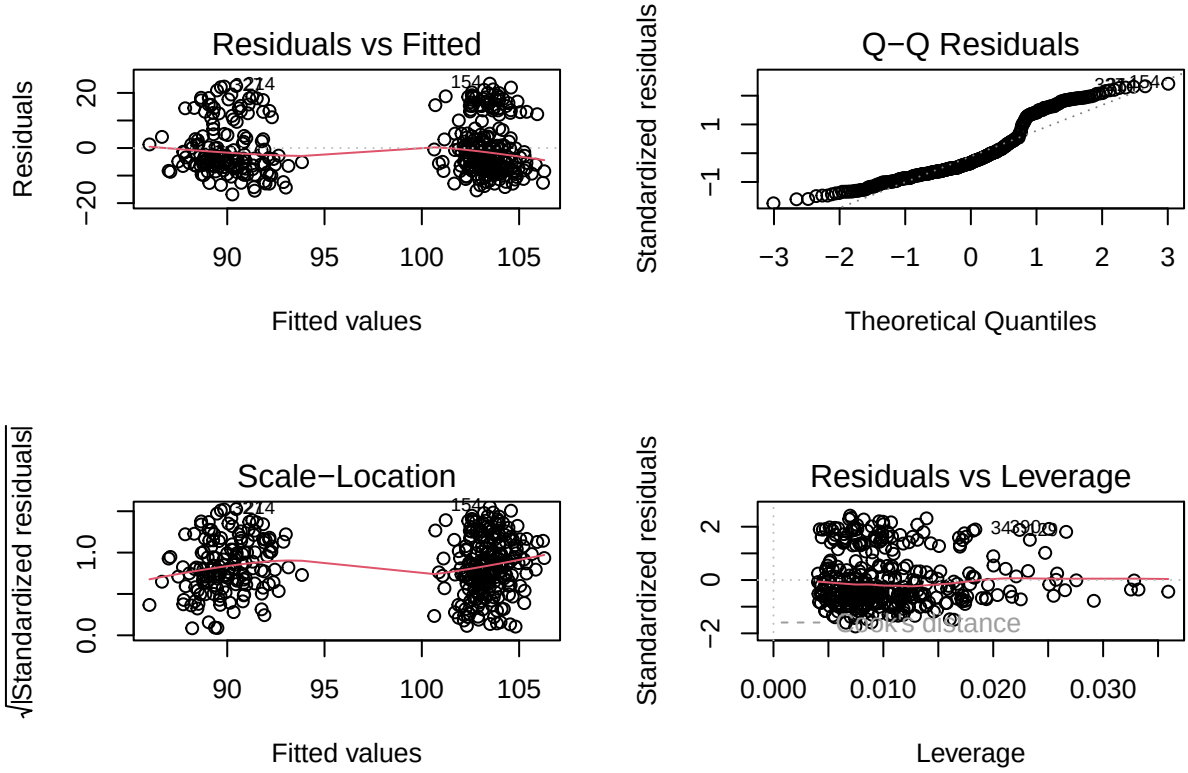
```
par(mfrow = c(2,2))
plot(model)
```


Residuals vs Fitted
Residuals
Fitted values
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance

```
plot(model2)
```


Residuals vs Fitted
Residuals
Fitted values
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage

```
plot(model3)
```




```
vif(model)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## Crop     1.020451  3        1.003380
## Lutum  220.195200  1       14.838976
## Sand  1290.303279  1       35.920792
## Silt   891.776351  1       29.862625
## pH       5.445307  1        2.333518
```

```
vif(model2)
```

```
##           GVIF Df GVIF^(1/(2*Df))
## Crop  1.015515  3        1.002569
## Lutum 1.301558  1        1.140858
## Silt  4.810677  1        2.193326
## pH    5.443813  1        2.333198
```

```
vif(model3)
```

```
##    Lutum     Silt       pH
## 1.299523 4.803564 5.421124
```

```r
#There is no interaction effect of Crop and Region Two-Way ANOVA Non-Additive model
anova_model <- aov(Cstock ~ Crop * Region, data = data_clean)
summary(anova_model)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Crop          3  30019   10006 808.492 <2e-16 ***
## Region        5  17712    3542 286.209 <2e-16 ***
## Crop:Region   3     35      12   0.938  0.422
## Residuals   368   4555      12
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
TukeyHSD(anova_model)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = Cstock ~ Crop * Region, data = data_clean)
##
## $Crop
##                diff        lwr        upr p adj
## PG-GM    23.536205  22.192426  24.879983 0e+00
## SM-GM     2.577796   1.331248   3.824343 1e-06
## SMGR-GM   6.669180   5.321009   8.017350 0e+00
## SM-PG   -20.958409 -22.266424 -19.650394 0e+00
## SMGR-PG -16.867025 -18.272225 -15.461826 0e+00
## SMGR-SM   4.091384   2.778857   5.403911 0e+00
##
## $Region
##                            diff          lwr       upr     p adj
## Kempeen-Keempen        5.732975  -8.52021704 19.986167 0.8589219
## Kempen-Keempen         1.930094  -8.18197358 12.042162 0.9941459
## Leemstreek-Keempen    15.801915   5.70111319 25.902716 0.0001435
## Leemstrek-Keempen     13.261384  -0.99180785 27.514576 0.0848008
## Lemstreek-Keempen     15.212975   0.95978296 29.466167 0.0286892
## Kempen-Keempen        -3.802880 -13.91494834  6.309187 0.8901915
## Leemstreek-Kempen     10.068940  -0.03186156 20.169742 0.0512641
## Leemstrek-Kempen       7.528409  -6.72478261 21.781601 0.6560184
## Lemstreek-Kempen       9.480000  -4.77319180 23.733192 0.4004268
## Leemstreek-Kempen     13.871821  12.81039088 14.933250 0.0000000
## Leemstrek-Kempen      11.331290   1.21922180 21.443358 0.0179522
## Lemstreek-Kempen      13.282880   3.17081261 23.394948 0.0026676
## Leemstrek-Leemstreek -2.540531 -12.64133243  7.560271 0.9793757
## Lemstreek-Leemstreek -0.588940 -10.68974162  9.511862 0.9999819
## Lemstreek-Leemstrek   1.951591 -12.30160098 16.204783 0.9987948
##
## $`Crop:Region`
##                                        diff         lwr         upr     p adj
## PG:Keempen-GM:Keempen                    NA          NA          NA        NA
## SM:Keempen-GM:Keempen                    NA          NA          NA        NA
## SMGR:Keempen-GM:Keempen                  NA          NA          NA        NA
## GM:Kempeen-GM:Keempen                    NA          NA          NA        NA
## PG:Kempeen-GM:Keempen                    NA          NA          NA        NA
```

```
## SM:Kempeen-GM:Keempen                NA          NA          NA          NA
## SMGR:Kempeen-GM:Keempen              NA          NA          NA          NA
## GM:Kempen-GM:Keempen                 NA          NA          NA          NA
## PG:Kempen-GM:Keempen                 NA          NA          NA          NA
## SM:Kempen-GM:Keempen                 NA          NA          NA          NA
## SMGR:Kempen-GM:Keempen               NA          NA          NA          NA
## GM:Leemstreek-GM:Keempen             NA          NA          NA          NA
## PG:Leemstreek-GM:Keempen             NA          NA          NA          NA
## SM:Leemstreek-GM:Keempen             NA          NA          NA          NA
## SMGR:Leemstreek-GM:Keempen           NA          NA          NA          NA
## GM:Leemstrek-GM:Keempen              NA          NA          NA          NA
## PG:Leemstrek-GM:Keempen              NA          NA          NA          NA
## SM:Leemstrek-GM:Keempen              NA          NA          NA          NA
## SMGR:Leemstrek-GM:Keempen            NA          NA          NA          NA
## GM:Lemstreek-GM:Keempen              NA          NA          NA          NA
## PG:Lemstreek-GM:Keempen              NA          NA          NA          NA
## SM:Lemstreek-GM:Keempen              NA          NA          NA          NA
## SMGR:Lemstreek-GM:Keempen            NA          NA          NA          NA
## SM:Keempen-PG:Keempen                NA          NA          NA          NA
## SMGR:Keempen-PG:Keempen              NA          NA          NA          NA
## GM:Kempeen-PG:Keempen                NA          NA          NA          NA
## PG:Kempeen-PG:Keempen                NA          NA          NA          NA
## SM:Kempeen-PG:Keempen                NA          NA          NA          NA
## SMGR:Kempeen-PG:Keempen              NA          NA          NA          NA
## GM:Kempen-PG:Keempen                 NA          NA          NA          NA
## PG:Kempen-PG:Keempen                 NA          NA          NA          NA
## SM:Kempen-PG:Keempen                 NA          NA          NA          NA
## SMGR:Kempen-PG:Keempen               NA          NA          NA          NA
## GM:Leemstreek-PG:Keempen             NA          NA          NA          NA
## PG:Leemstreek-PG:Keempen             NA          NA          NA          NA
## SM:Leemstreek-PG:Keempen             NA          NA          NA          NA
## SMGR:Leemstreek-PG:Keempen           NA          NA          NA          NA
## GM:Leemstrek-PG:Keempen              NA          NA          NA          NA
## PG:Leemstrek-PG:Keempen              NA          NA          NA          NA
## SM:Leemstrek-PG:Keempen              NA          NA          NA          NA
## SMGR:Leemstrek-PG:Keempen            NA          NA          NA          NA
## GM:Lemstreek-PG:Keempen              NA          NA          NA          NA
## PG:Lemstreek-PG:Keempen              NA          NA          NA          NA
## SM:Lemstreek-PG:Keempen              NA          NA          NA          NA
## SMGR:Lemstreek-PG:Keempen            NA          NA          NA          NA
## SMGR:Keempen-SM:Keempen              NA          NA          NA          NA
## GM:Kempeen-SM:Keempen                NA          NA          NA          NA
## PG:Kempeen-SM:Keempen                NA          NA          NA          NA
## SM:Kempeen-SM:Keempen                NA          NA          NA          NA
## SMGR:Kempeen-SM:Keempen              NA          NA          NA          NA
## GM:Kempen-SM:Keempen                 NA          NA          NA          NA
## PG:Kempen-SM:Keempen                 NA          NA          NA          NA
## SM:Kempen-SM:Keempen                 NA          NA          NA          NA
## SMGR:Kempen-SM:Keempen               NA          NA          NA          NA
## GM:Leemstreek-SM:Keempen             NA          NA          NA          NA
## PG:Leemstreek-SM:Keempen             NA          NA          NA          NA
## SM:Leemstreek-SM:Keempen             NA          NA          NA          NA
## SMGR:Leemstreek-SM:Keempen           NA          NA          NA          NA
## GM:Leemstrek-SM:Keempen              NA          NA          NA          NA
```

```
## PG:Leemstrek-SM:Keempen                NA          NA          NA        NA
## SM:Leemstrek-SM:Keempen                NA          NA          NA        NA
## SMGR:Leemstrek-SM:Keempen              NA          NA          NA        NA
## GM:Lemstreek-SM:Keempen                NA          NA          NA        NA
## PG:Lemstreek-SM:Keempen                NA          NA          NA        NA
## SM:Lemstreek-SM:Keempen                NA          NA          NA        NA
## SMGR:Lemstreek-SM:Keempen              NA          NA          NA        NA
## GM:Kempeen-SMGR:Keempen                NA          NA          NA        NA
## PG:Kempeen-SMGR:Keempen        22.6000000   4.3636704  40.8363296 0.0018056
## SM:Kempeen-SMGR:Keempen                NA          NA          NA        NA
## SMGR:Kempeen-SMGR:Keempen              NA          NA          NA        NA
## GM:Kempen-SMGR:Keempen         -5.2590244 -18.3103658   7.7923170 0.9987774
## PG:Kempen-SMGR:Keempen         18.6009375   5.5059703  31.6959047 0.0000838
## SM:Kempen-SMGR:Keempen         -2.9352632 -15.9988651  10.1283388 1.0000000
## SMGR:Kempen-SMGR:Keempen        3.3917949  -9.6675120  16.4511017 0.9999994
## GM:Leemstreek-SMGR:Keempen      9.7235593  -3.2802938  22.7274124 0.4863242
## PG:Leemstreek-SMGR:Keempen     32.5582000  19.5348558  45.5815442 0.0000000
## SM:Leemstreek-SMGR:Keempen     10.7875676  -2.1943010  23.7694361 0.2718161
## SMGR:Leemstreek-SMGR:Keempen   16.7083721   3.6642593  29.7524849 0.0009263
## GM:Leemstrek-SMGR:Keempen              NA          NA          NA        NA
## PG:Leemstrek-SMGR:Keempen              NA          NA          NA        NA
## SM:Leemstrek-SMGR:Keempen       9.1700000  -9.0663296  27.4063296 0.9770353
## SMGR:Leemstrek-SMGR:Keempen            NA          NA          NA        NA
## GM:Lemstreek-SMGR:Keempen              NA          NA          NA        NA
## PG:Lemstreek-SMGR:Keempen      32.0800000  13.8436704  50.3163296 0.0000001
## SM:Lemstreek-SMGR:Keempen              NA          NA          NA        NA
## SMGR:Lemstreek-SMGR:Keempen            NA          NA          NA        NA
## PG:Kempeen-GM:Kempeen                  NA          NA          NA        NA
## SM:Kempeen-GM:Kempeen                  NA          NA          NA        NA
## SMGR:Kempeen-GM:Kempeen                NA          NA          NA        NA
## GM:Kempen-GM:Kempeen                   NA          NA          NA        NA
## PG:Kempen-GM:Kempeen                   NA          NA          NA        NA
## SM:Kempen-GM:Kempeen                   NA          NA          NA        NA
## SMGR:Kempen-GM:Kempeen                 NA          NA          NA        NA
## GM:Leemstreek-GM:Kempeen               NA          NA          NA        NA
## PG:Leemstreek-GM:Kempeen               NA          NA          NA        NA
## SM:Leemstreek-GM:Kempeen               NA          NA          NA        NA
## SMGR:Leemstreek-GM:Kempeen             NA          NA          NA        NA
## GM:Leemstrek-GM:Kempeen                NA          NA          NA        NA
## PG:Leemstrek-GM:Kempeen                NA          NA          NA        NA
## SM:Leemstrek-GM:Kempeen                NA          NA          NA        NA
## SMGR:Leemstrek-GM:Kempeen              NA          NA          NA        NA
## GM:Lemstreek-GM:Kempeen                NA          NA          NA        NA
## PG:Lemstreek-GM:Kempeen                NA          NA          NA        NA
## SM:Lemstreek-GM:Kempeen                NA          NA          NA        NA
## SMGR:Lemstreek-GM:Kempeen              NA          NA          NA        NA
## SM:Kempeen-PG:Kempeen                  NA          NA          NA        NA
## SMGR:Kempeen-PG:Kempeen                NA          NA          NA        NA
## GM:Kempen-PG:Kempeen          -27.8590244 -40.9103658 -14.8076830 0.0000000
## PG:Kempen-PG:Kempeen           -3.9990625 -17.0940297   9.0959047 0.9999875
## SM:Kempen-PG:Kempeen          -25.5352632 -38.5988651 -12.4716612 0.0000000
## SMGR:Kempen-PG:Kempeen        -19.2082051 -32.2675120  -6.1488983 0.0000332
## GM:Leemstreek-PG:Kempeen      -12.8764407 -25.8802938   0.1274124 0.0560799
## PG:Leemstreek-PG:Kempeen        9.9582000  -3.0651442  22.9815442 0.4385856
```

```
## SM:Leemstreek-PG:Kempeen       -11.8124324 -24.7943010   1.1694361 0.1333870
## SMGR:Leemstreek-PG:Kempeen      -5.8916279 -18.9357407   7.1524849 0.9939070
## GM:Leemstrek-PG:Kempeen                NA          NA          NA        NA
## PG:Leemstrek-PG:Kempeen                NA          NA          NA        NA
## SM:Leemstrek-PG:Kempeen       -13.4300000 -31.6663296   4.8063296 0.5188834
## SMGR:Leemstrek-PG:Kempeen              NA          NA          NA        NA
## GM:Lemstreek-PG:Kempeen                NA          NA          NA        NA
## PG:Lemstreek-PG:Kempeen         9.4800000  -8.7563296  27.7163296 0.9666940
## SM:Lemstreek-PG:Kempeen                NA          NA          NA        NA
## SMGR:Lemstreek-PG:Kempeen              NA          NA          NA        NA
## SMGR:Kempeen-SM:Kempeen                NA          NA          NA        NA
## GM:Kempen-SM:Kempeen                   NA          NA          NA        NA
## PG:Kempen-SM:Kempeen                   NA          NA          NA        NA
## SM:Kempen-SM:Kempeen                   NA          NA          NA        NA
## SMGR:Kempen-SM:Kempeen                 NA          NA          NA        NA
## GM:Leemstreek-SM:Kempeen               NA          NA          NA        NA
## PG:Leemstreek-SM:Kempeen               NA          NA          NA        NA
## SM:Leemstreek-SM:Kempeen               NA          NA          NA        NA
## SMGR:Leemstreek-SM:Kempeen             NA          NA          NA        NA
## GM:Leemstrek-SM:Kempeen                NA          NA          NA        NA
## PG:Leemstrek-SM:Kempeen                NA          NA          NA        NA
## SM:Leemstrek-SM:Kempeen                NA          NA          NA        NA
## SMGR:Leemstrek-SM:Kempeen              NA          NA          NA        NA
## GM:Lemstreek-SM:Kempeen                NA          NA          NA        NA
## PG:Lemstreek-SM:Kempeen                NA          NA          NA        NA
## SM:Lemstreek-SM:Kempeen                NA          NA          NA        NA
## SMGR:Lemstreek-SM:Kempeen              NA          NA          NA        NA
## GM:Kempen-SMGR:Kempeen                 NA          NA          NA        NA
## PG:Kempen-SMGR:Kempeen                 NA          NA          NA        NA
## SM:Kempen-SMGR:Kempeen                 NA          NA          NA        NA
## SMGR:Kempen-SMGR:Kempeen               NA          NA          NA        NA
## GM:Leemstreek-SMGR:Kempeen             NA          NA          NA        NA
## PG:Leemstreek-SMGR:Kempeen             NA          NA          NA        NA
## SM:Leemstreek-SMGR:Kempeen             NA          NA          NA        NA
## SMGR:Leemstreek-SMGR:Kempeen           NA          NA          NA        NA
## GM:Leemstrek-SMGR:Kempeen              NA          NA          NA        NA
## PG:Leemstrek-SMGR:Kempeen              NA          NA          NA        NA
## SM:Leemstrek-SMGR:Kempeen              NA          NA          NA        NA
## SMGR:Leemstrek-SMGR:Kempeen            NA          NA          NA        NA
## GM:Lemstreek-SMGR:Kempeen              NA          NA          NA        NA
## PG:Lemstreek-SMGR:Kempeen              NA          NA          NA        NA
## SM:Lemstreek-SMGR:Kempeen              NA          NA          NA        NA
## SMGR:Lemstreek-SMGR:Kempeen            NA          NA          NA        NA
## PG:Kempen-GM:Kempen            23.8599619  20.8182579  26.9016659 0.0000000
## SM:Kempen-GM:Kempen             2.3237612  -0.5799422   5.2274646 0.3440359
## SMGR:Kempen-GM:Kempen           8.6508193   5.7665009  11.5351376 0.0000000
## GM:Leemstreek-GM:Kempen        14.9825837  12.3607539  17.6044135 0.0000000
## PG:Leemstreek-GM:Kempen        37.8172244  35.1003710  40.5340778 0.0000000
## SM:Leemstreek-GM:Kempen        16.0465920  13.5360726  18.5571113 0.0000000
## SMGR:Leemstreek-GM:Kempen      21.9673965  19.1526718  24.7821212 0.0000000
## GM:Leemstrek-GM:Kempen                 NA          NA          NA        NA
## PG:Leemstrek-GM:Kempen                 NA          NA          NA        NA
## SM:Leemstrek-GM:Kempen         14.4290244   1.3776830  27.4803658 0.0129777
## SMGR:Leemstrek-GM:Kempen               NA          NA          NA        NA
```

```
## GM:Lemstreek-GM:Kempen                NA          NA          NA        NA
## PG:Lemstreek-GM:Kempen          37.3390244  24.2876830  50.3903658 0.0000000
## SM:Lemstreek-GM:Kempen                NA          NA          NA        NA
## SMGR:Lemstreek-GM:Kempen              NA          NA          NA        NA
## SM:Kempen-PG:Kempen          -21.5362007 -24.6300892 -18.4423121 0.0000000
## SMGR:Kempen-PG:Kempen        -15.2091426 -18.2848451 -12.1334402 0.0000000
## GM:Leemstreek-PG:Kempen       -8.8773782 -11.7083930  -6.0463634 0.0000000
## PG:Leemstreek-PG:Kempen       13.9572625  11.0380254  16.8764996 0.0000000
## SM:Leemstreek-PG:Kempen       -7.8133699 -10.5416222  -5.0851177 0.0000000
## SMGR:Leemstreek-PG:Kempen     -1.8925654  -4.9031016   1.1179708 0.8103240
## GM:Leemstrek-PG:Kempen                NA          NA          NA        NA
## PG:Leemstrek-PG:Kempen                NA          NA          NA        NA
## SM:Leemstrek-PG:Kempen        -9.4309375 -22.5259047   3.6640297 0.5660915
## SMGR:Leemstrek-PG:Kempen              NA          NA          NA        NA
## GM:Lemstreek-PG:Kempen                NA          NA          NA        NA
## PG:Lemstreek-PG:Kempen        13.4790625   0.3840953  26.5740297 0.0350750
## SM:Lemstreek-PG:Kempen                NA          NA          NA        NA
## SMGR:Lemstreek-PG:Kempen              NA          NA          NA        NA
## SMGR:Kempen-SM:Kempen          6.3270580   3.3877595   9.2663565 0.0000000
## GM:Leemstreek-SM:Kempen       12.6588225   9.9766266  15.3410184 0.0000000
## PG:Leemstreek-SM:Kempen       35.4934632  32.7183099  38.2686164 0.0000000
## SM:Leemstreek-SM:Kempen       13.7228307  11.1493330  16.2963284 0.0000000
## SMGR:Leemstreek-SM:Kempen     19.6436353  16.7725974  22.5146731 0.0000000
## GM:Leemstrek-SM:Kempen                NA          NA          NA        NA
## PG:Leemstrek-SM:Kempen                NA          NA          NA        NA
## SM:Leemstrek-SM:Kempen        12.1052632  -0.9583388  25.1688651 0.1126225
## SMGR:Leemstrek-SM:Kempen              NA          NA          NA        NA
## GM:Lemstreek-SM:Kempen                NA          NA          NA        NA
## PG:Lemstreek-SM:Kempen        35.0152632  21.9516612  48.0788651 0.0000000
## SM:Lemstreek-SM:Kempen                NA          NA          NA        NA
## SMGR:Lemstreek-SM:Kempen              NA          NA          NA        NA
## GM:Leemstreek-SMGR:Kempen      6.3317645   3.6705666   8.9929623 0.0000000
## PG:Leemstreek-SMGR:Kempen     29.1664051  26.4115413  31.9212689 0.0000000
## SM:Leemstreek-SMGR:Kempen      7.3957727   4.8441674   9.9473780 0.0000000
## SMGR:Leemstreek-SMGR:Kempen   13.3165772  10.4651465  16.1680080 0.0000000
## GM:Leemstrek-SMGR:Kempen              NA          NA          NA        NA
## PG:Leemstrek-SMGR:Kempen              NA          NA          NA        NA
## SM:Leemstrek-SMGR:Kempen       5.7782051  -7.2811017  18.8375120 0.9953754
## SMGR:Leemstrek-SMGR:Kempen            NA          NA          NA        NA
## GM:Lemstreek-SMGR:Kempen              NA          NA          NA        NA
## PG:Lemstreek-SMGR:Kempen      28.6882051  15.6288983  41.7475120 0.0000000
## SM:Lemstreek-SMGR:Kempen              NA          NA          NA        NA
## SMGR:Lemstreek-SMGR:Kempen            NA          NA          NA        NA
## PG:Leemstreek-GM:Leemstreek   22.8346407  20.3559366  25.3133447 0.0000000
## SM:Leemstreek-GM:Leemstreek    1.0640082  -1.1866336   3.3146501 0.9890403
## SMGR:Leemstreek-GM:Leemstreek  6.9848128   4.3992071   9.5704184 0.0000000
## GM:Leemstrek-GM:Leemstreek            NA          NA          NA        NA
## PG:Leemstrek-GM:Leemstreek            NA          NA          NA        NA
## SM:Leemstrek-GM:Leemstreek    -0.5535593 -13.5574124  12.4502938 1.0000000
## SMGR:Leemstrek-GM:Leemstreek          NA          NA          NA        NA
## GM:Lemstreek-GM:Leemstreek            NA          NA          NA        NA
## PG:Lemstreek-GM:Leemstreek    22.3564407   9.3525876  35.3602938 0.0000002
## SM:Lemstreek-GM:Leemstreek            NA          NA          NA        NA
## SMGR:Lemstreek-GM:Leemstreek          NA          NA          NA        NA
```

```
## SM:Leemstreek-PG:Leemstreek    -21.7706324 -24.1312869 -19.4099780 0.0000000
## SMGR:Leemstreek-PG:Leemstreek  -15.8498279 -18.5317409 -13.1679149 0.0000000
## GM:Leemstrek-PG:Leemstreek              NA          NA          NA        NA
## PG:Leemstrek-PG:Leemstreek              NA          NA          NA        NA
## SM:Leemstrek-PG:Leemstreek     -23.3882000 -36.4115442 -10.3648558 0.0000000
## SMGR:Leemstrek-PG:Leemstreek            NA          NA          NA        NA
## GM:Lemstreek-PG:Leemstreek              NA          NA          NA        NA
## PG:Lemstreek-PG:Leemstreek      -0.4782000 -13.5015442  12.5451442 1.0000000
## SM:Lemstreek-PG:Leemstreek              NA          NA          NA        NA
## SMGR:Lemstreek-PG:Leemstreek            NA          NA          NA        NA
## SMGR:Leemstreek-SM:Leemstreek    5.9208045   3.4481395   8.3934695 0.0000000
## GM:Leemstrek-SM:Leemstreek              NA          NA          NA        NA
## PG:Leemstrek-SM:Leemstreek              NA          NA          NA        NA
## SM:Leemstrek-SM:Leemstreek      -1.6175676 -14.5994361  11.3643010 1.0000000
## SMGR:Leemstrek-SM:Leemstreek            NA          NA          NA        NA
## GM:Lemstreek-SM:Leemstreek              NA          NA          NA        NA
## PG:Lemstreek-SM:Leemstreek      21.2924324   8.3105639  34.2743010 0.0000012
## SM:Lemstreek-SM:Leemstreek              NA          NA          NA        NA
## SMGR:Lemstreek-SM:Leemstreek            NA          NA          NA        NA
## GM:Leemstrek-SMGR:Leemstreek            NA          NA          NA        NA
## PG:Leemstrek-SMGR:Leemstreek            NA          NA          NA        NA
## SM:Leemstrek-SMGR:Leemstreek    -7.5383721 -20.5824849   5.5057407 0.9046043
## SMGR:Leemstrek-SMGR:Leemstreek          NA          NA          NA        NA
## GM:Lemstreek-SMGR:Leemstreek            NA          NA          NA        NA
## PG:Lemstreek-SMGR:Leemstreek    15.3716279   2.3275151  28.4157407 0.0045820
## SM:Lemstreek-SMGR:Leemstreek            NA          NA          NA        NA
## SMGR:Lemstreek-SMGR:Leemstreek          NA          NA          NA        NA
## PG:Leemstrek-GM:Leemstrek               NA          NA          NA        NA
## SM:Leemstrek-GM:Leemstrek               NA          NA          NA        NA
## SMGR:Leemstrek-GM:Leemstrek             NA          NA          NA        NA
## GM:Lemstreek-GM:Leemstrek               NA          NA          NA        NA
## PG:Lemstreek-GM:Leemstrek               NA          NA          NA        NA
## SM:Lemstreek-GM:Leemstrek               NA          NA          NA        NA
## SMGR:Lemstreek-GM:Leemstrek             NA          NA          NA        NA
## SM:Leemstrek-PG:Leemstrek               NA          NA          NA        NA
## SMGR:Leemstrek-PG:Leemstrek             NA          NA          NA        NA
## GM:Lemstreek-PG:Leemstrek               NA          NA          NA        NA
## PG:Lemstreek-PG:Leemstrek               NA          NA          NA        NA
## SM:Lemstreek-PG:Leemstrek               NA          NA          NA        NA
## SMGR:Lemstreek-PG:Leemstrek             NA          NA          NA        NA
## SMGR:Leemstrek-SM:Leemstrek             NA          NA          NA        NA
## GM:Lemstreek-SM:Leemstrek               NA          NA          NA        NA
## PG:Lemstreek-SM:Leemstrek       22.9100000   4.6736704  41.1463296 0.0013788
## SM:Lemstreek-SM:Leemstrek               NA          NA          NA        NA
## SMGR:Lemstreek-SM:Leemstrek             NA          NA          NA        NA
## GM:Lemstreek-SMGR:Leemstrek             NA          NA          NA        NA
## PG:Lemstreek-SMGR:Leemstrek             NA          NA          NA        NA
## SM:Lemstreek-SMGR:Leemstrek             NA          NA          NA        NA
## SMGR:Lemstreek-SMGR:Leemstrek           NA          NA          NA        NA
## PG:Lemstreek-GM:Lemstreek               NA          NA          NA        NA
## SM:Lemstreek-GM:Lemstreek               NA          NA          NA        NA
## SMGR:Lemstreek-GM:Lemstreek             NA          NA          NA        NA
## SM:Lemstreek-PG:Lemstreek               NA          NA          NA        NA
## SMGR:Lemstreek-PG:Lemstreek             NA          NA          NA        NA
```

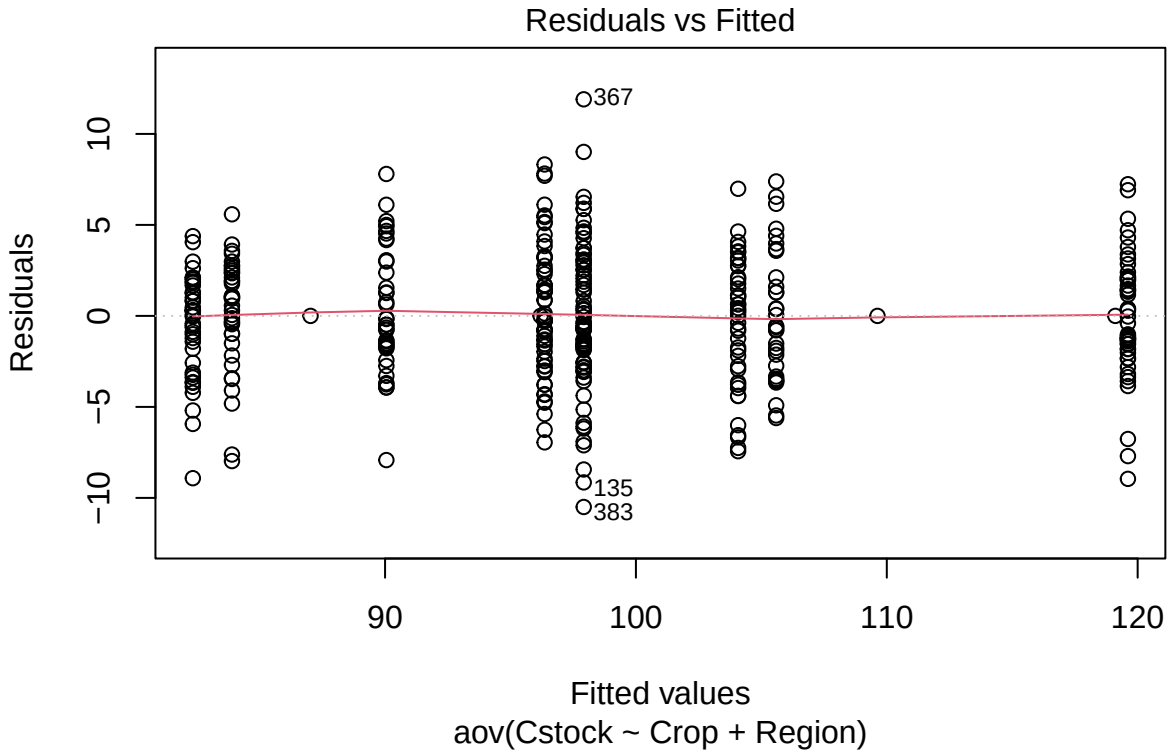
```
## SMGR:Lemstreek-SM:Lemstreek             NA          NA          NA        NA
```

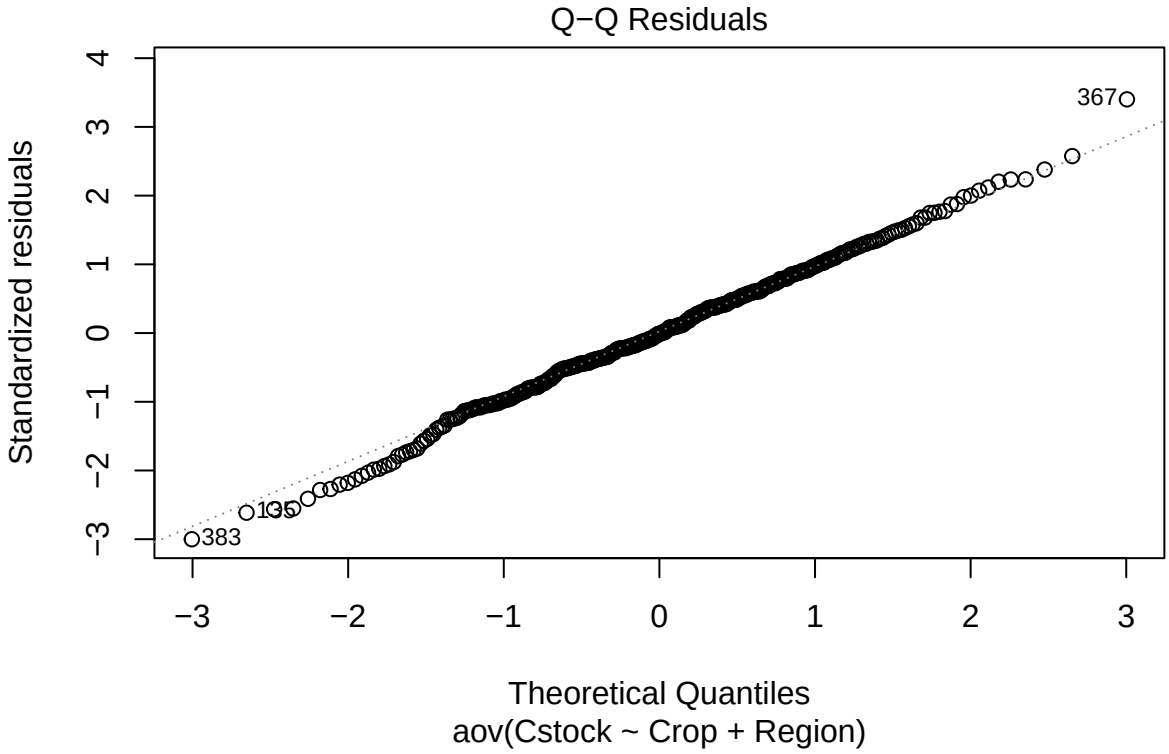
```
#Two-Way ANOVA Additive model
anova_model2 <- aov(Cstock ~ Crop + Region, data = data_clean)
summary(anova_model2)
```

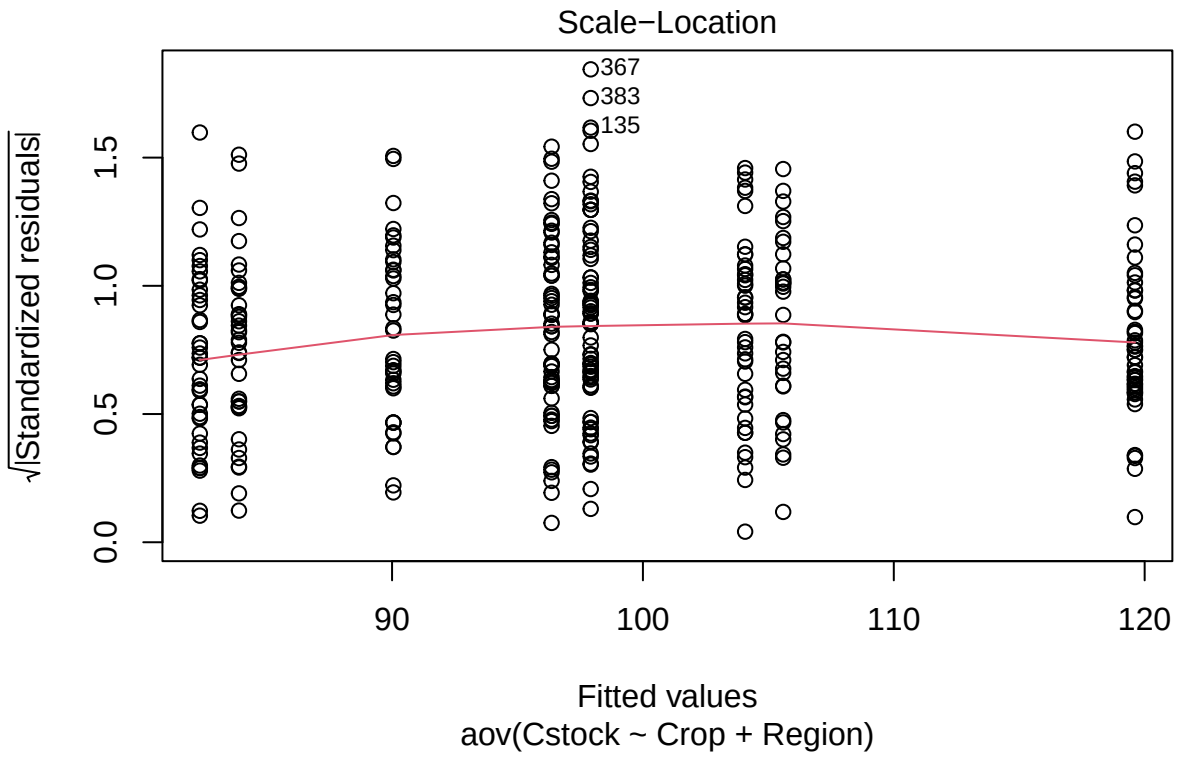
```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Crop          3  30019   10006   808.9 <2e-16 ***
## Region        5  17712    3542   286.4 <2e-16 ***
## Residuals   371   4589      12
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

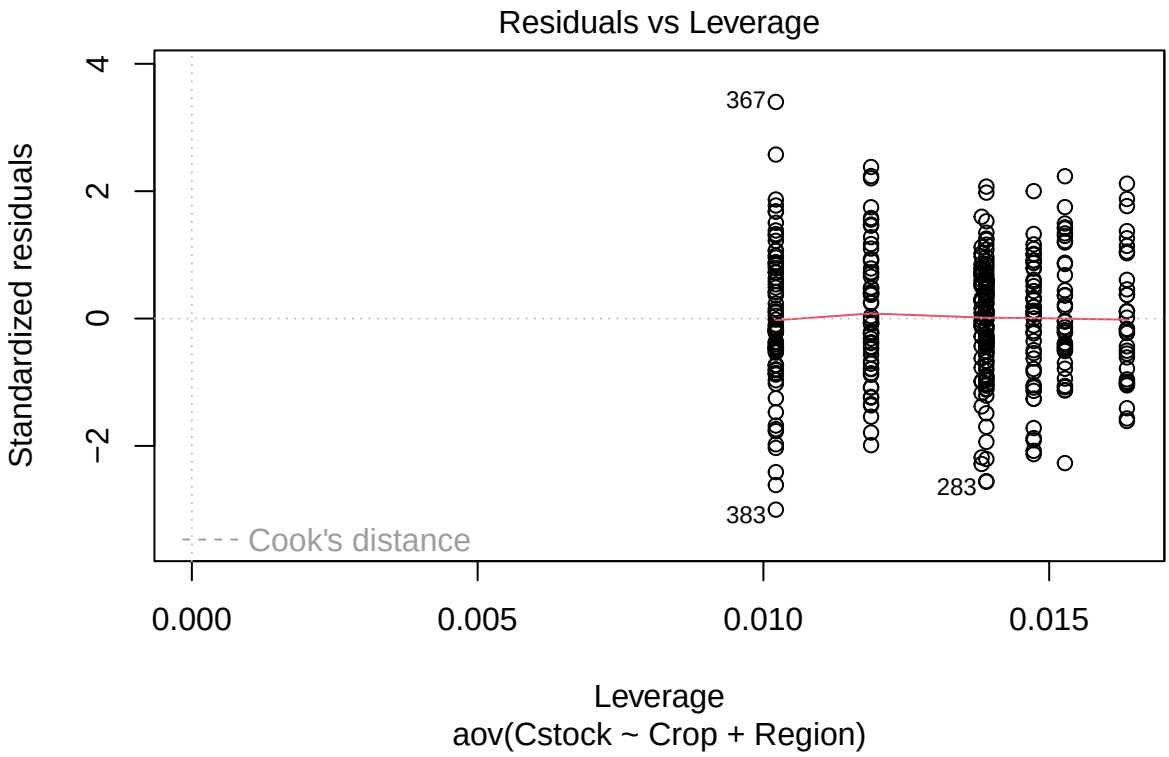
```
#There are some effect of Crop type and Region on Cstock
plot(anova_model2)
```

```
## Warning: not plotting observations with leverage one:
##   25, 82, 212, 320
```




Q−Q Residuals
Standardized residuals
Theoretical Quantiles
aov(Cstock ~ Crop + Region)
367
195
383
−3
−2
−1
0
1
2
3
4


Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
367
383
135
90
100
110
120
Fitted values
aov(Cstock ~ Crop + Region)


Residuals vs Leverage
Standardized residuals
4
2
0
−2
367
383
283
Cook's distance
0.000
0.005
0.010
0.015
Leverage
aov(Cstock ~ Crop + Region)